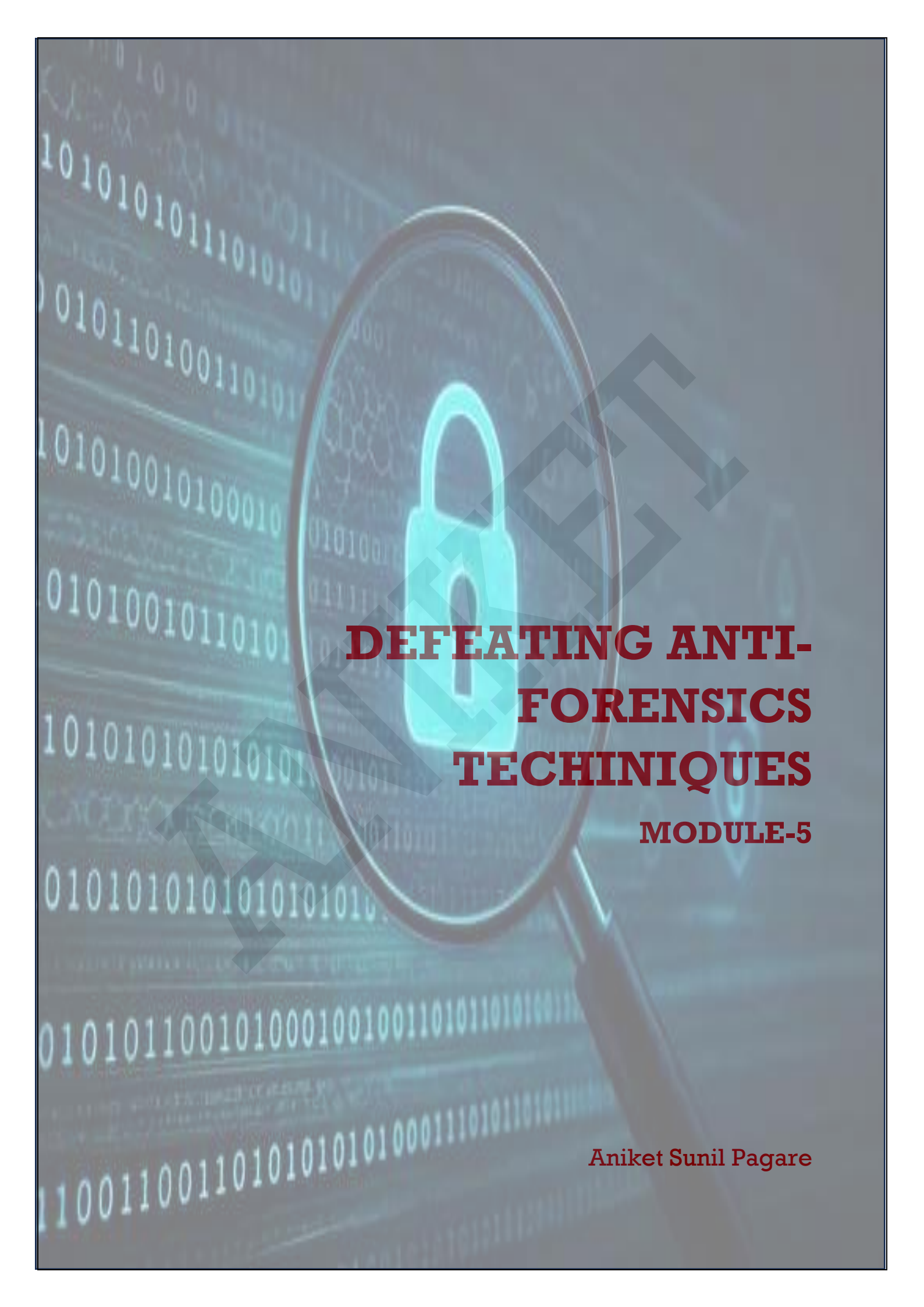




DEFEATING ANTI- FORENSICS TECHNIQUES

MODULE-5

Aniket Sunil Pagare

Defeating Anti-Forensics means detecting, countering, or bypassing techniques used by attackers to **hide, alter, or destroy digital evidence** to **avoid detection** during a forensic investigation.

What is Anti-Forensics?

Anti-Forensics refers to ****methods attackers use to:**

- Hide their presence
 - Obstruct forensic investigations
 - Destroy or manipulate evidence**
-

Defeating Anti-Forensics: What It Involves

Goal	Action
 Detect hidden or deleted data	Use forensic tools to recover deleted files, hidden partitions, or encrypted containers
 Identify manipulation	Analyze timestamps, logs, file metadata for inconsistencies
 Restore overwritten data	Use disk carving or memory analysis to recover overwritten artifacts
 Uncover obfuscation	Deobfuscate code, decode hidden payloads, detect steganography or alternate data streams
 Validate file integrity	Use hash verification (MD5, SHA1) to find tampered files
 Trace attacker actions	Rebuild activity timelines despite log tampering

Common Anti-Forensics Techniques and How They're Defeated

Anti-Forensics Technique	How to Defeat It
Data deletion	File carving, \$MFT analysis, unallocated space recovery
Timestomping (modifying timestamps)	Cross-reference file system metadata (e.g., MFT vs LogFile)
Log cleaning	Correlate logs from other sources (firewalls, network, system events)
Steganography	Use stego-detectors (like steghide, zsteg, or binwalk)
Encryption	Attempt brute-force, dictionary attacks, or memory dump analysis
Alternate Data Streams (ADS)	Use tools like Streams, FTK Imager, or X-Ways to view hidden streams
Rootkits	Use live forensic tools, memory analysis, or boot from trusted media
Obfuscation or Packing	Unpack executables, static/dynamic analysis, signature detection

Lab-1-: SSD File Carving On A Windows File System

SSD Carving refers to the process of **recovering deleted or hidden files** from a **Solid State Drive (SSD)** using **file carving techniques** — *without relying on the file system's metadata.*

A) SSD File Carving On A Windows File System Using Autopsy

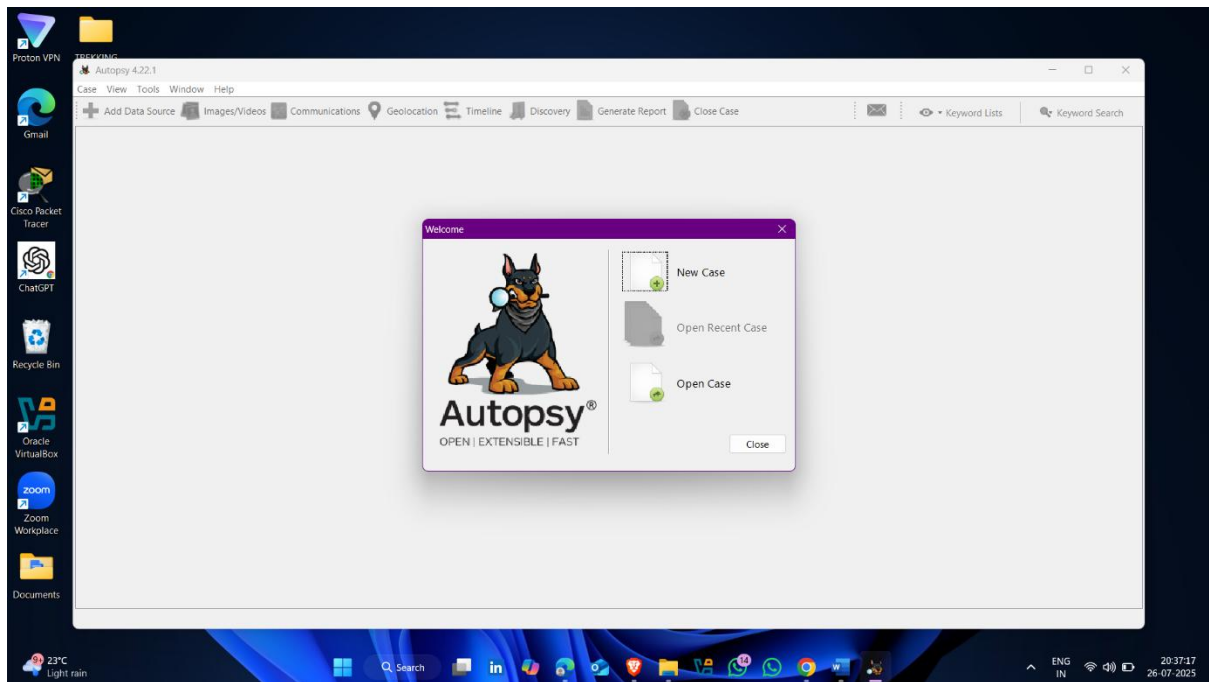
Autopsy is a **free and open-source digital forensics platform** used to analyze **hard drives, smartphones, memory images**, and other digital evidence. It provides a **graphical interface** for **The Sleuth Kit (TSK)** — a powerful set of command-line forensic tools.

How to use it -:

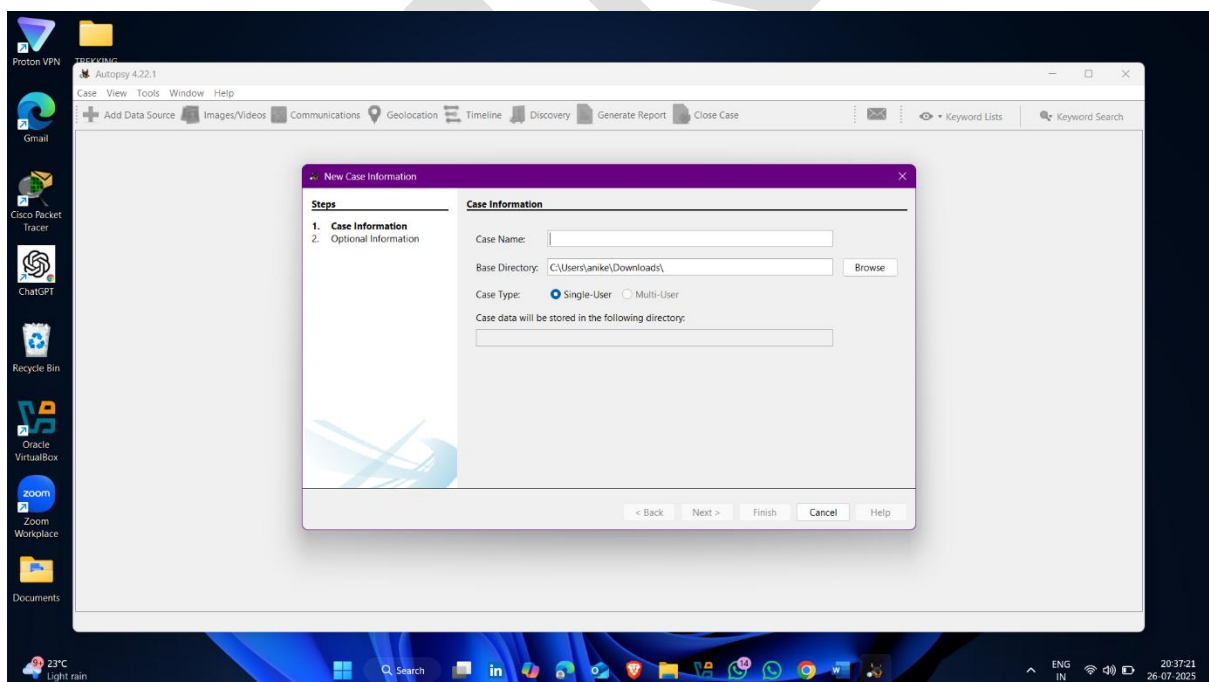
- Open Autopsy



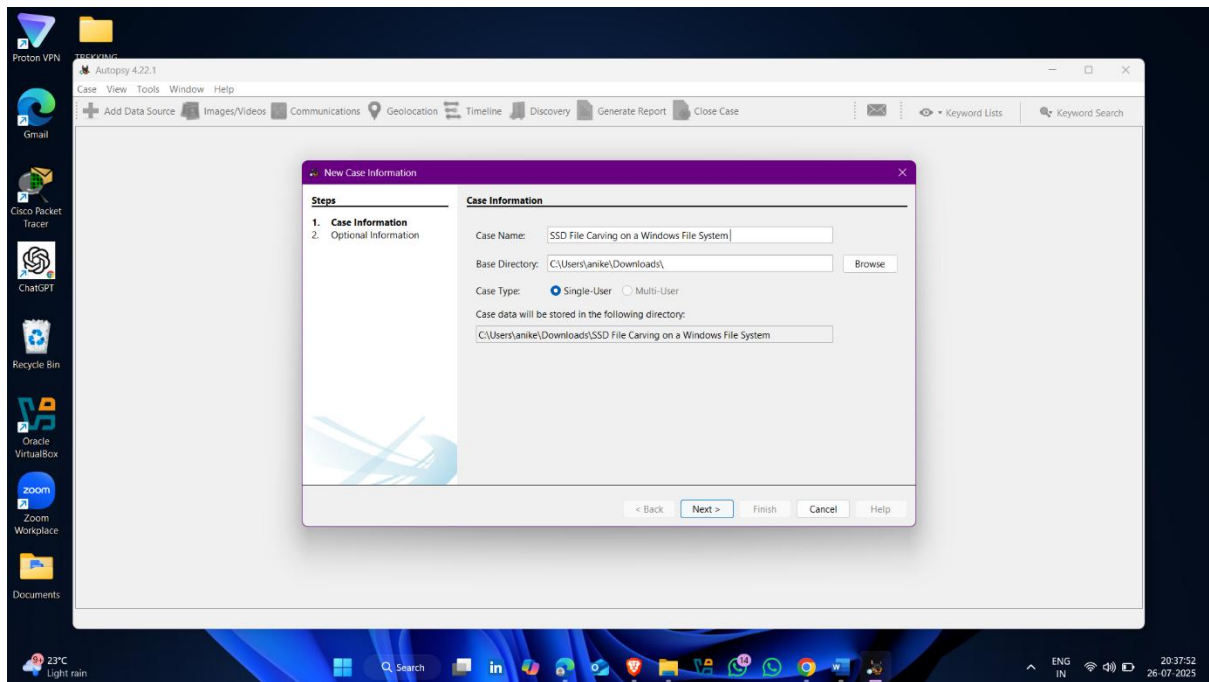
- click on **new case**



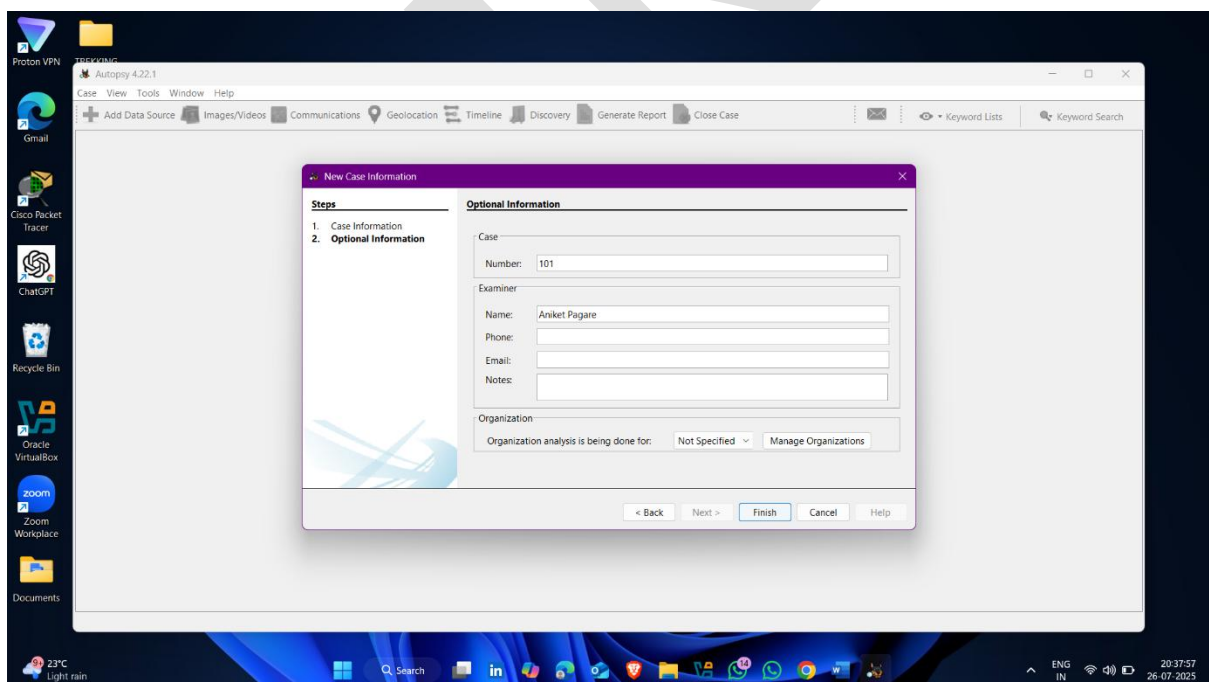
- enter case name



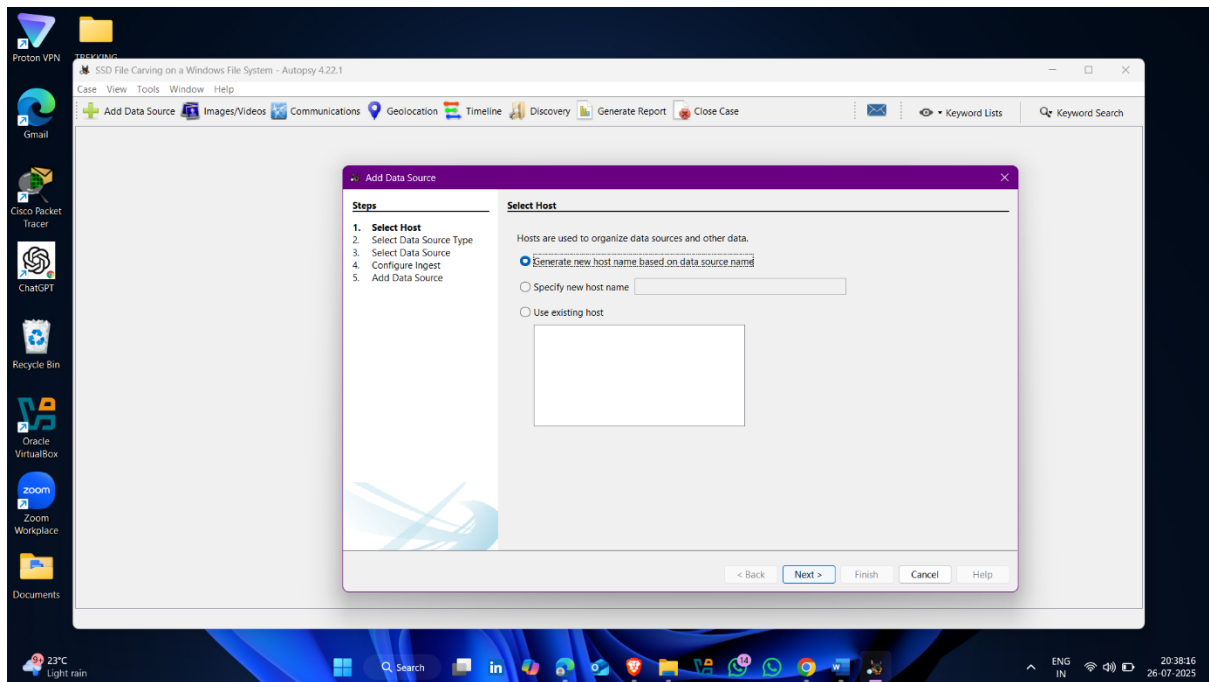
- click on **next**



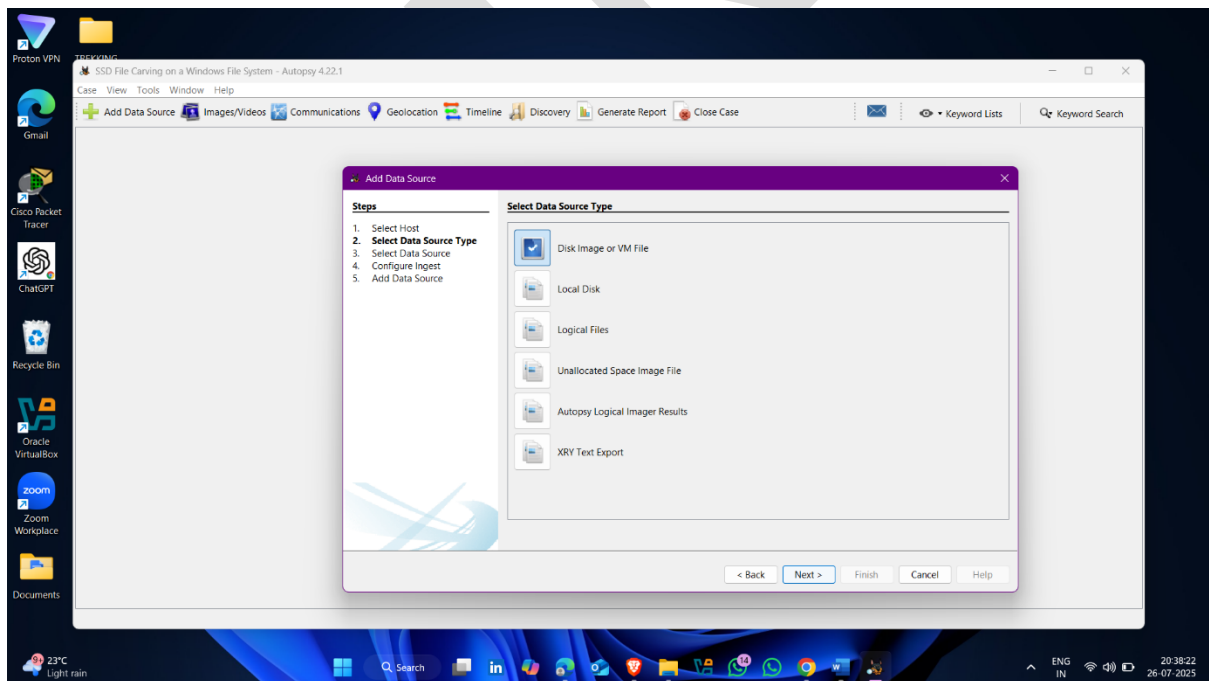
- enter case number and other details and click on finish



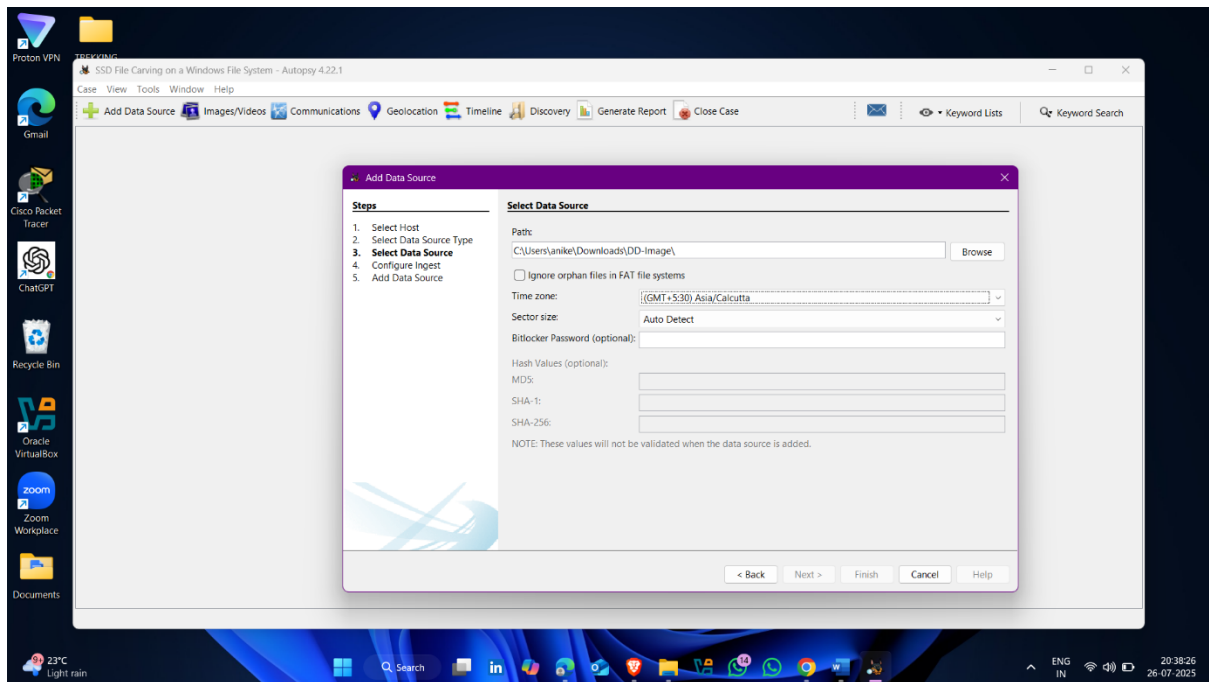
- select first option and click on next



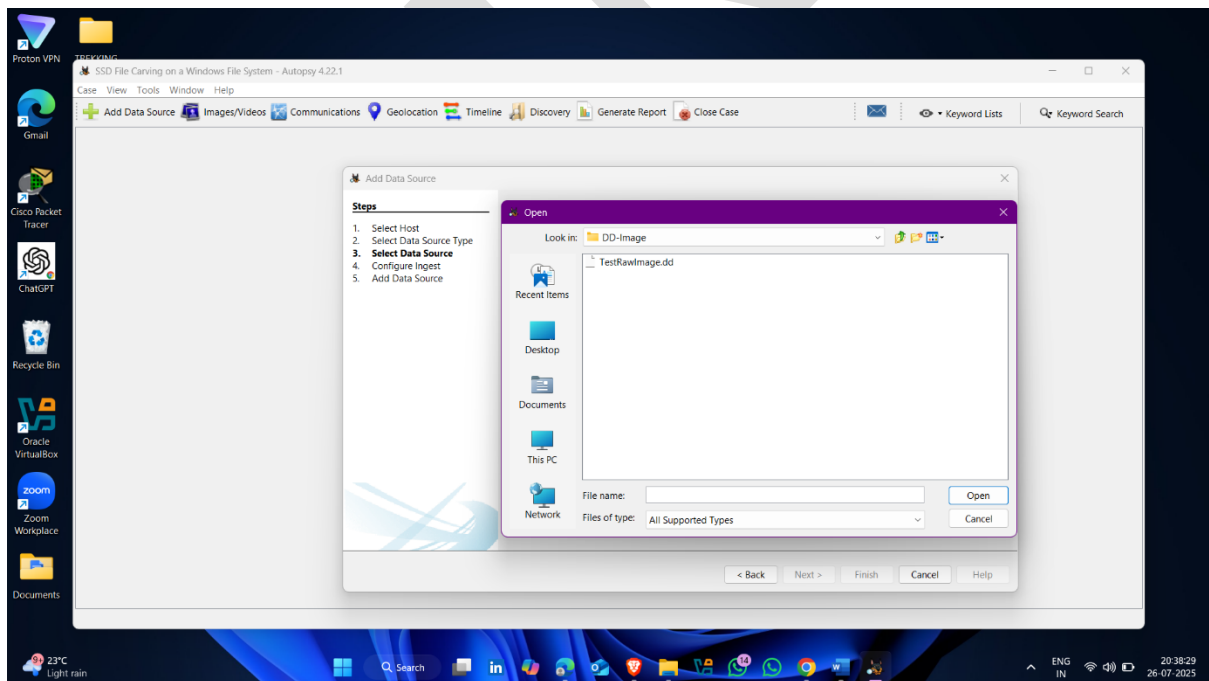
- select option first and click on next



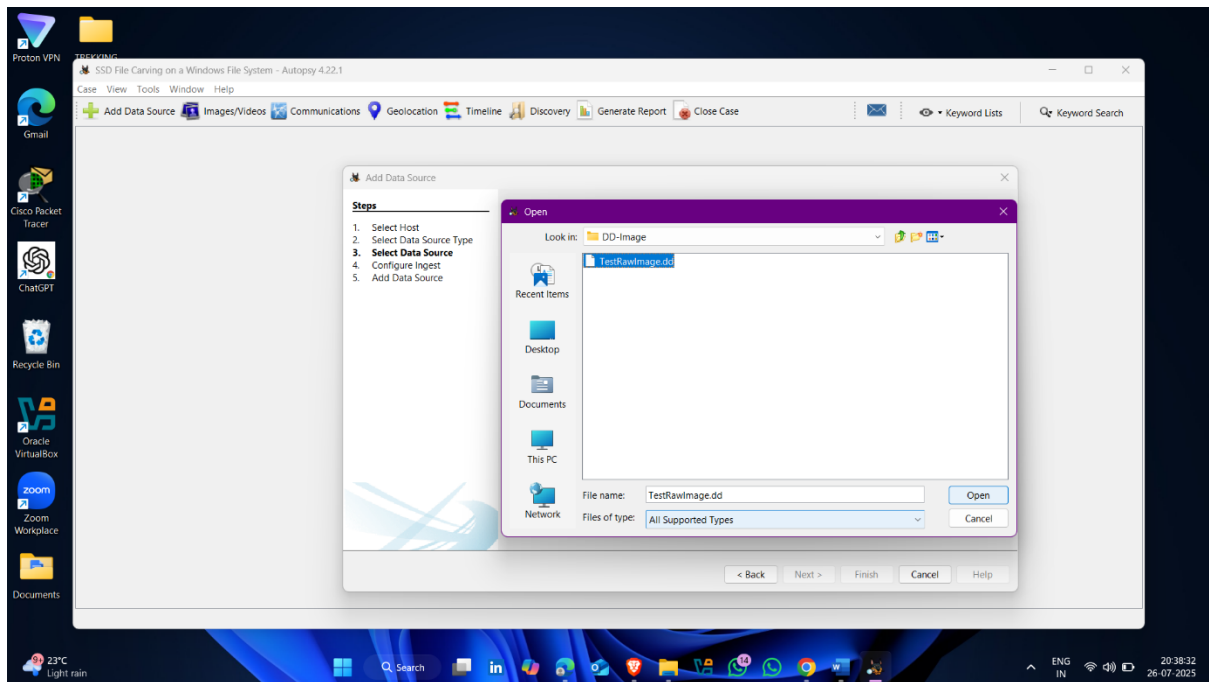
- click on Browse option



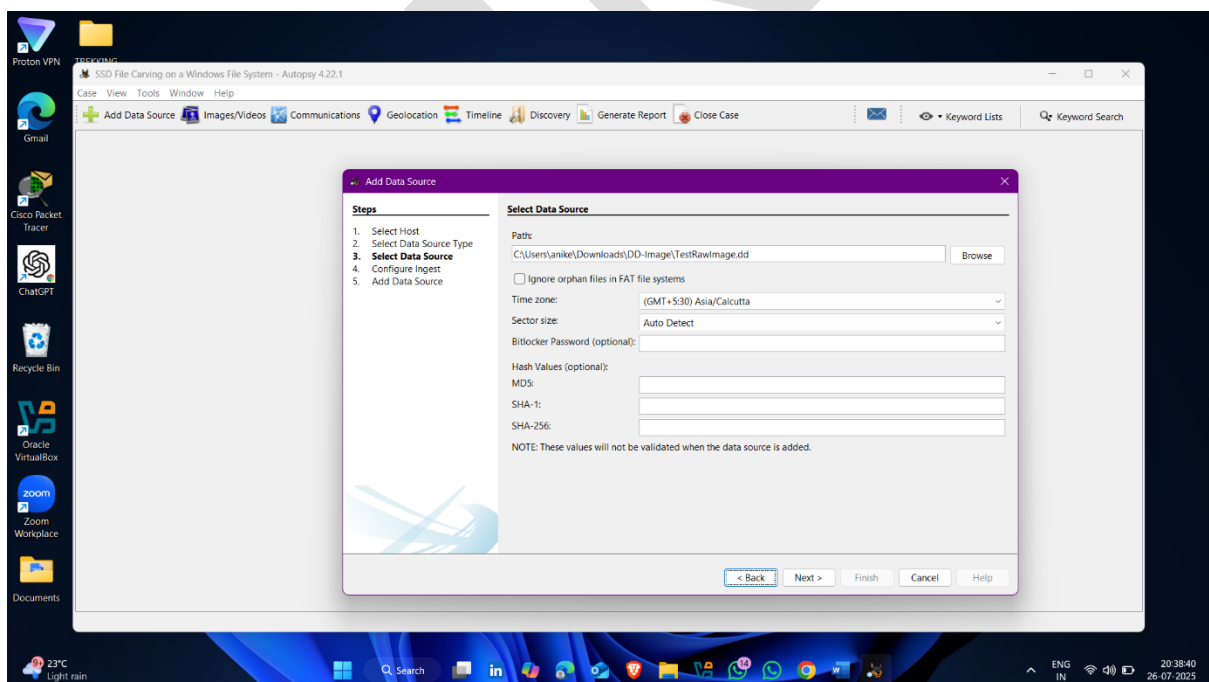
- and select image file



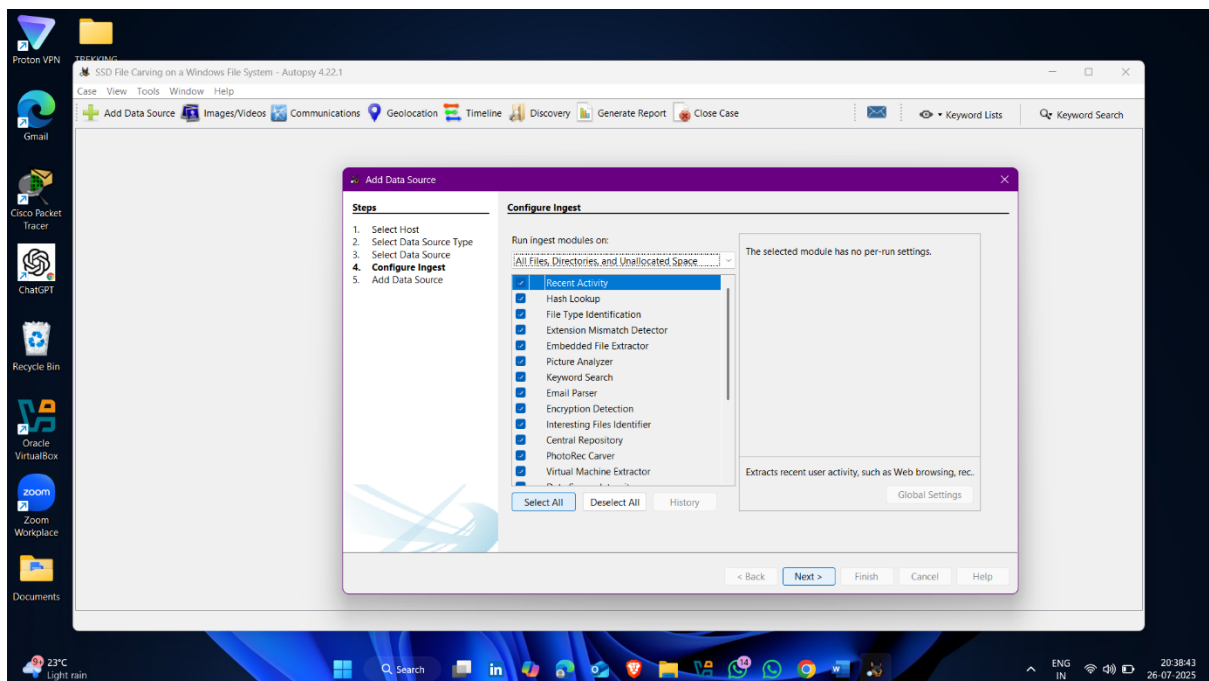
- select file and click on open



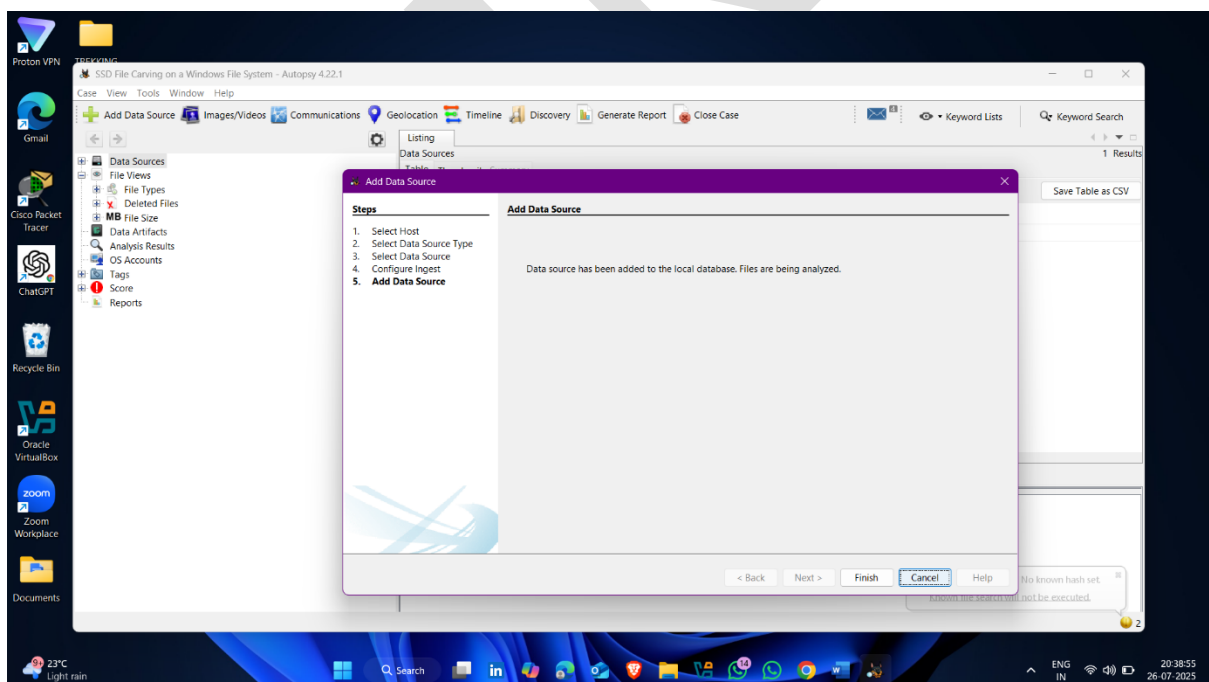
- click on **Next**



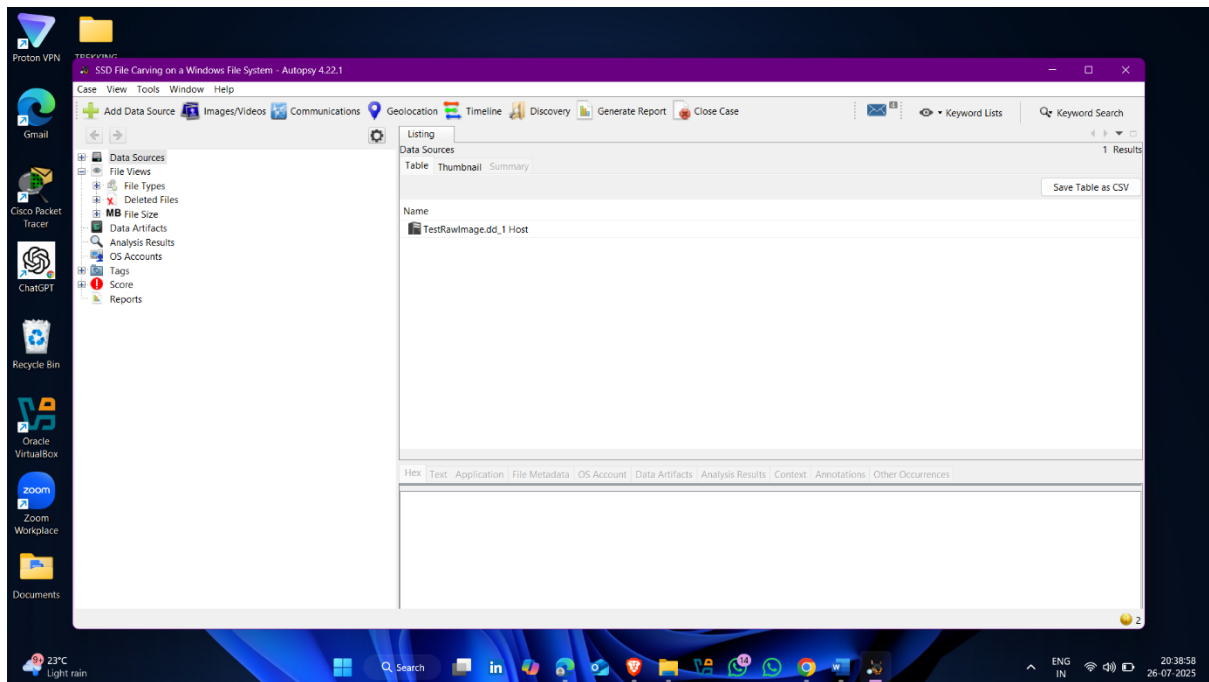
- once again click on next



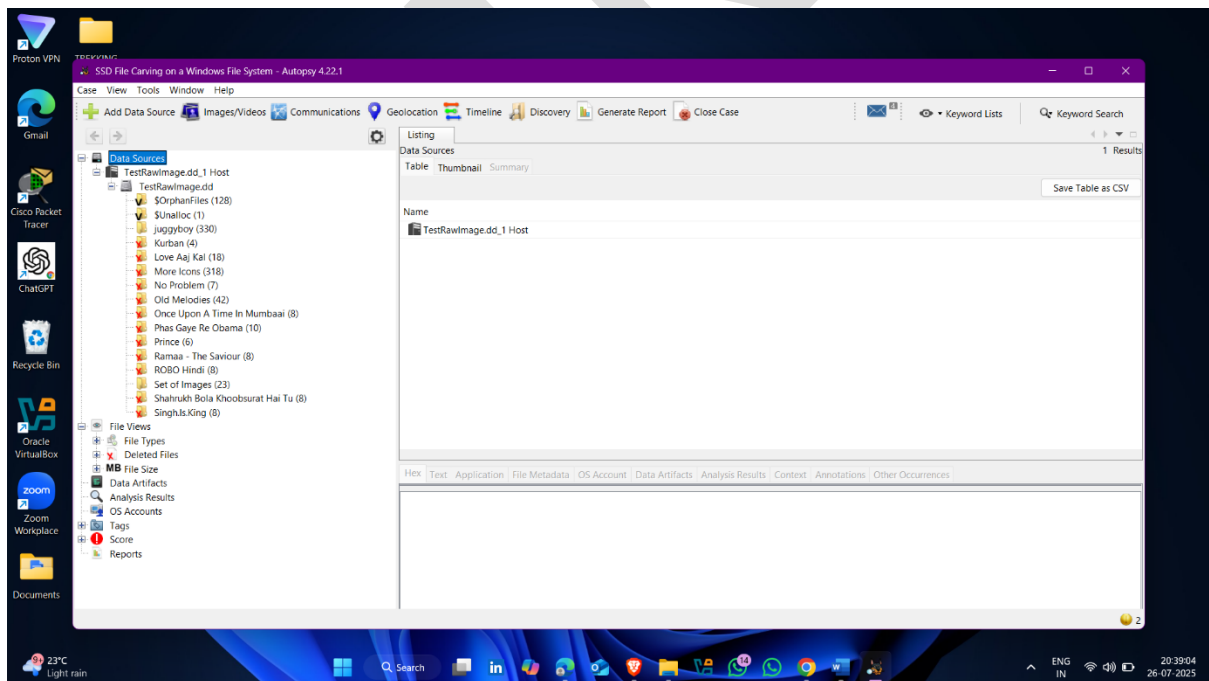
- click on finish

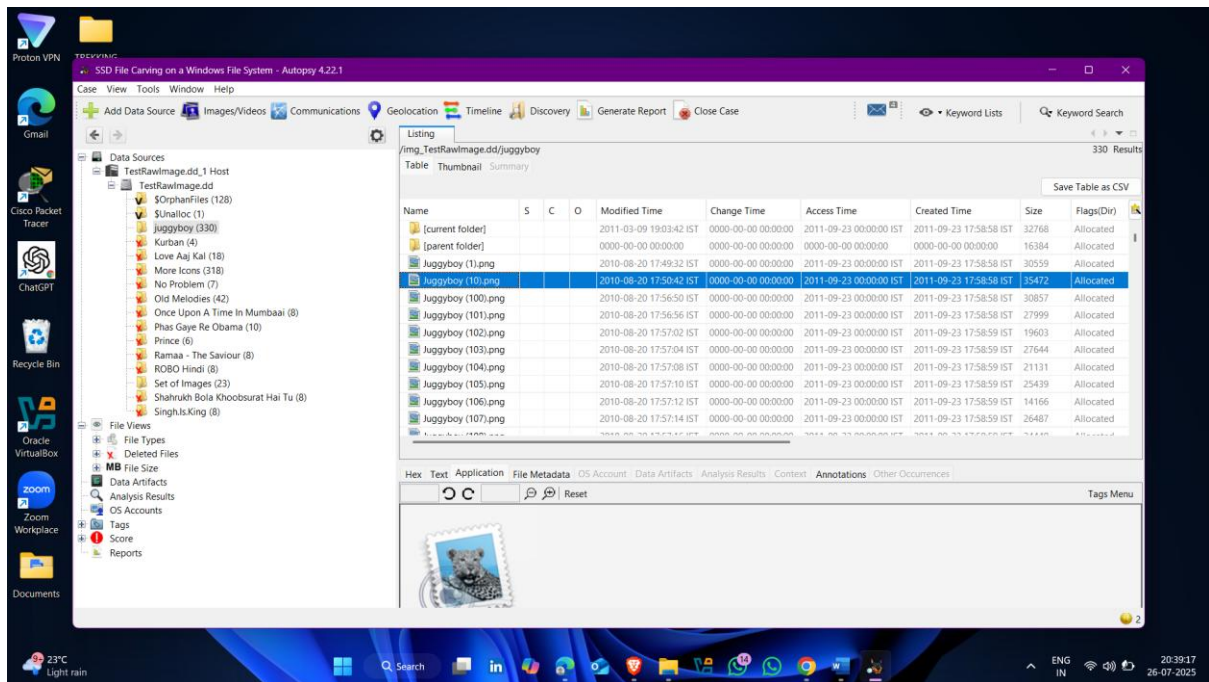


- click on data source

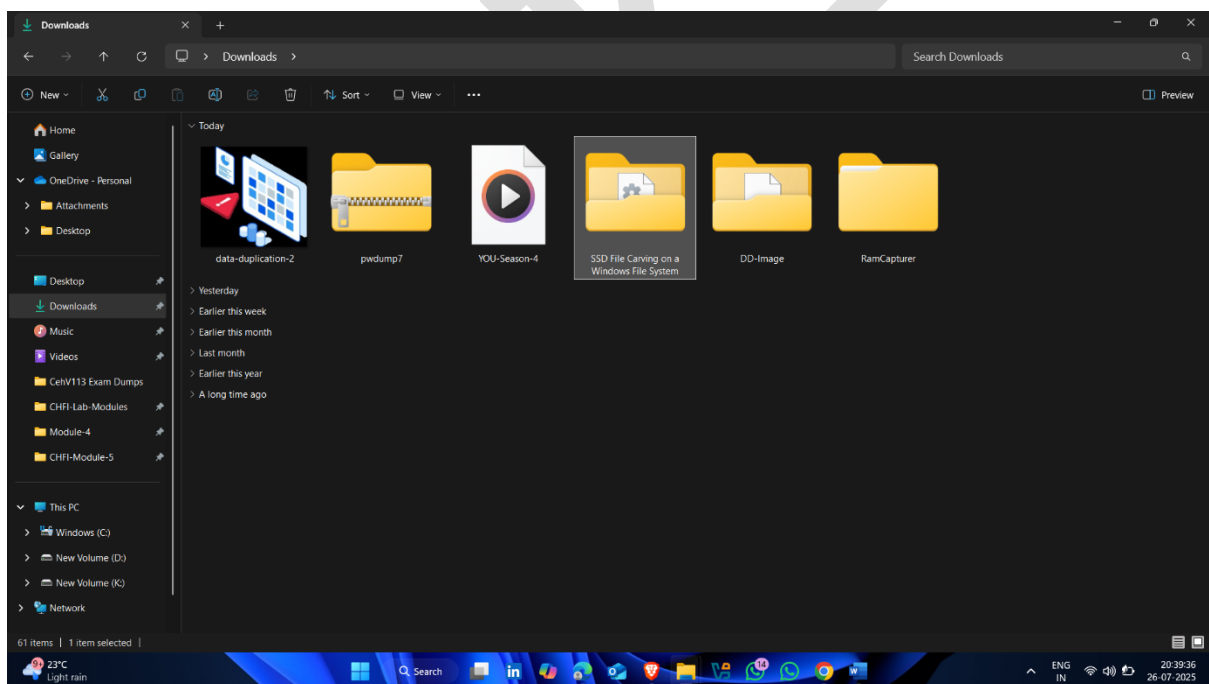


- all data are retrieve on image file

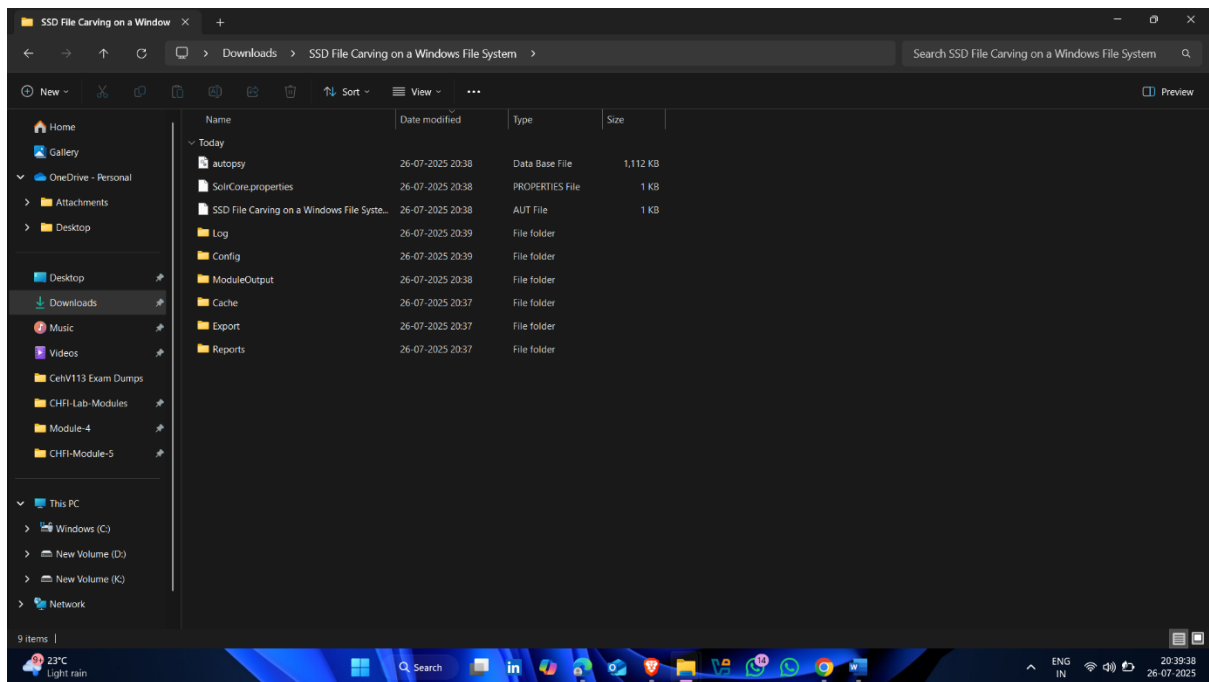




- also create a folder ✓👉



- all data recover ✓👉

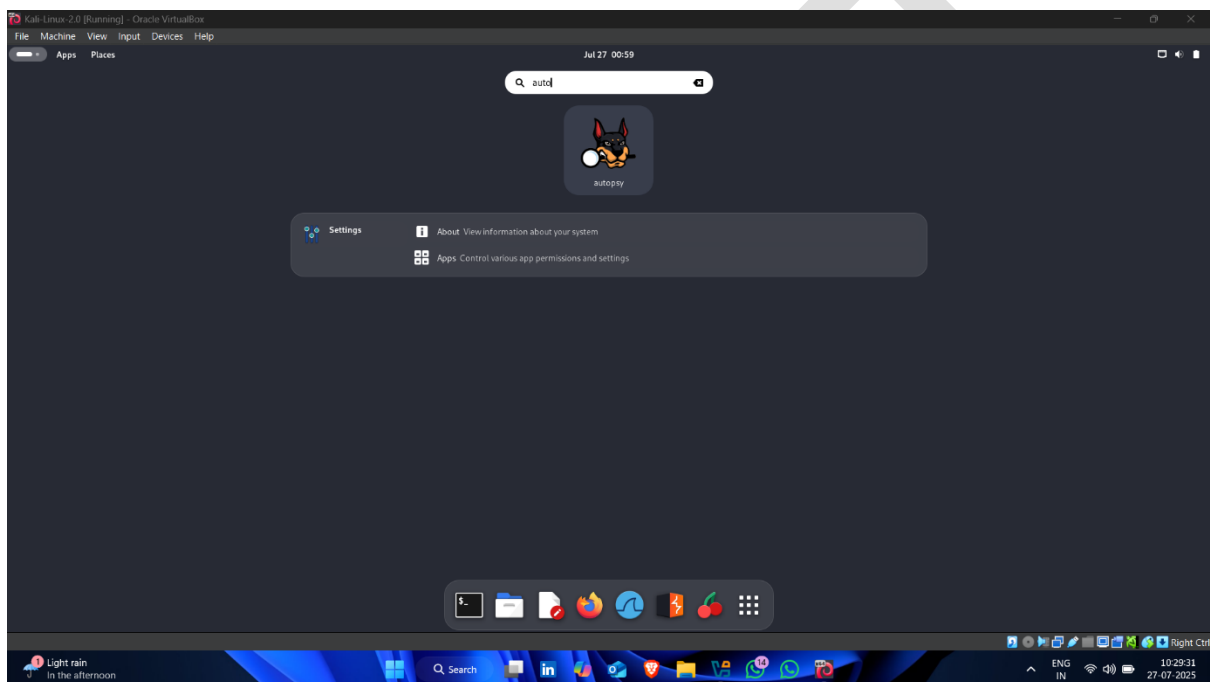


Lab-2-: SSD File Carving On A Linux File System

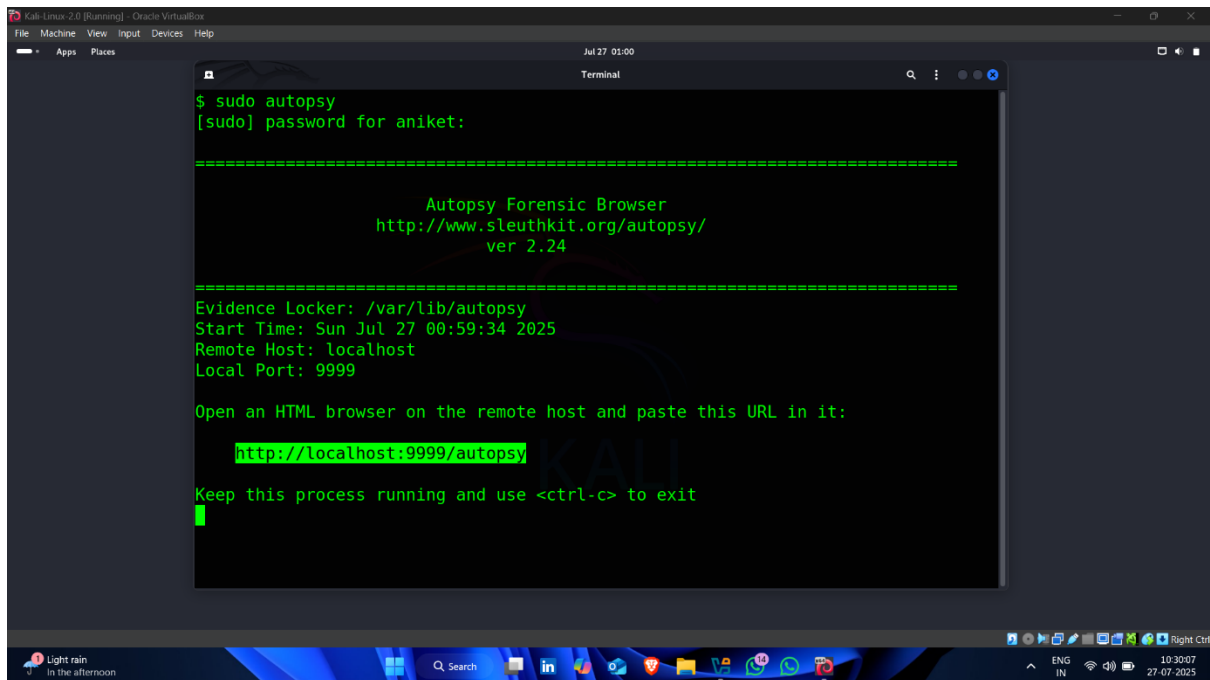
A) SSD File Carving On A Linux File System Using Autopsy

How to use it -:

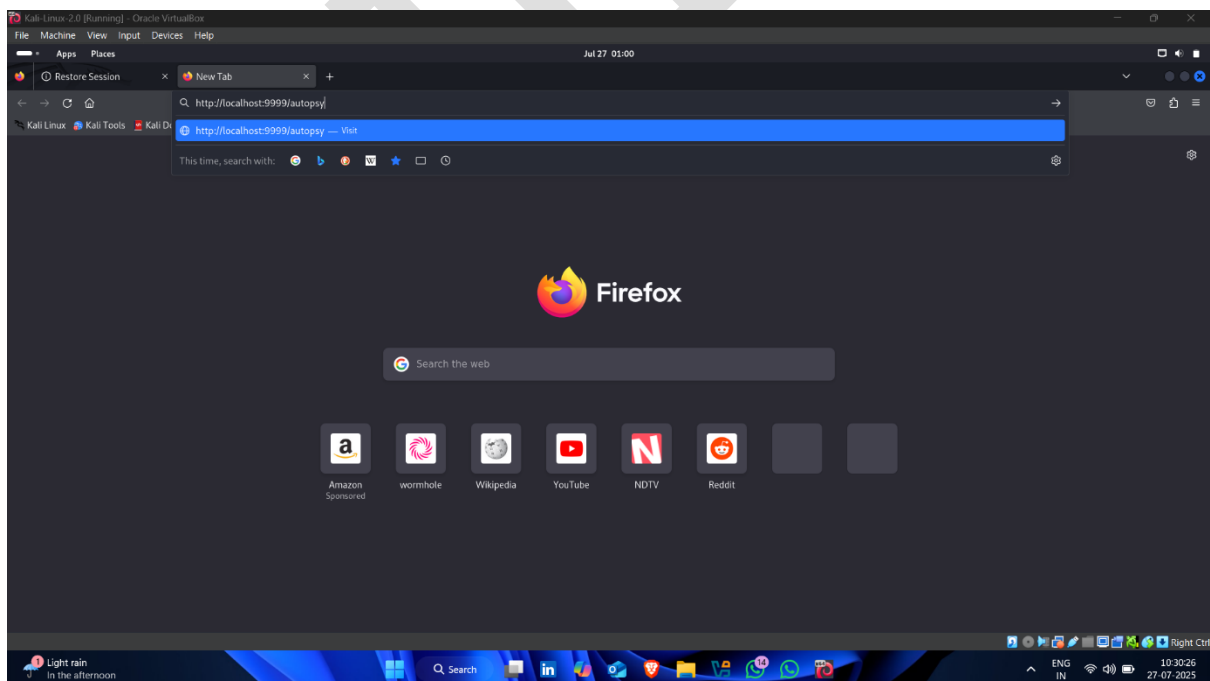
- Open Kali linux and search **Autopsy**



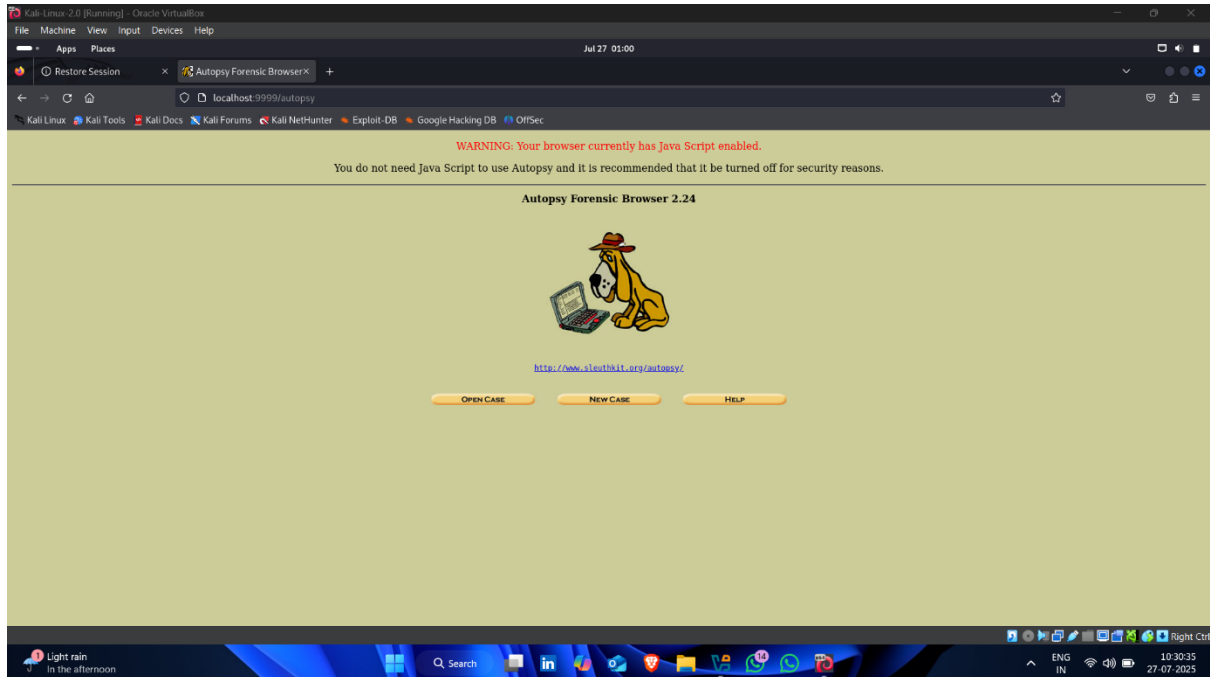
- Copy This url



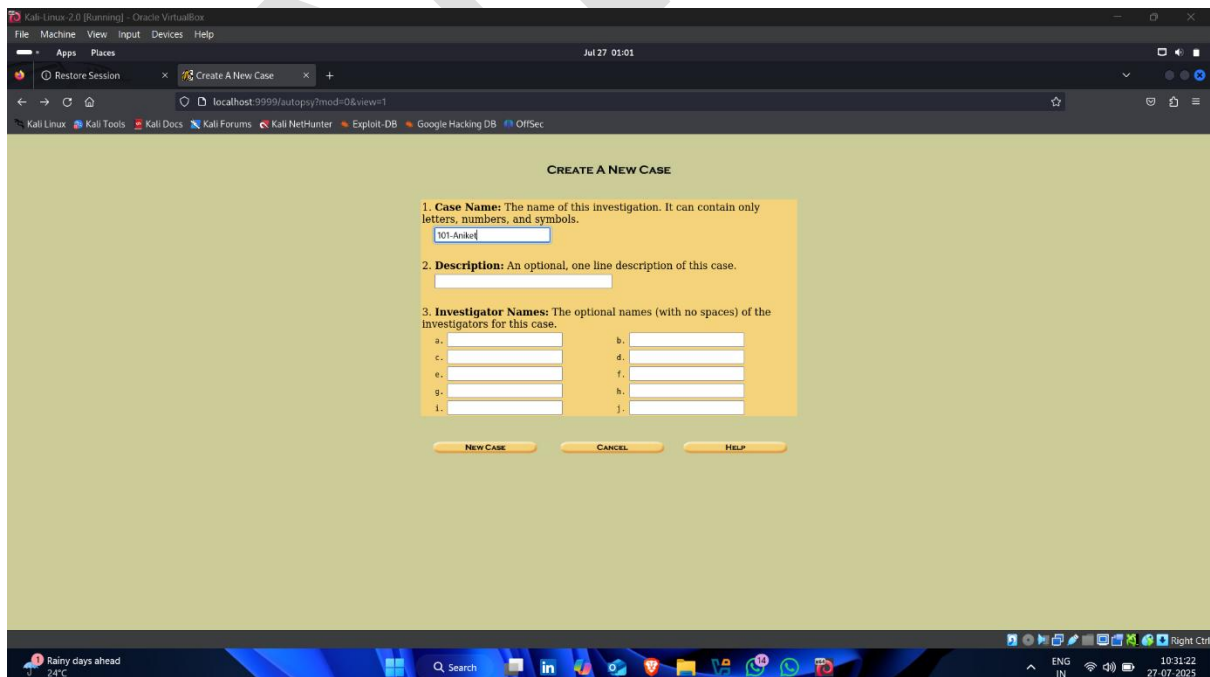
- Open Browser and paste in url section



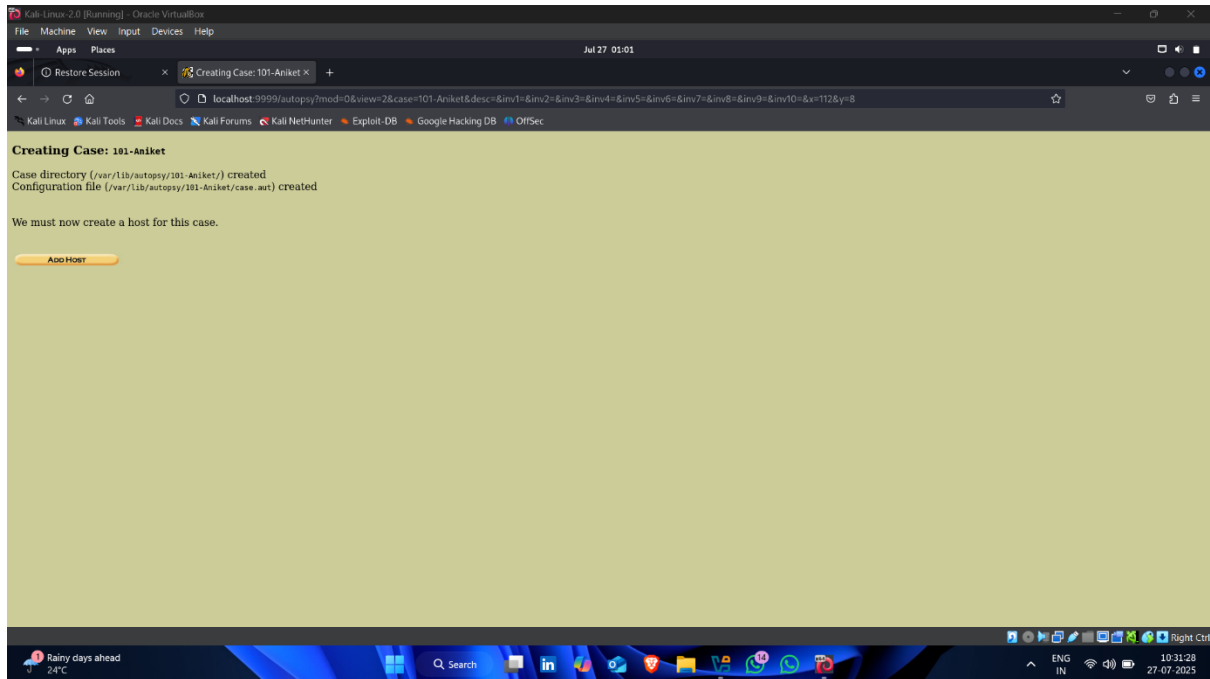
- Autopsy started
- Click on **New case**



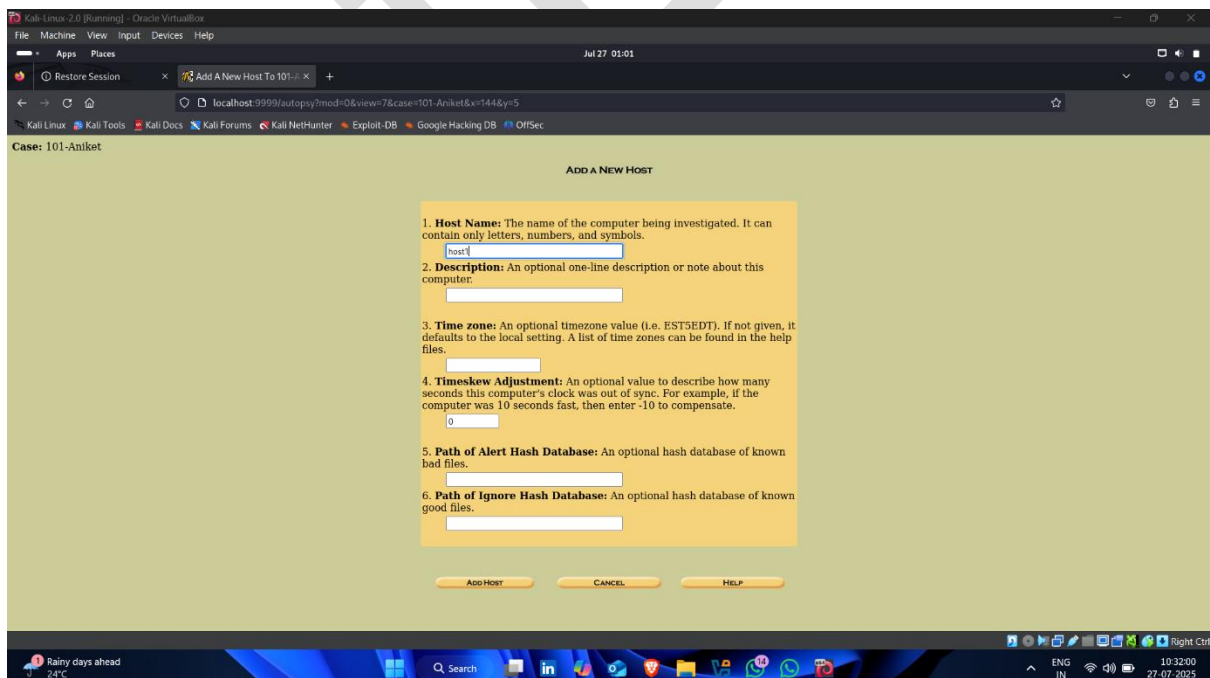
- Enter case number and name



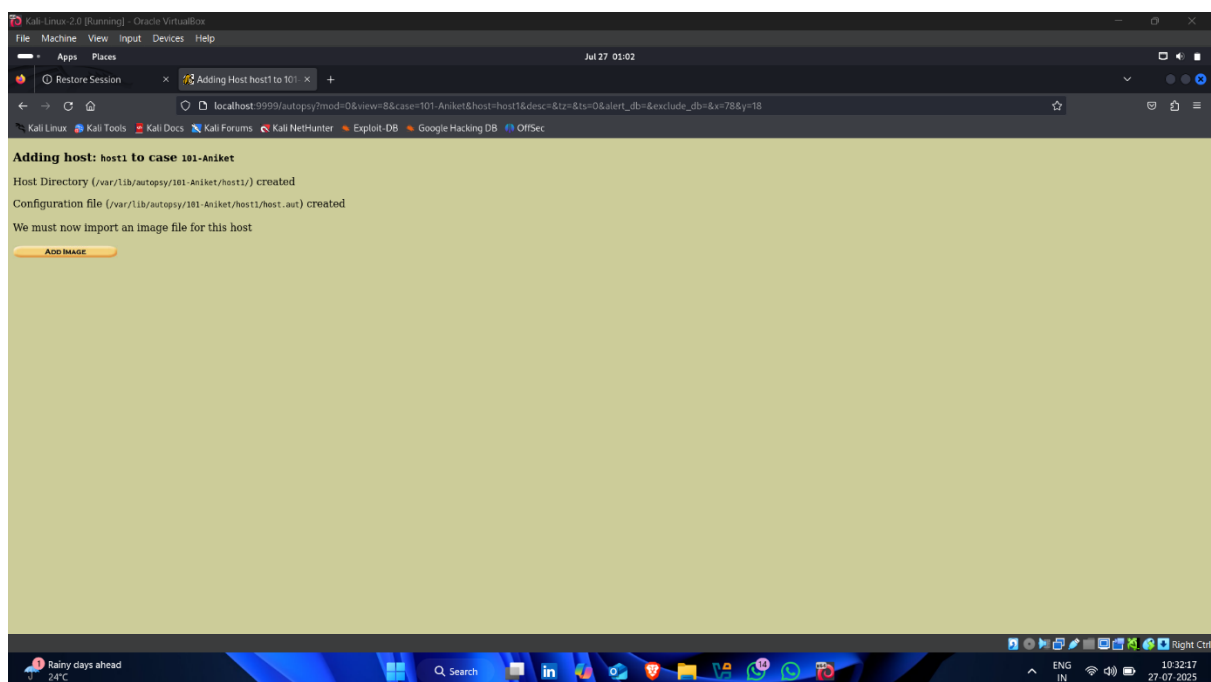
- Now click on **Add Host**



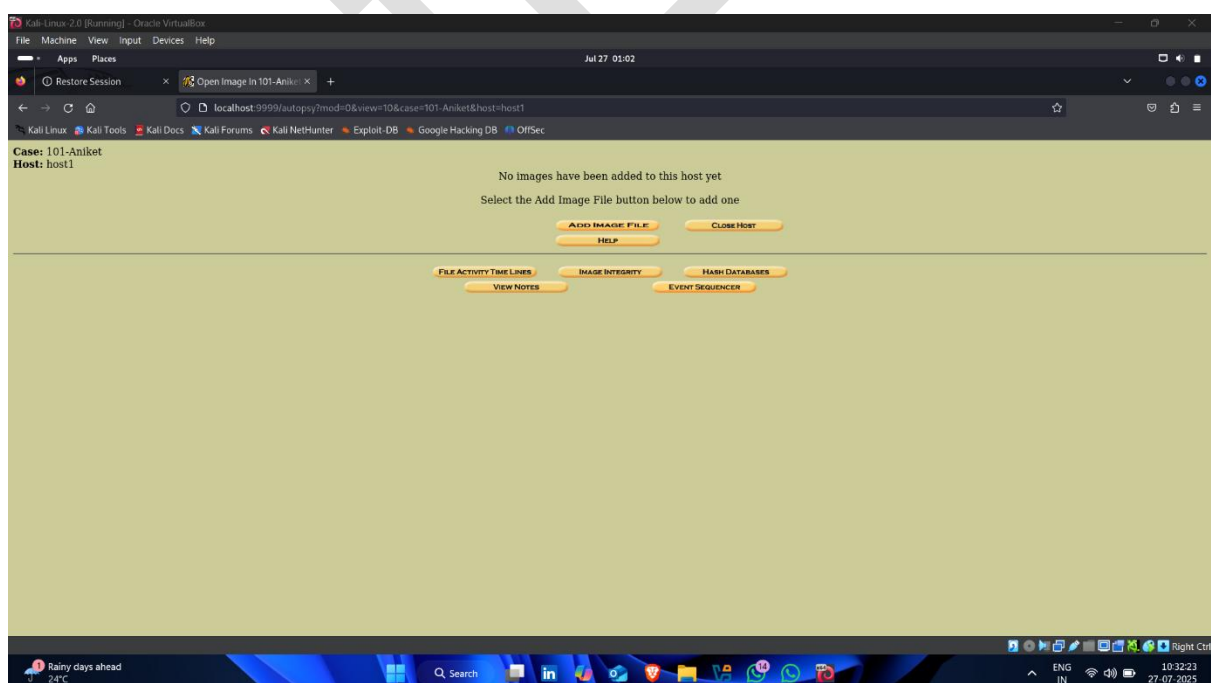
- Click on **Add Host**



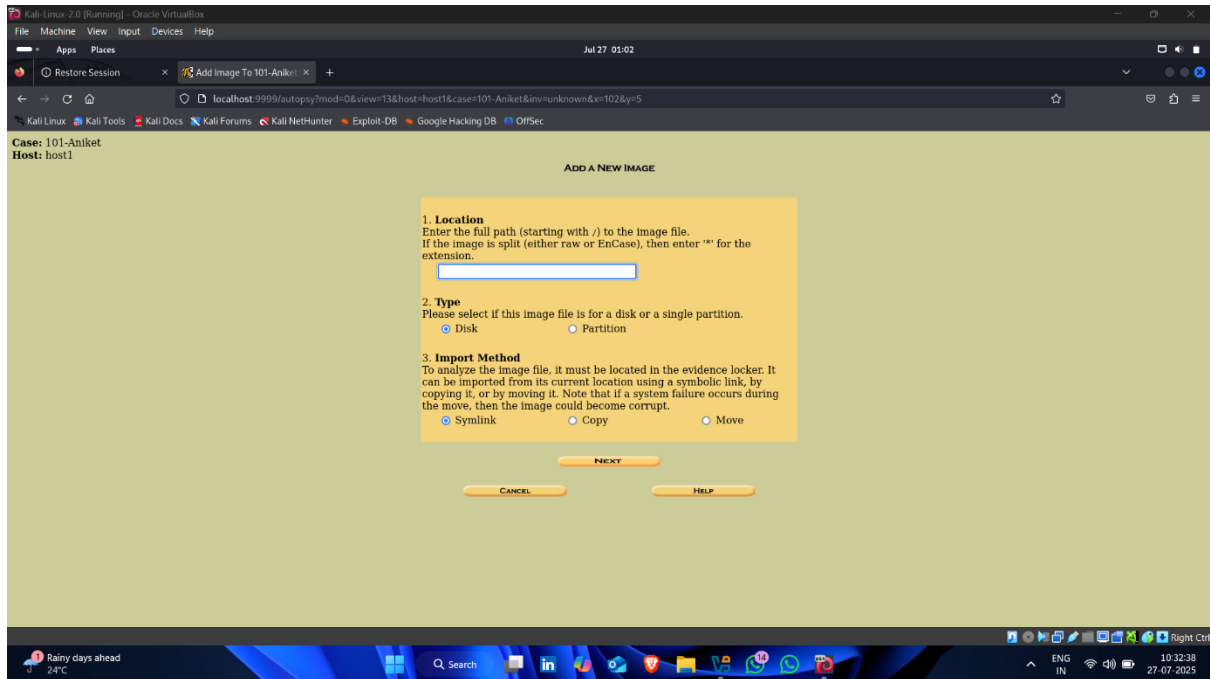
- Click on **Add Image**



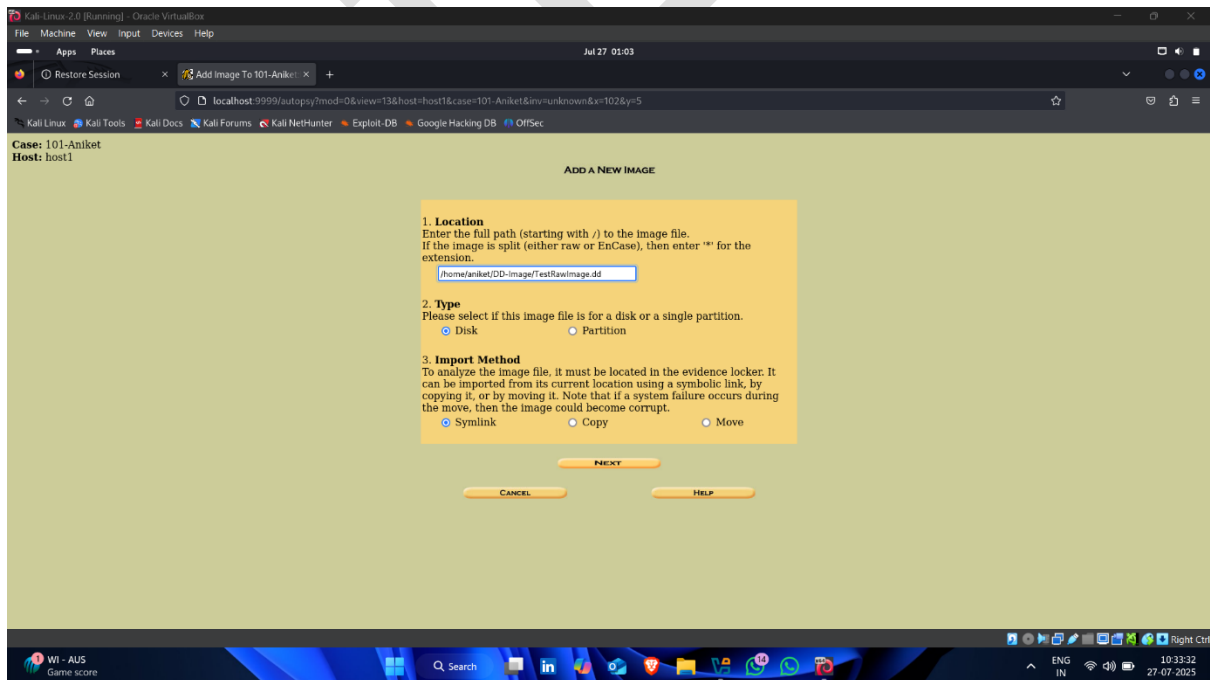
- Click on **Add Image File**



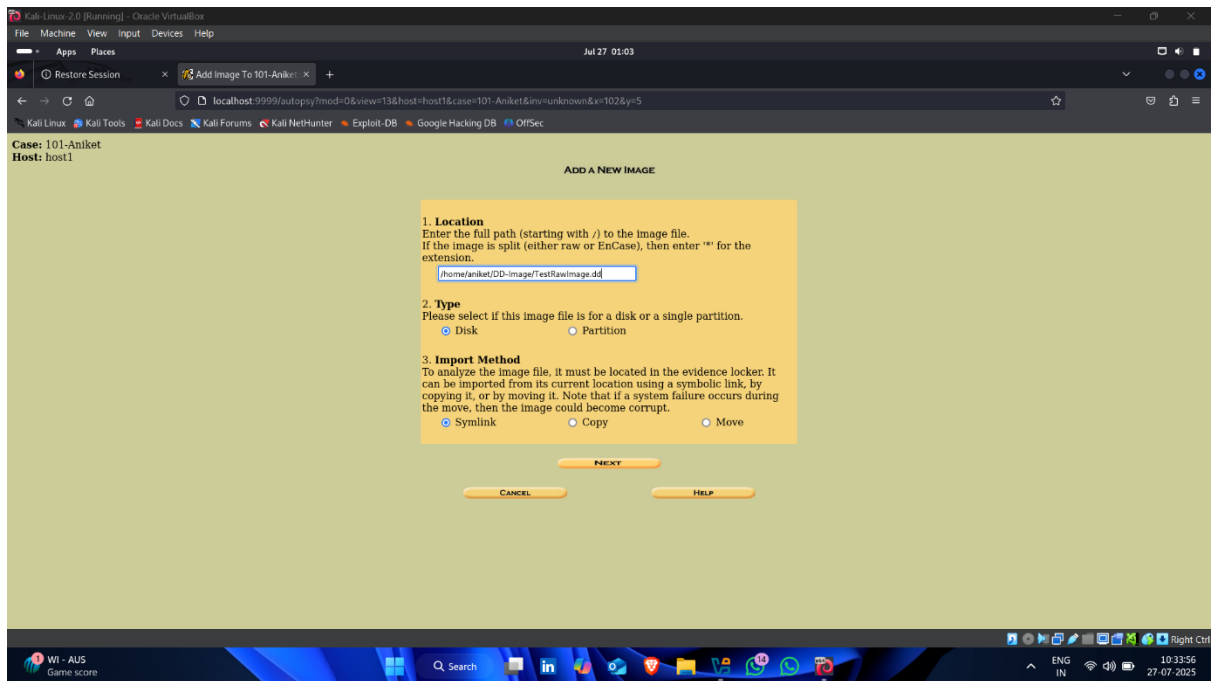
- Enter the full path of image file



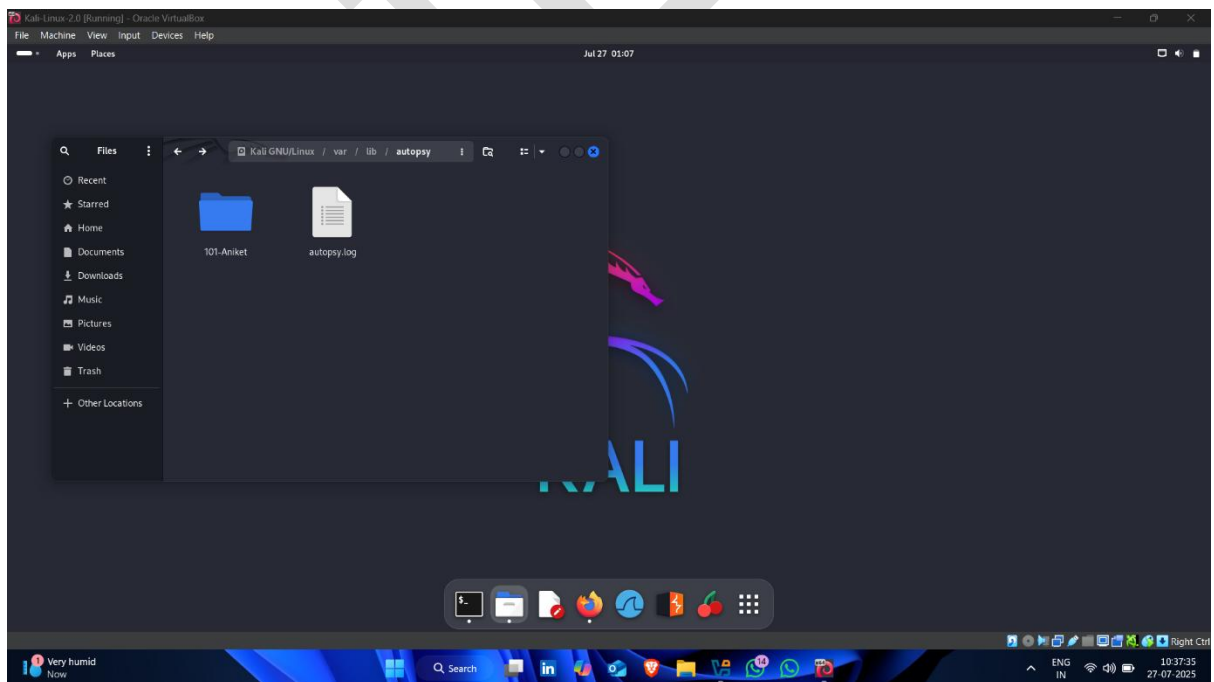
- Path of image file 📎 ✅



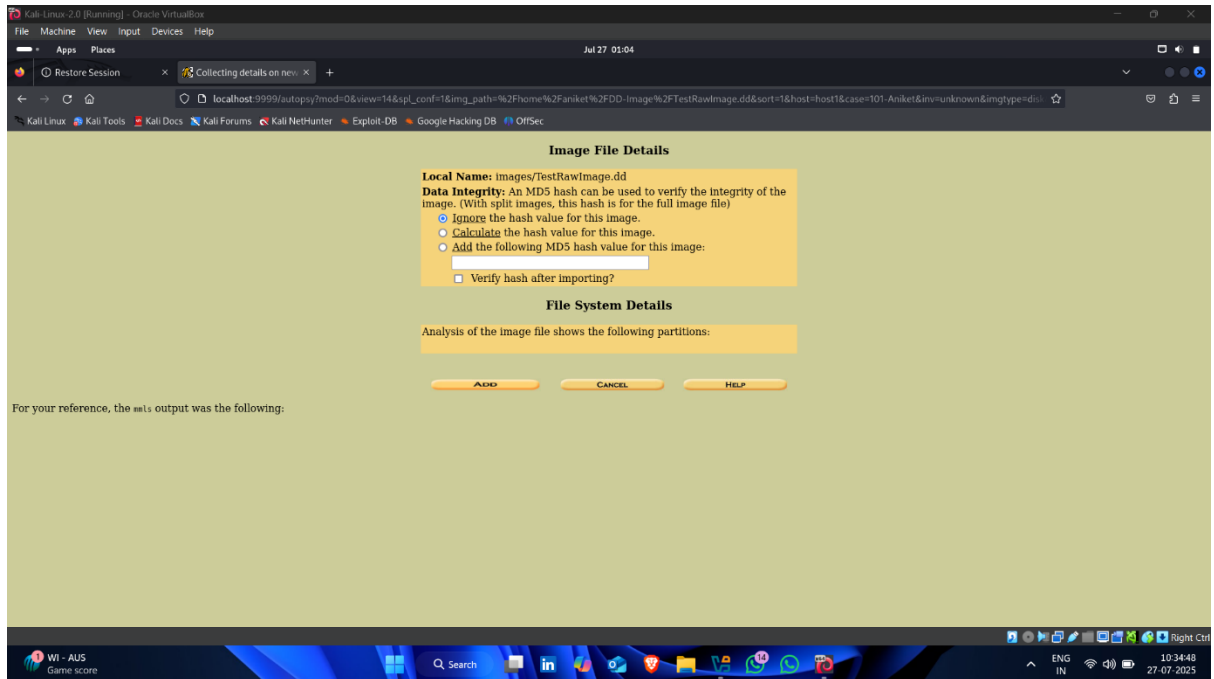
- Click on next



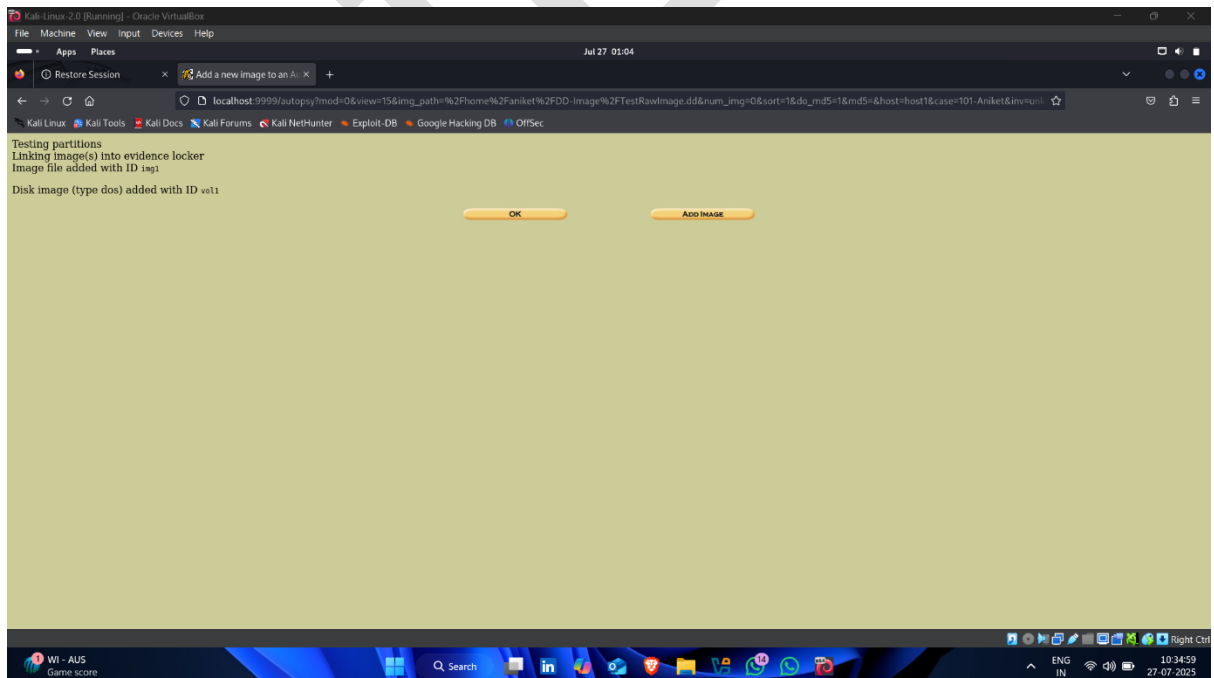
- Case folder created ✓👉



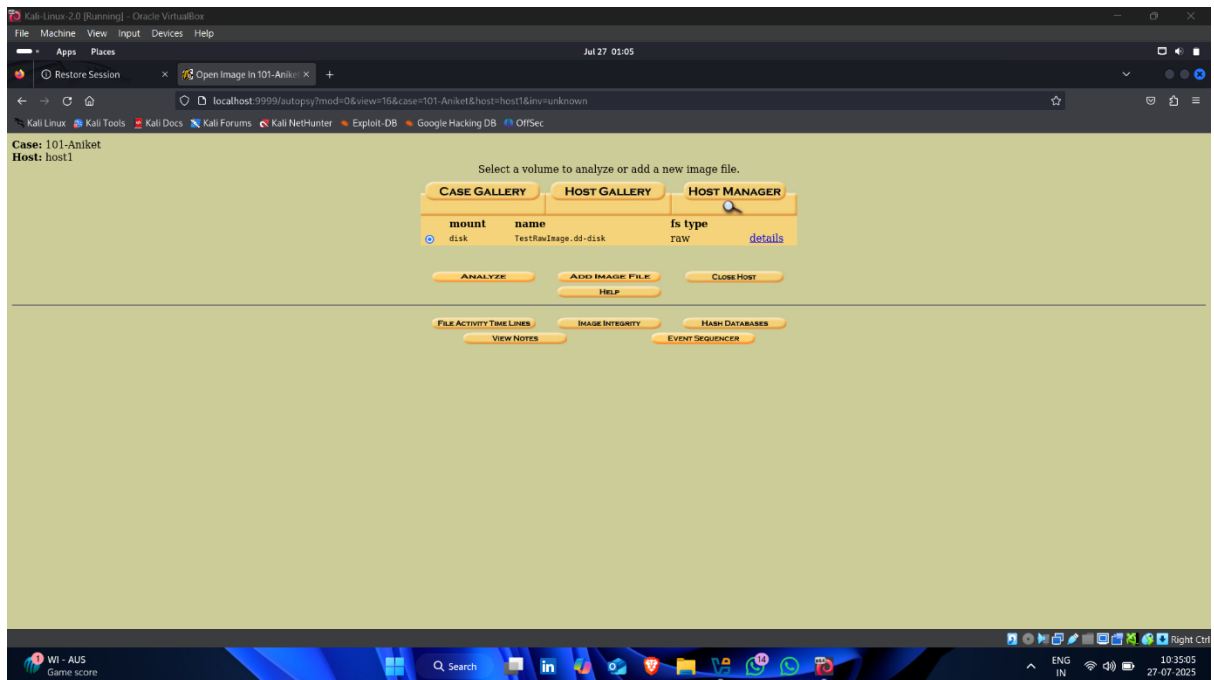
- Click on **Add**



- Click on **ok**



- Result 📌✅



Lab-3 -: Recover Data From Lost /Deleted Disk Partition

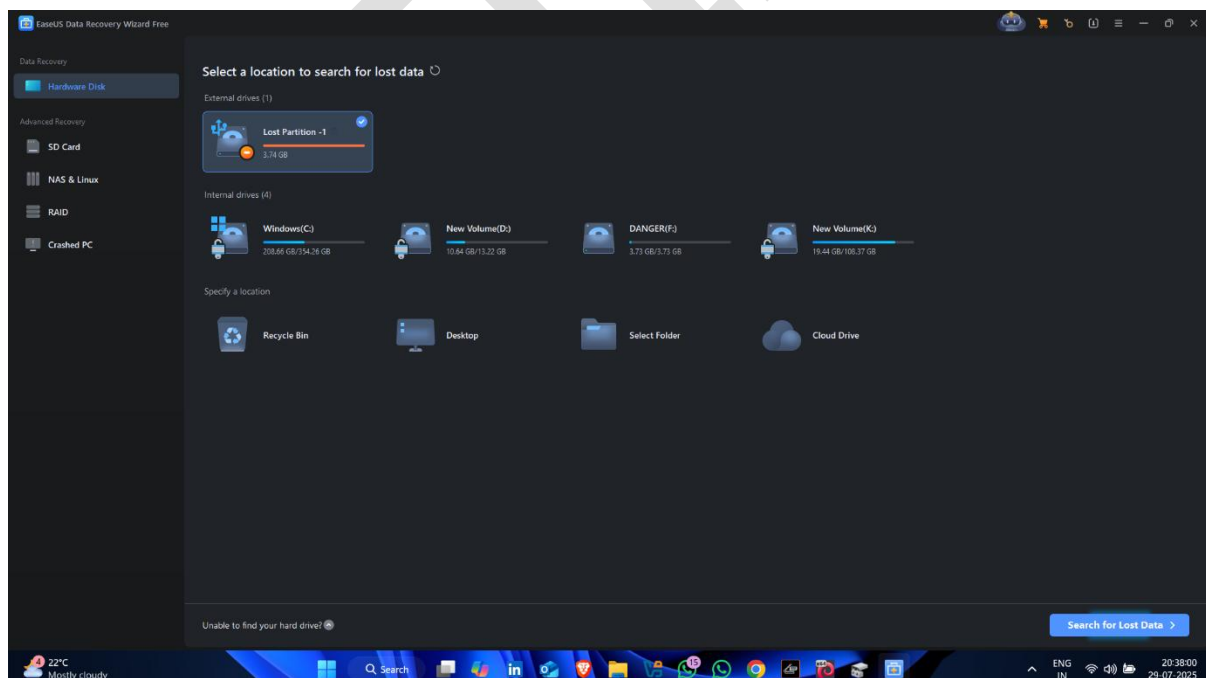
When a **disk partition** is **accidentally deleted**, becomes **corrupted**, or is **lost due to formatting**, the data inside it **isn't immediately erased**—only the **reference (partition table)** is removed or damaged. You can often **recover** the data if the space hasn't been overwritten.

A) Recover Data From Lost /Deleted Disk Partition Using EaseUP Tool

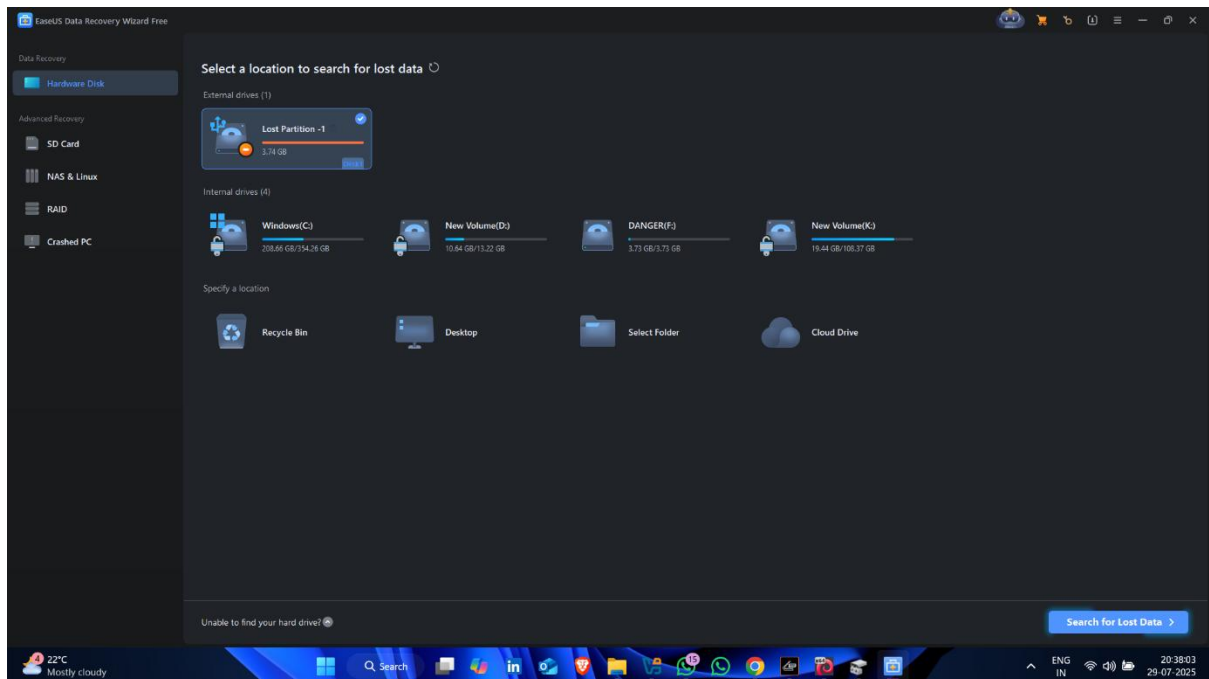
The **EaseUS** (not "Easeup") tool refers to a family of **Windows utilities** developed by **EaseUS**, mainly used for **data recovery**, **partition management**, and **backup**. Here's a breakdown of what **EaseUS tools** are and what they do:

How to use it -:

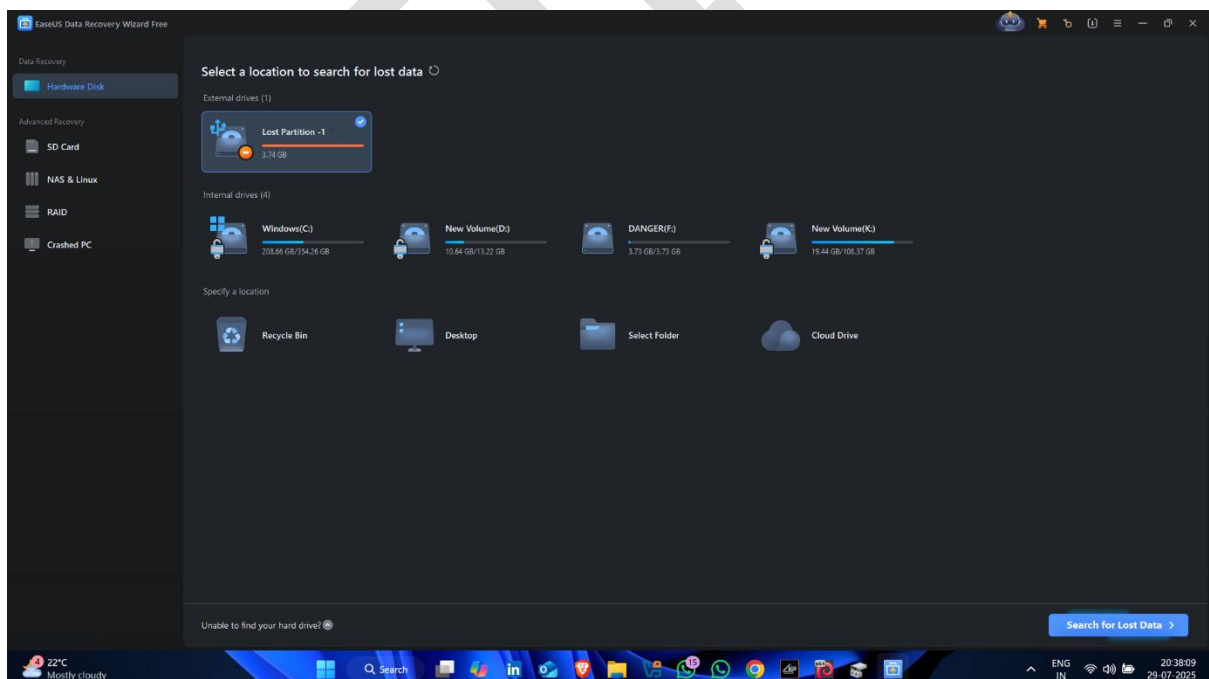
- Open EaseUp Application



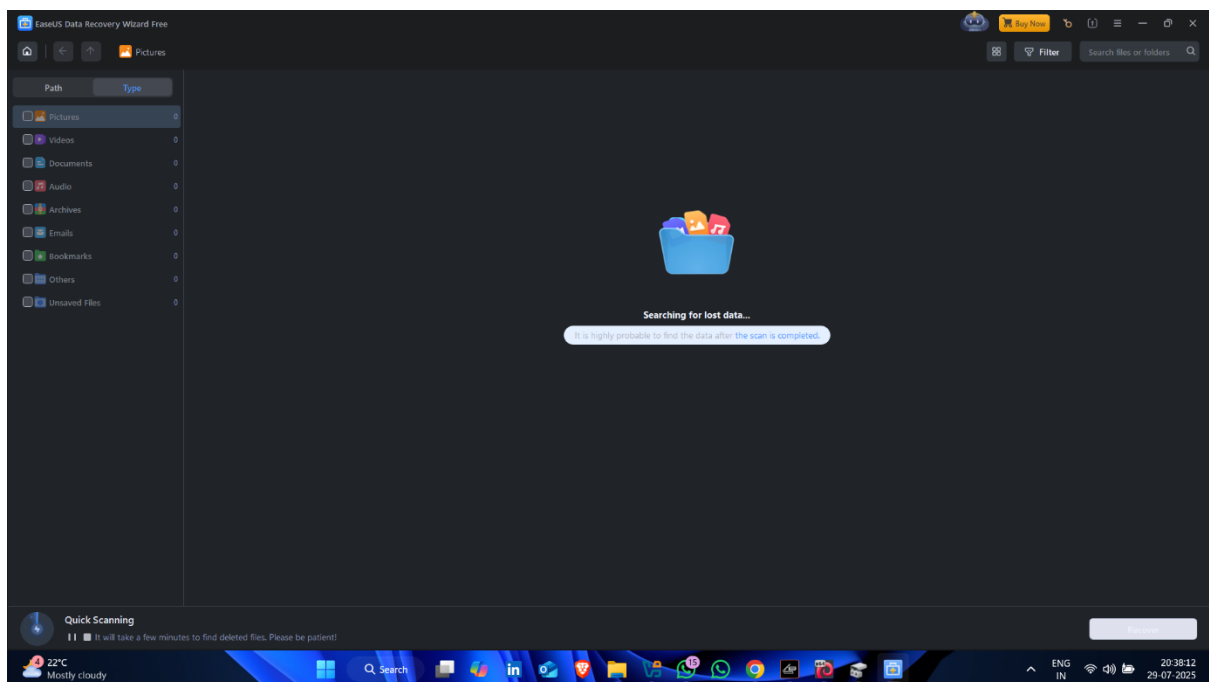
- EaseUp Automatically Detect Deleted Drive 🖱️
- Click on **Deleted Drive (First Drive)**



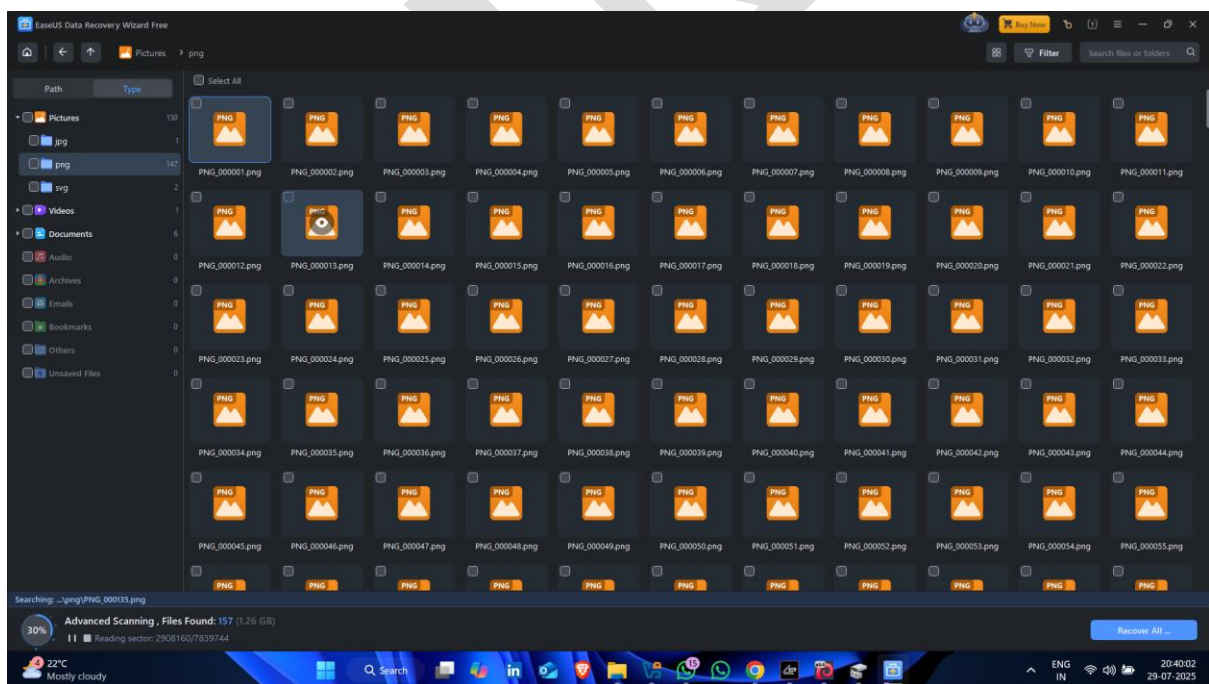
- Click on **Search For lost data**



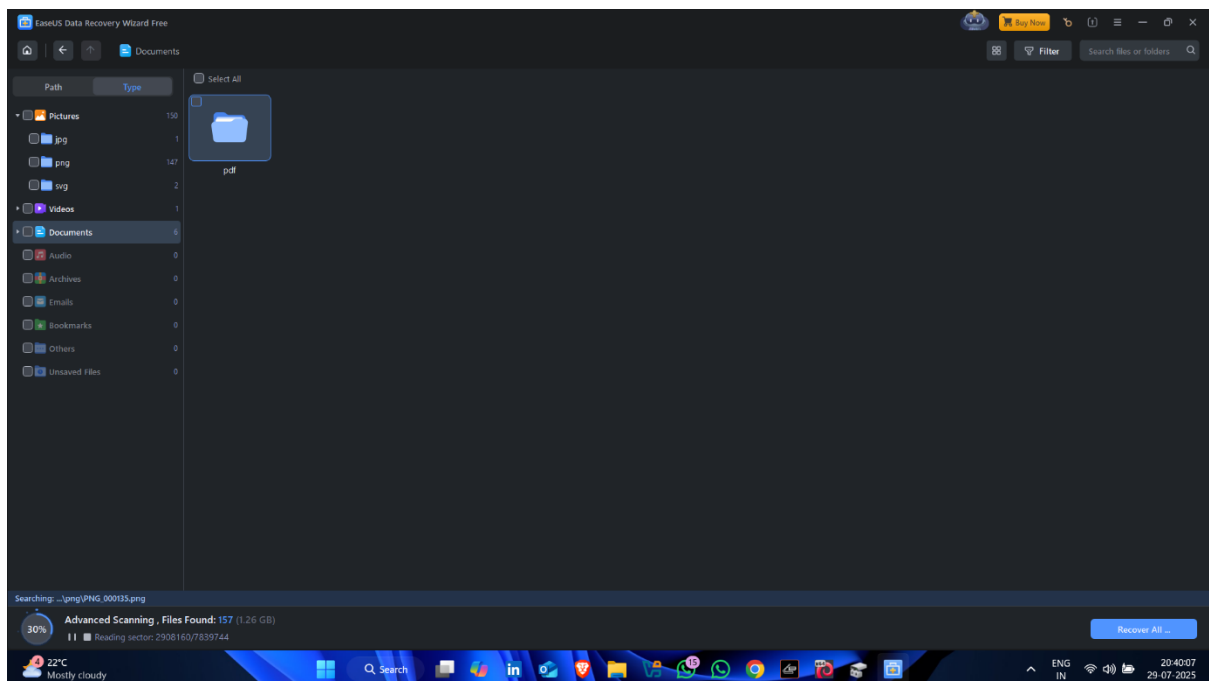
- Data Recovery started 🖱️



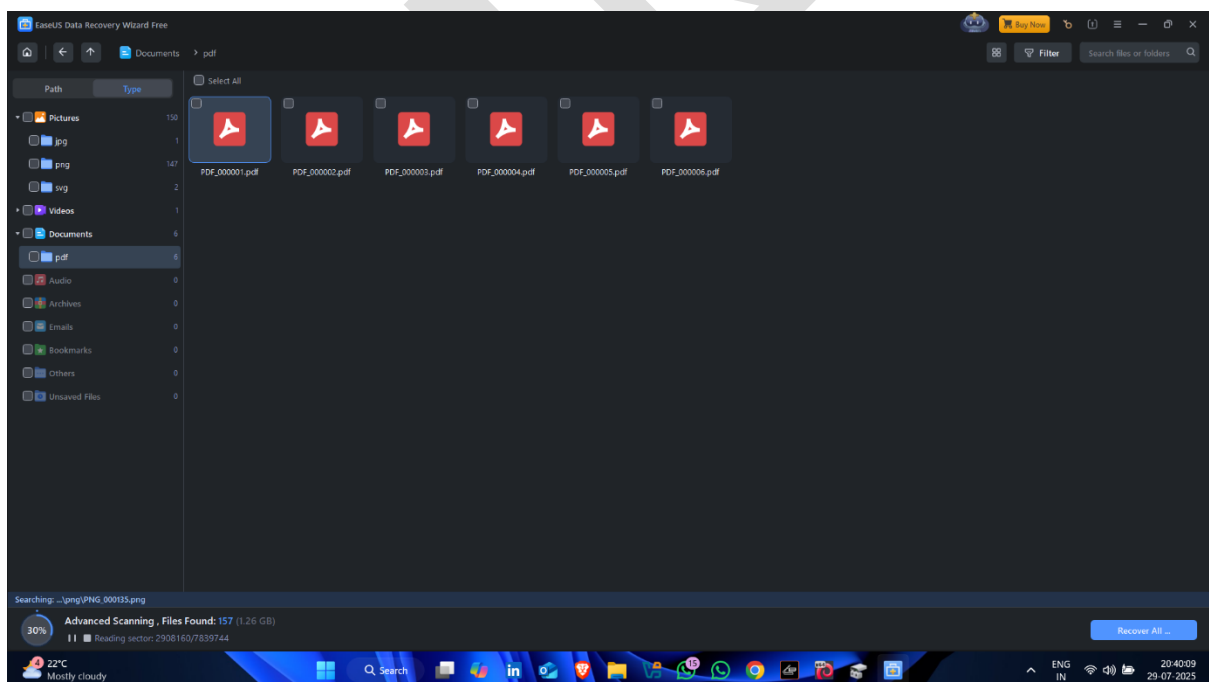
- Deleted Images on Drive



- Some Documents Recover , click on Pdf Folder



- Documents 📁 ✅



Lab-4 - : Recover Data From Partition That Is Deleted And Merged Into Another Partition

When a partition is **deleted** and its space is **merged with another partition**, the original file system and data **do not get instantly erased** — only the **partition table changes**. **Recovery is possible** if no significant data has been written over the deleted partition's area.

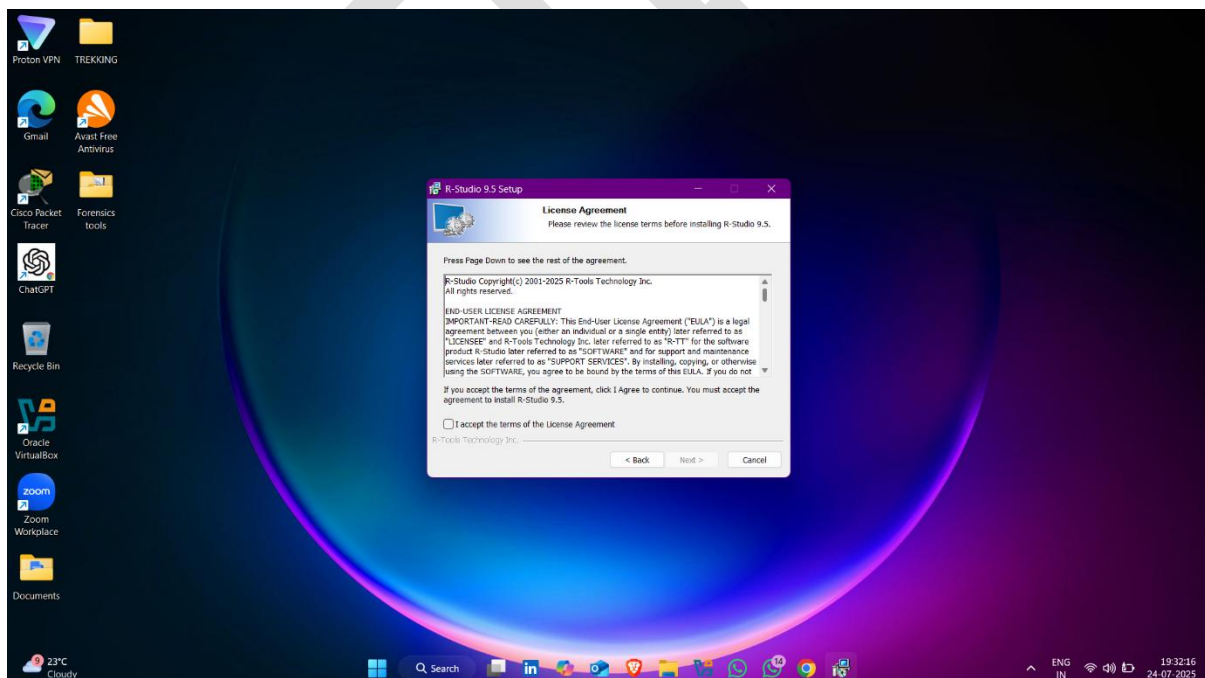
A) Recover Data From Partition That Is Deleted And Merged Into Another Partition Using R-Studio

R-Studio is a **professional data recovery tool** used for recovering lost or deleted files from **hard drives, SSDs, memory cards, USB drives**, and even **damaged or reformatted partitions**. It's widely used in both **digital forensics** and **data recovery industries**.

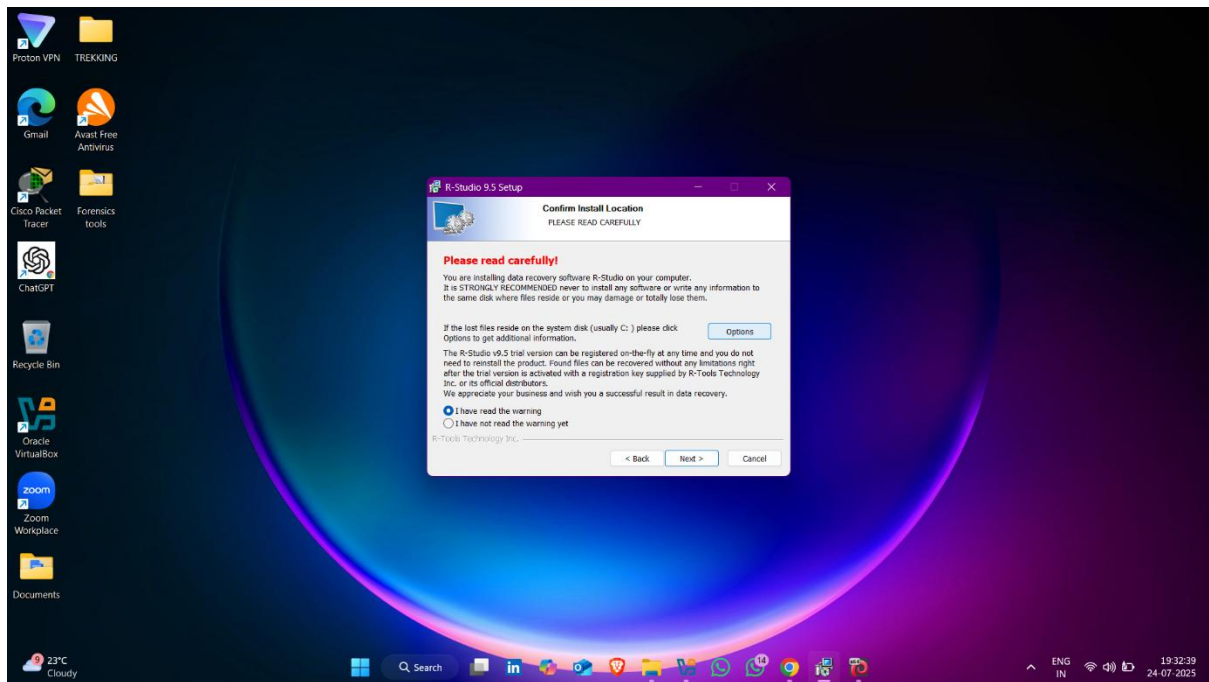
Download Link -: <https://www.r-studio.com/>

How to use it -:

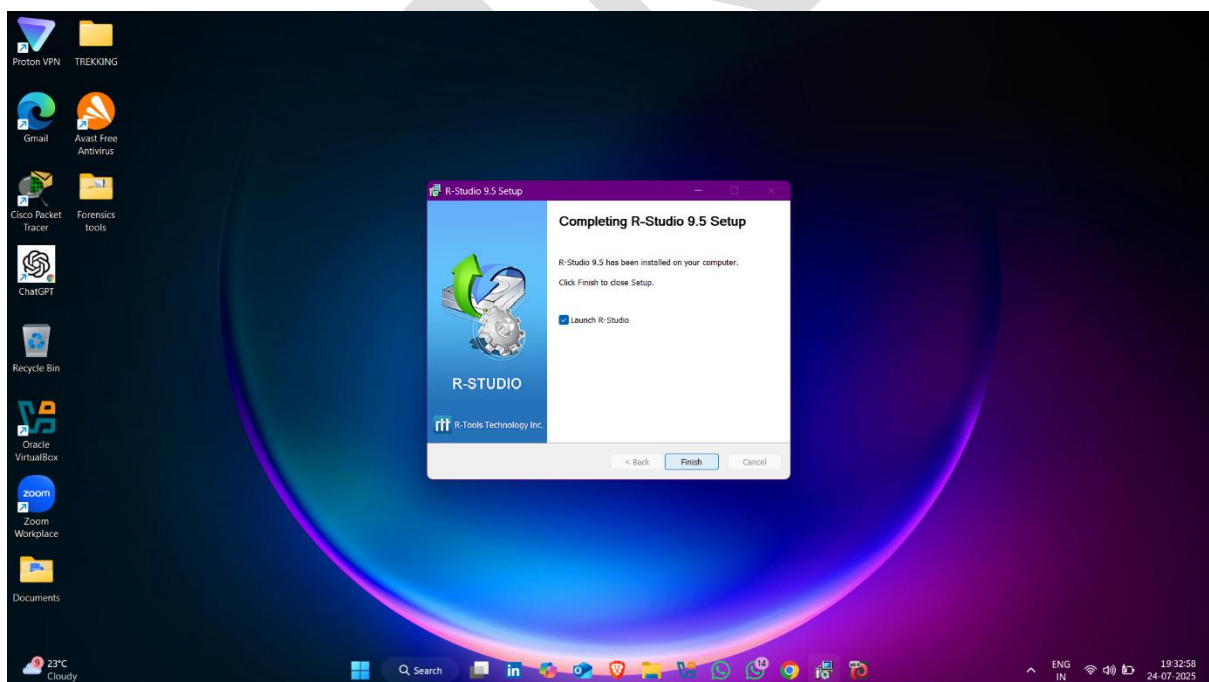
- Open R-Studio ✓👉
- Click on **accept term of the license agreement**



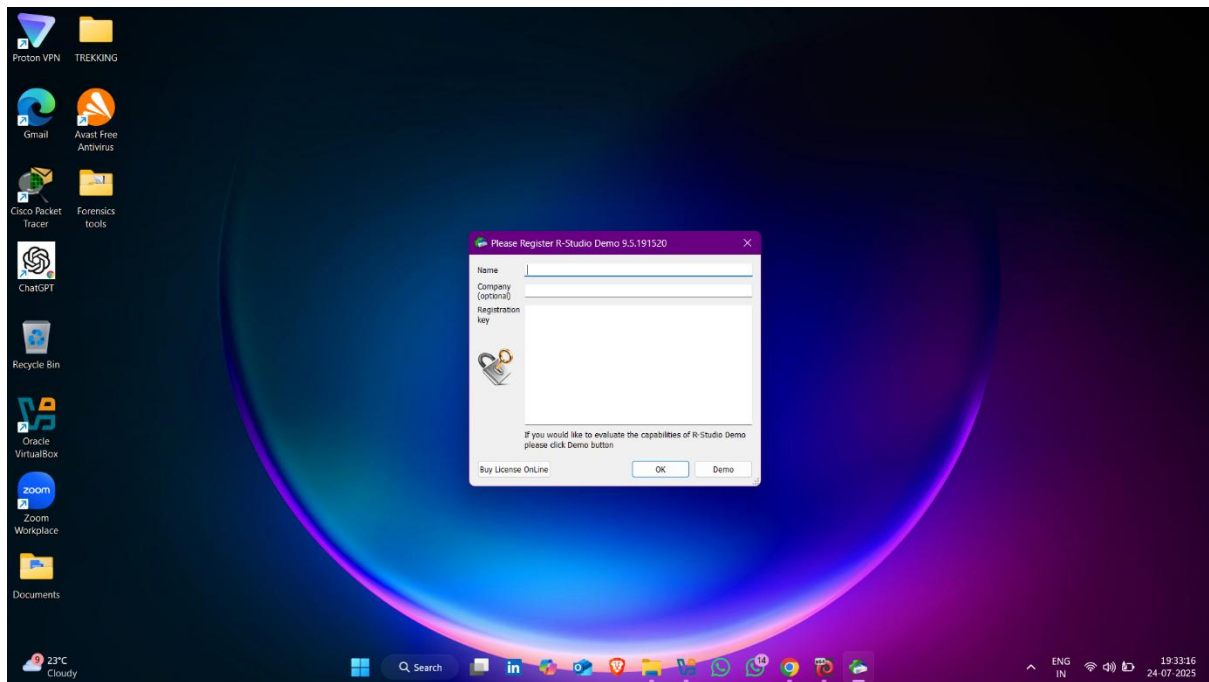
- Click on **next**



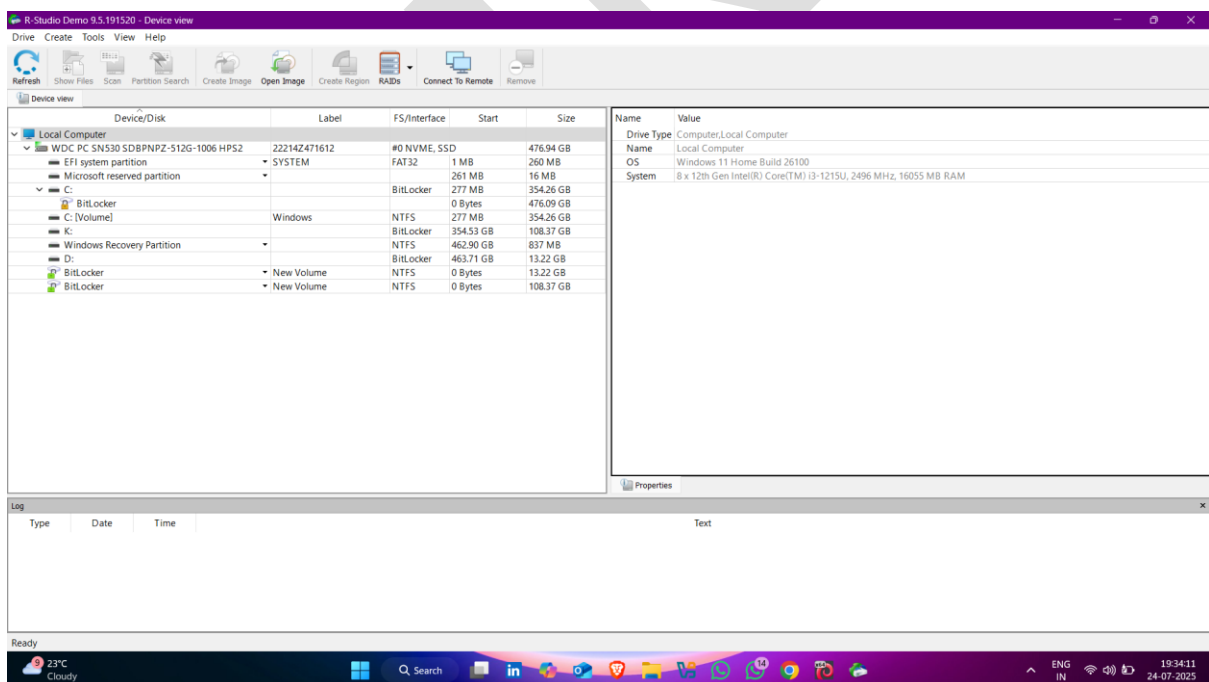
- Click on **Finish**



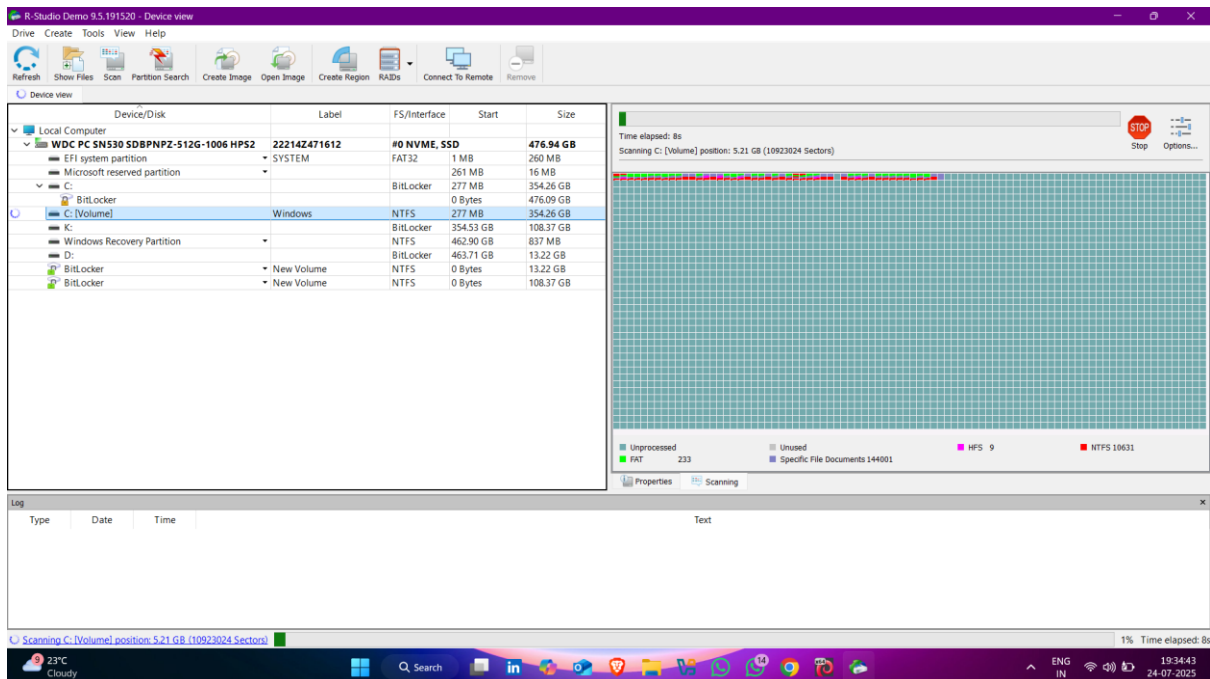
- Enter a name and then click on **ok**



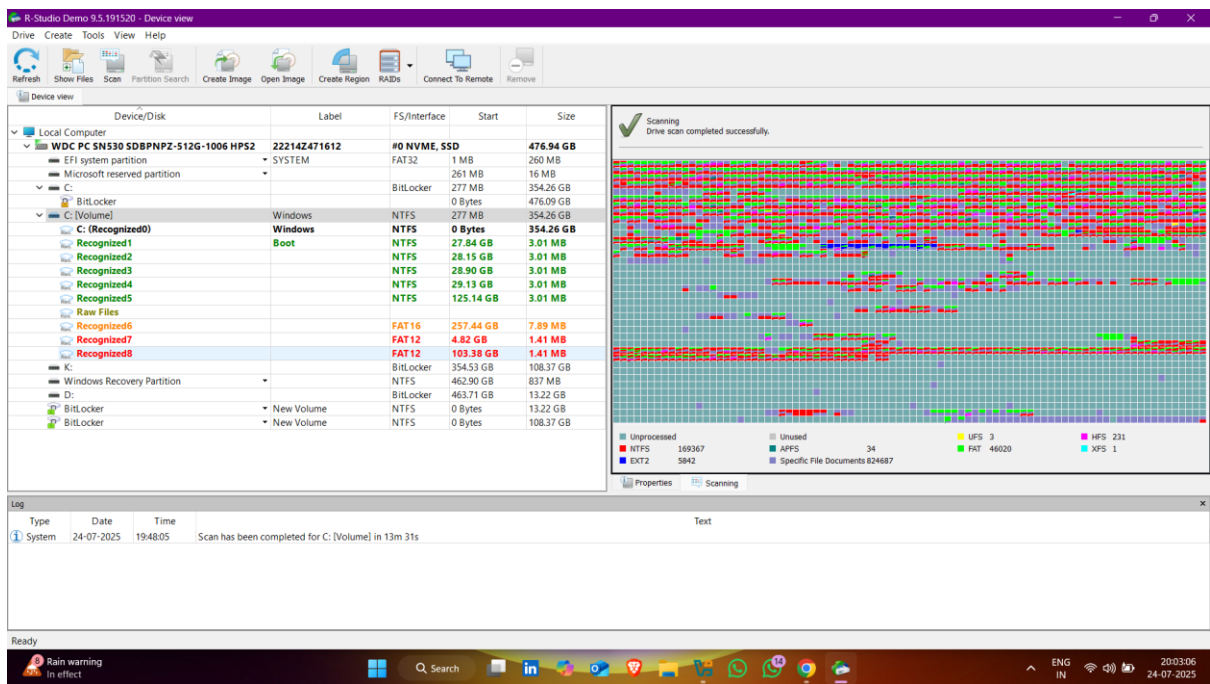
- Select Drive



- click on **Scan** (top navigation bar)



- **Result** 📌 ✅
- The drive was scanned **successfully**, and **multiple partitions were recognized**, including some **deleted or lost** ones.
- **Recoverable files** were identified, and many **specific file types** (e.g., documents) were located.
- The **block map** shows a **fragmented** disk with a large number of recoverable segments.
- The drive contains **encrypted volumes (BitLocker)** which may require passwords or keys for further access.



Lab-5 -: Windows Password Hash Cracking

Password hash cracking is the process of recovering the original password **from its hashed value**, usually by using **computational techniques**. It's commonly used in cybersecurity (e.g., in penetration testing, digital forensics, or cybercrime investigations).

A) Windows Password Hash Cracking Using Pwdump7

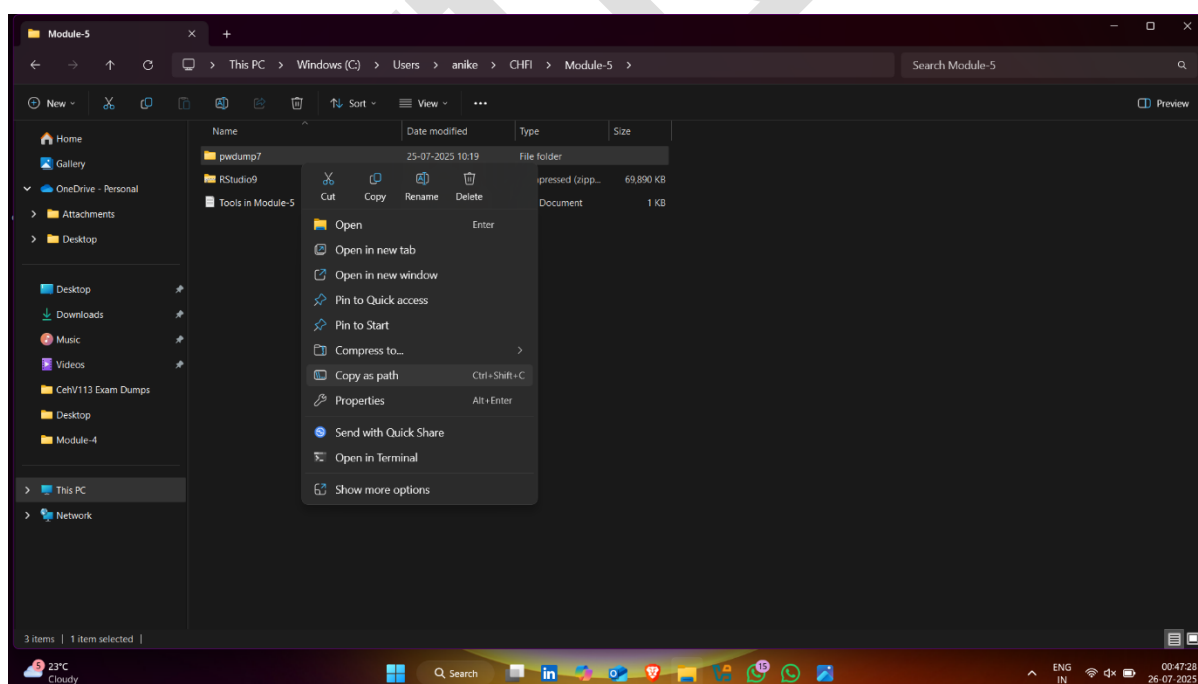
Pwdump7 is a **Windows password hash extraction tool**. It is used to **extract NTLM and LM password hashes** from the **Security Accounts Manager (SAM) database** on a Windows system.

Download Link -:

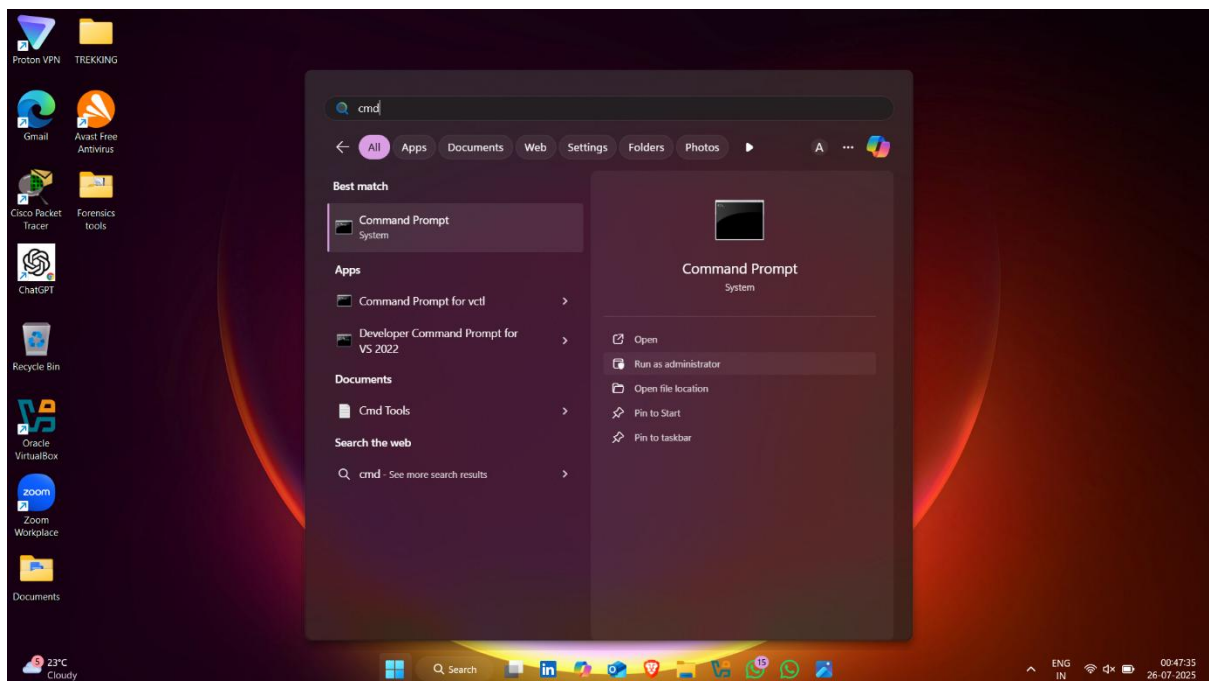
<https://www.openwall.com/passwords/windows-pwdump>

How to use it -:

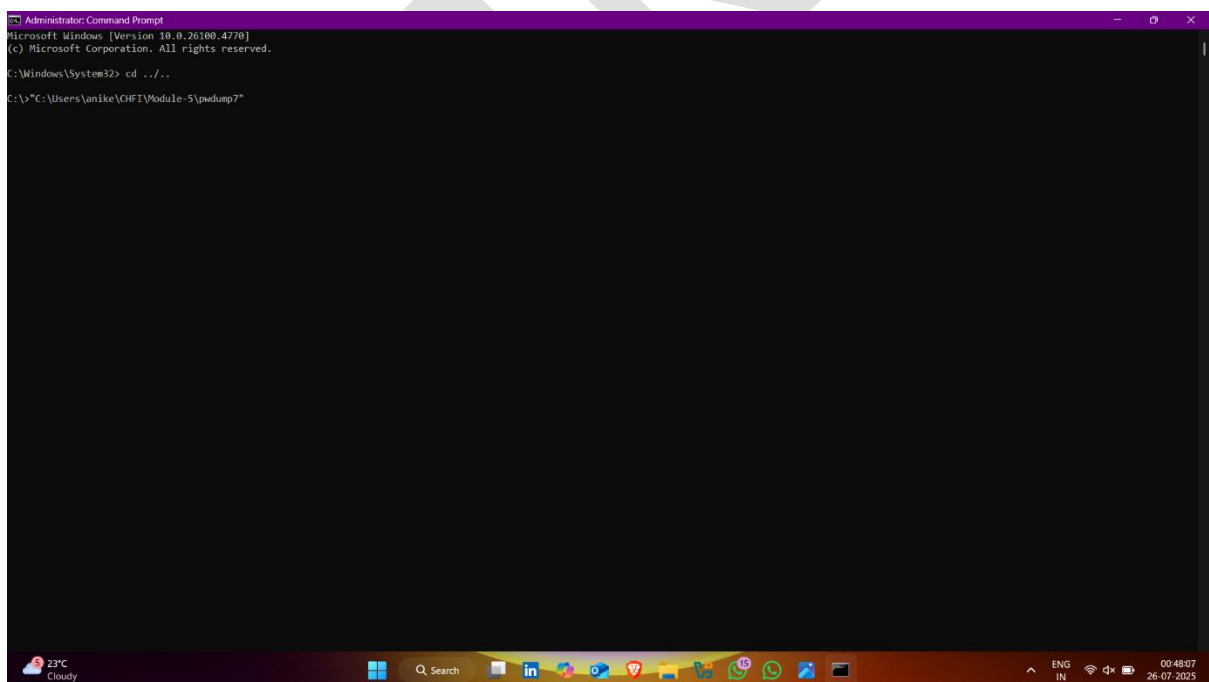
- Right click on **pwdump7** folder and click on **copy as path**



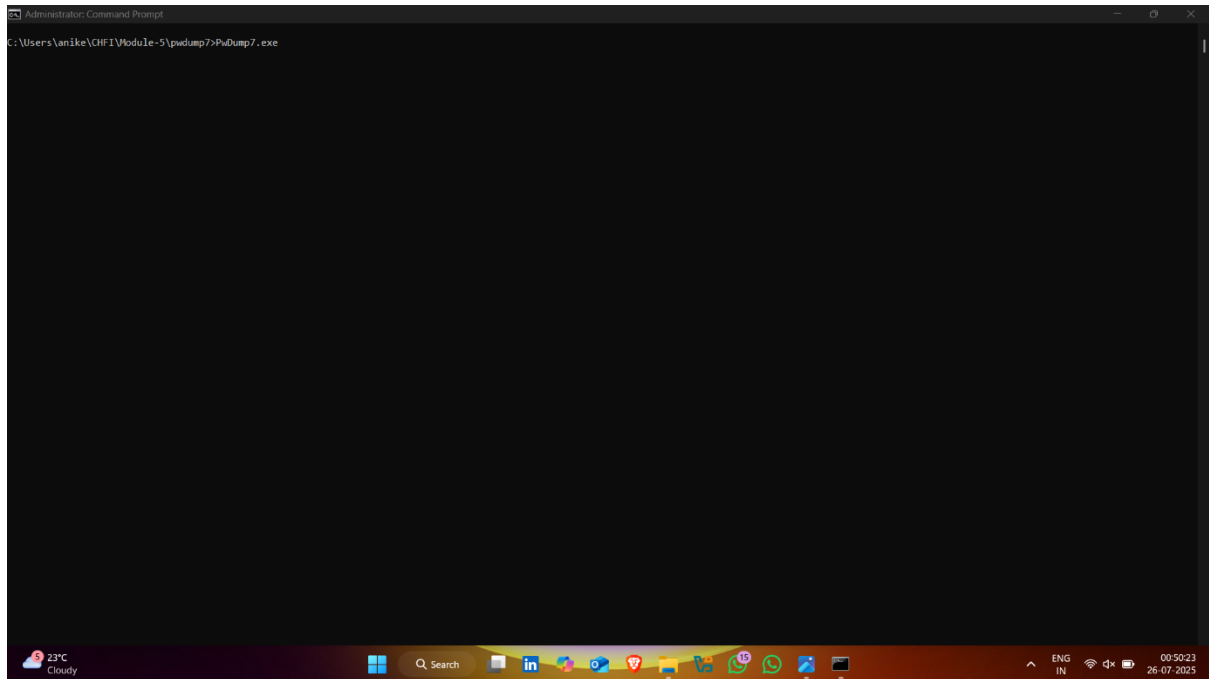
- Now open **Command Prompt** and **run as a administrator**



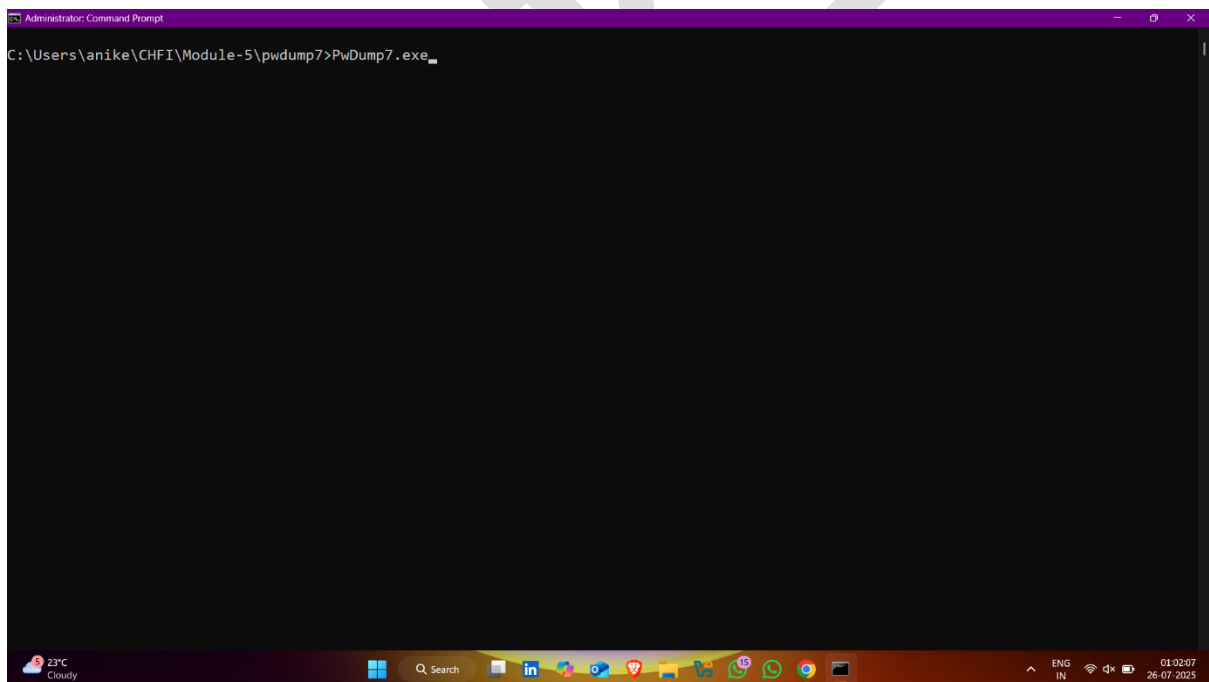
- Paste **pwdump7** path



- Now run `pwdump7.exe` file 🖱️



A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt shows the directory `C:\Users\anike\CHFI\Module-5\pwdump7>` and the command `PwDump7.exe` has been entered. The window is dark-themed. The Windows taskbar at the bottom shows the date and time as 00:50:23 on 26-07-2025, along with various system icons and a weather widget showing 23°C Cloudy.



A second screenshot of a Windows Command Prompt window, identical to the first one, showing the same command prompt and taskbar information. The time in the taskbar is now 01:02:07 on 26-07-2025.

• RESULT 🙌 ✅

```
Administrator: Command Prompt
C:\Users\anike\CHFI\Module-5\pwdump7>PwDump7.exe
PwDump v7.1 - raw password extractor
Author: Andres Tarasco Acuna
url: http://www.514.es

Administrator:500:535E726EE673A875C7A7EE87A609164F:E617CA377F276F803C8583ED522FA923:::
Guest:501:E61ED86631A0808379BDF981E2A30702:CE4599D77A4E660FA1F923F13B235104:::
J:503:61064A77D50EBB578679C852F845B9D8:5F9BF01C20FC78C10B48711702E85D93:::
J:504:C496598A265507AC175522D38416BC73:3EFE6A8312D6841FD27A6172BFD926D7:::
Aniket:1001:40FD93E0499FEB42C0C2FD84542C255A:B7C2AF854A5B76AE054FB3A180F5DBA1:::
J:1004:AD1E9F1FDCA467B3A761B007973A6995:F7091BD952736FD0A186CA8C836681DD:::
J:1005:B9CB9AF1F4F0D7E020F621D7C6416AC3:C858527FC27026E873BFCDD039C0E64D:::

C:\Users\anike\CHFI\Module-5\pwdump7>
```

Lab-6 -: Crack Application Password

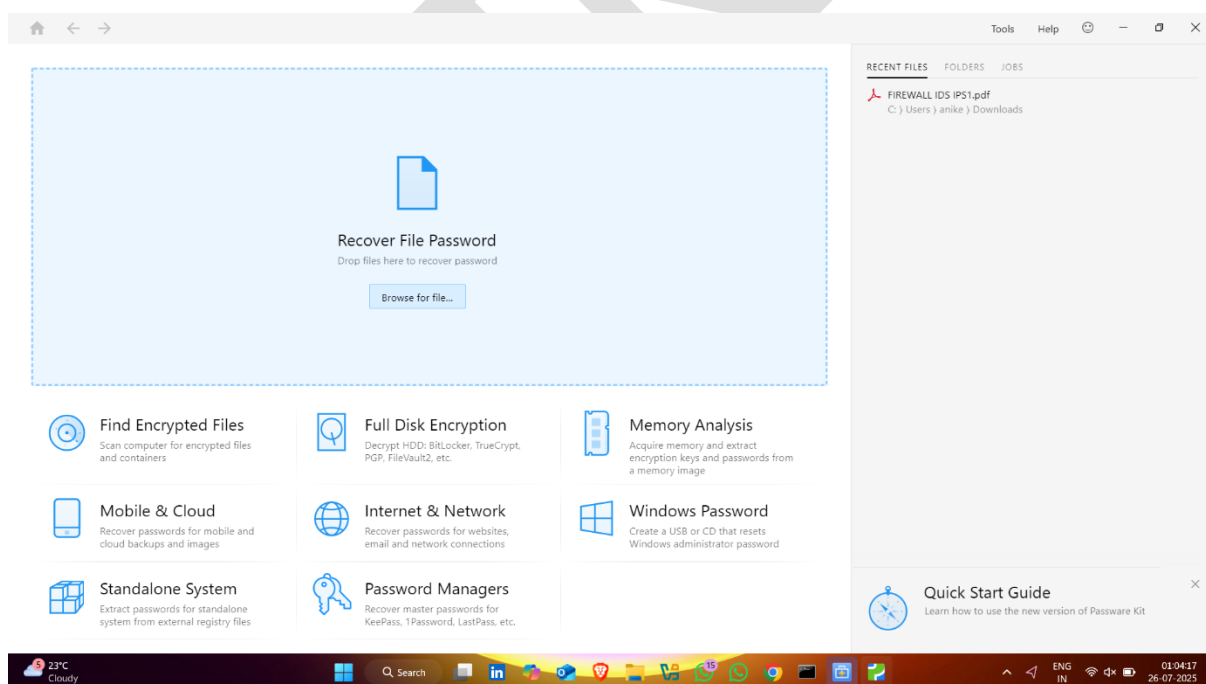
Application password cracking in this context means **recovering or bypassing passwords set on files** created by applications like Microsoft Office, Adobe Reader, WinRAR, etc.

A) Crack Application Password Using Passware Forensics Tool Kit

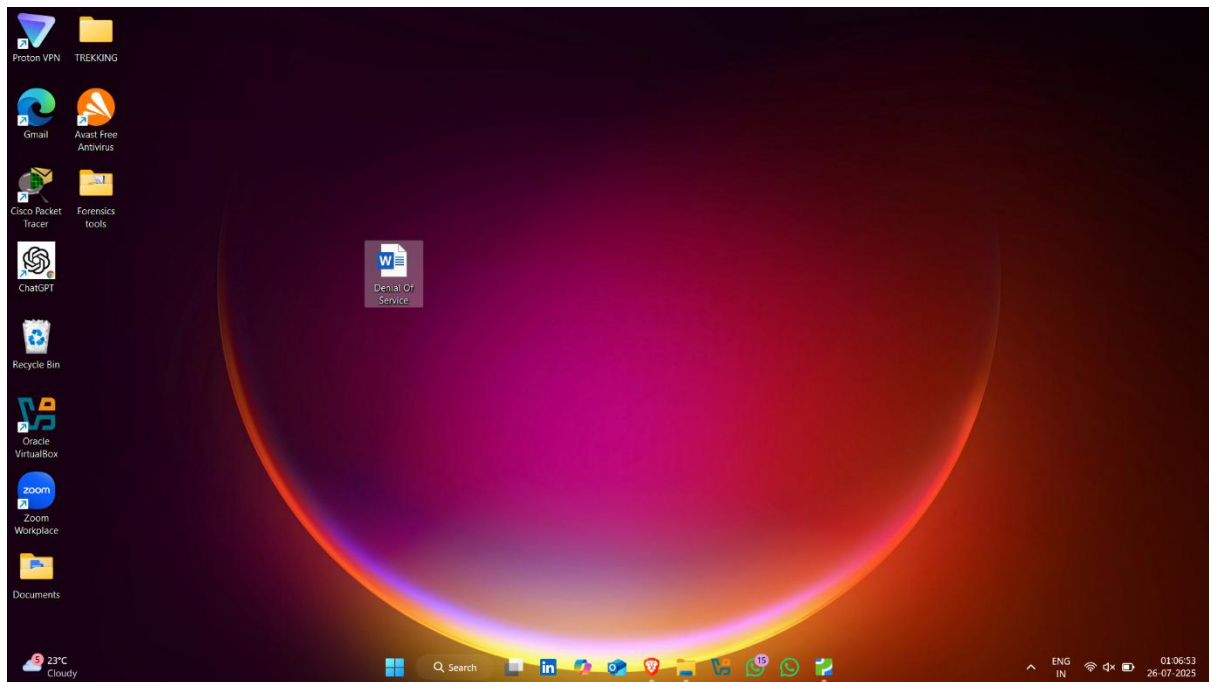
The **Passware Forensic Toolkit** is a powerful **commercial password recovery and decryption suite** designed for **digital forensic investigators**. It's primarily used by law enforcement, forensic analysts, and cybersecurity professionals to **unlock and extract data** from password-protected or encrypted files, systems, and devices.

How to use it -:

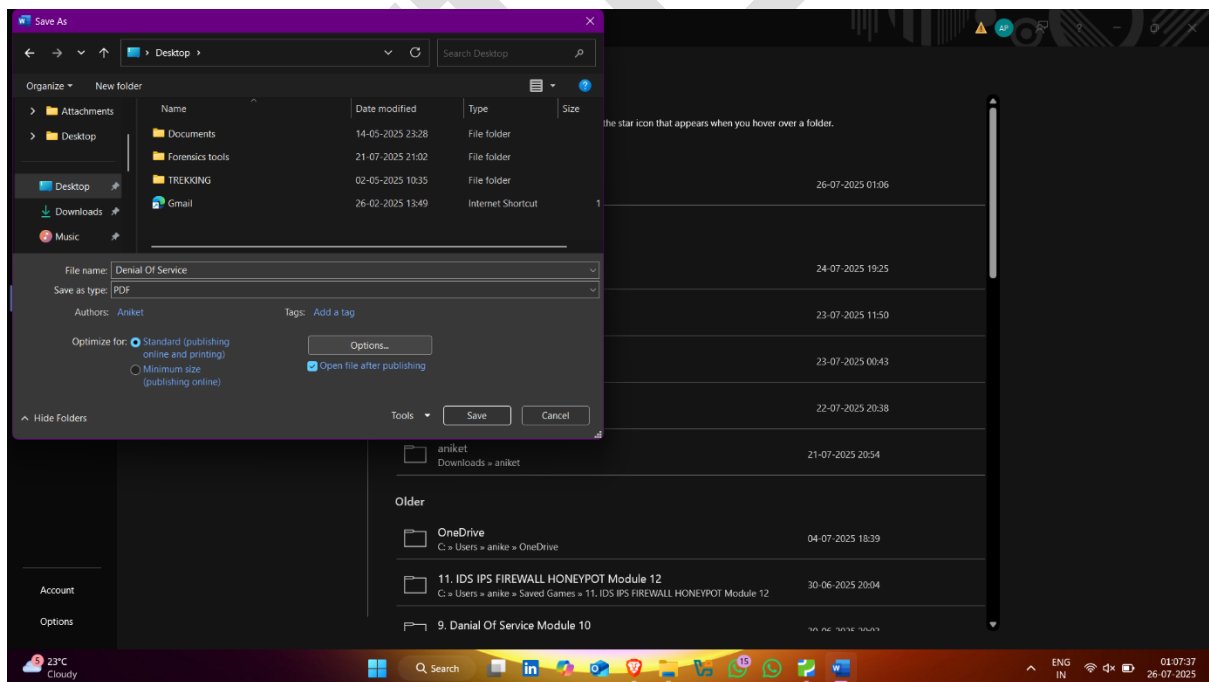
- Open Application



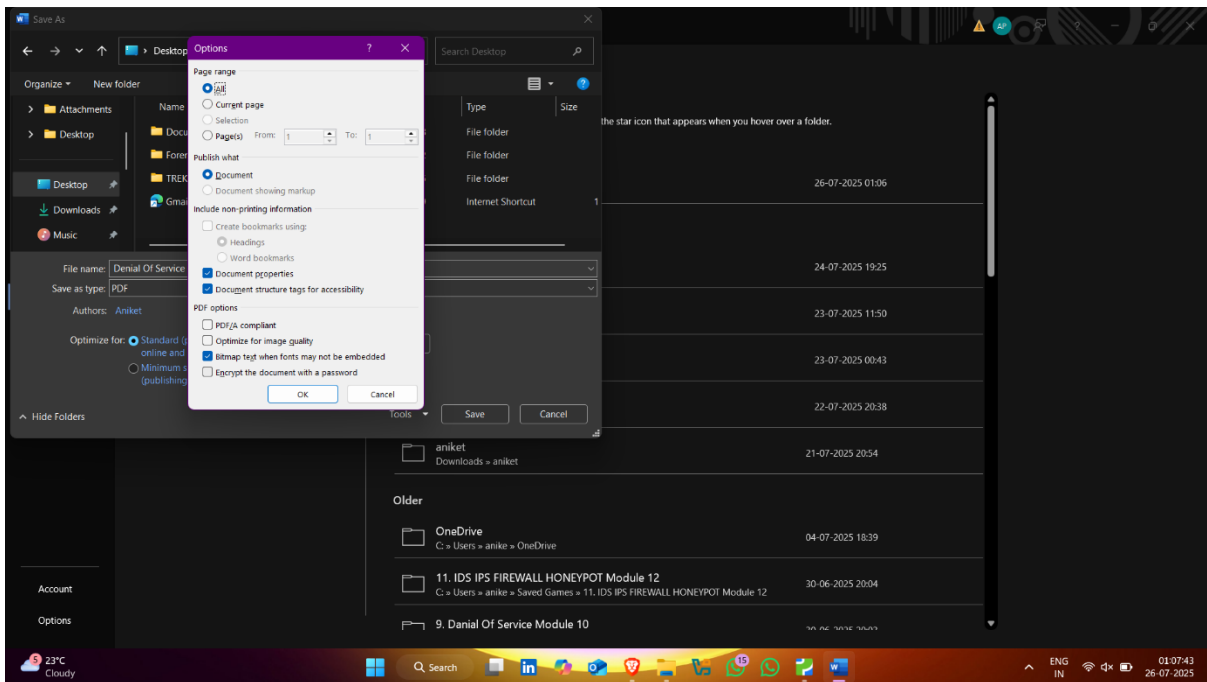
- Open file



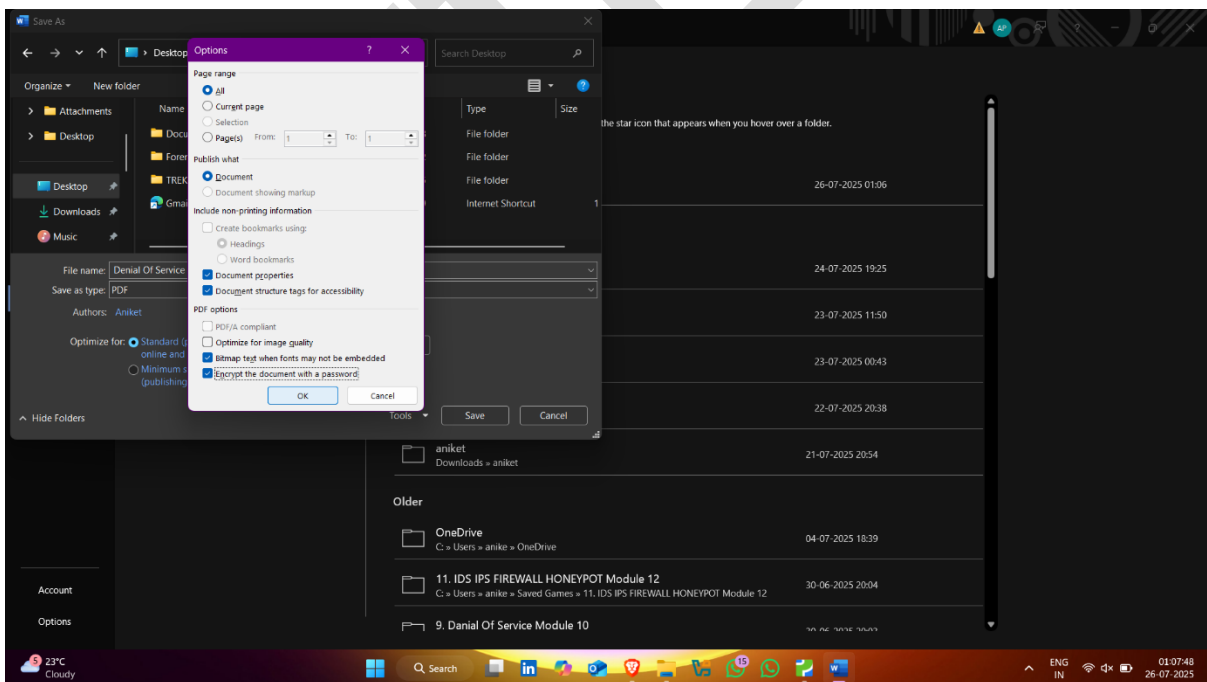
- Click on **options**



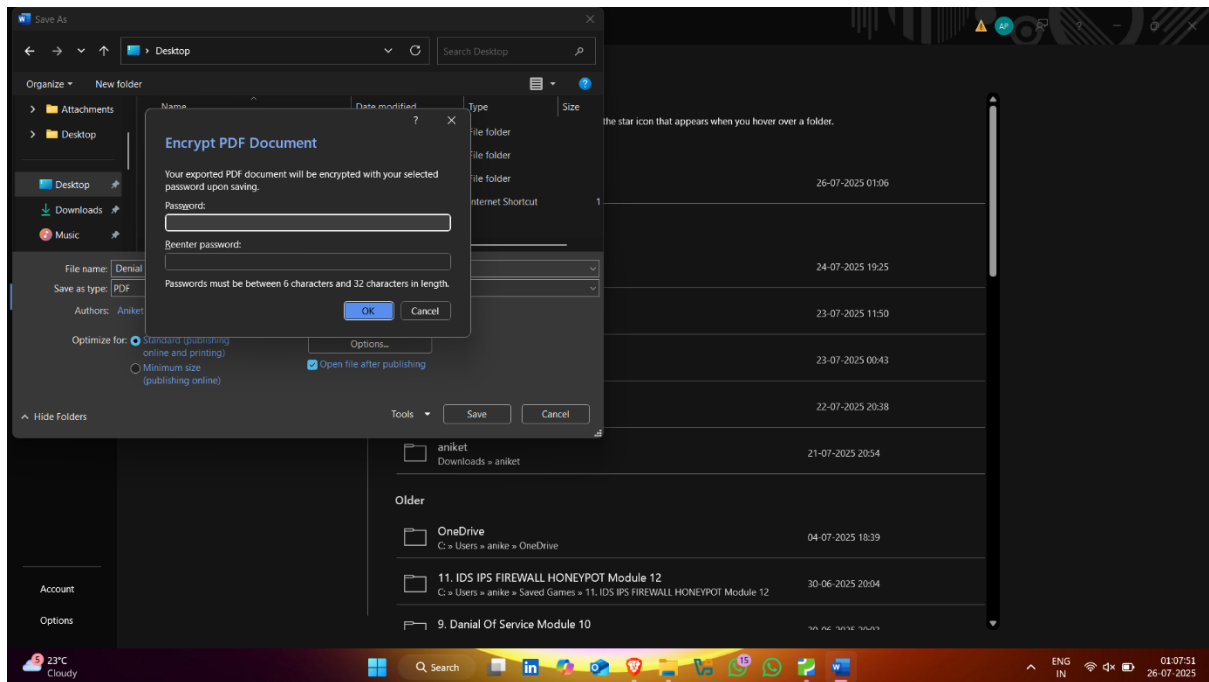
- Select last option – **Encrypt the document with a password**



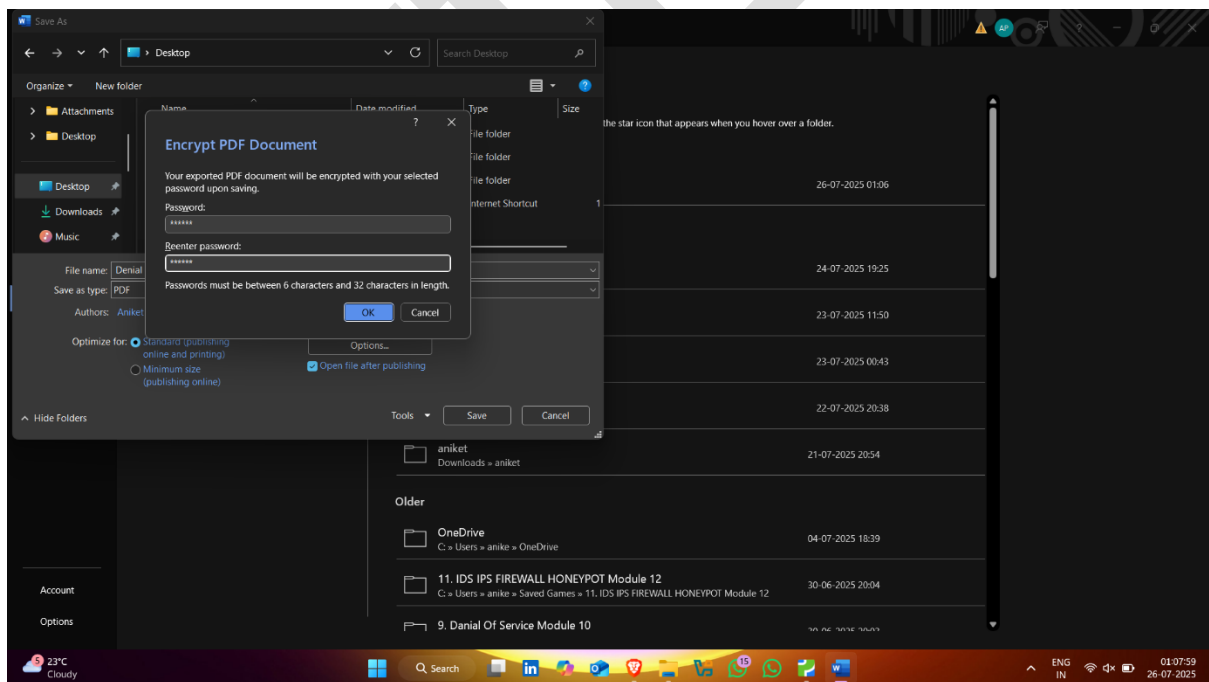
- Click on **ok**



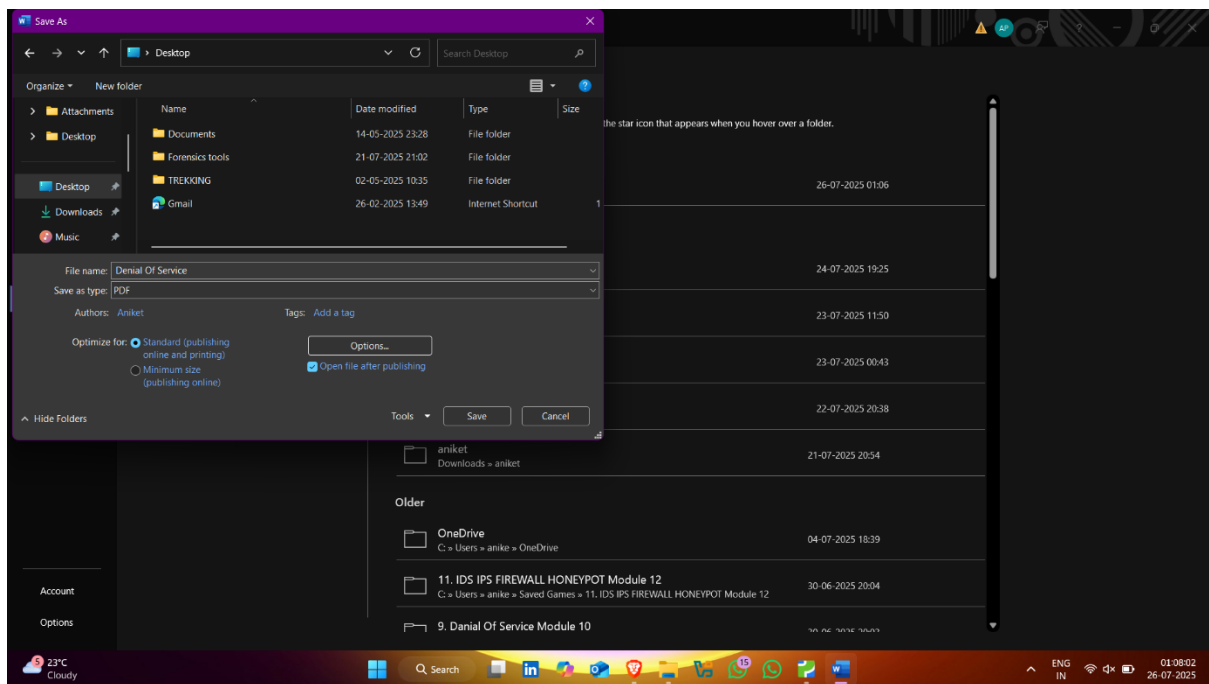
- Enter password



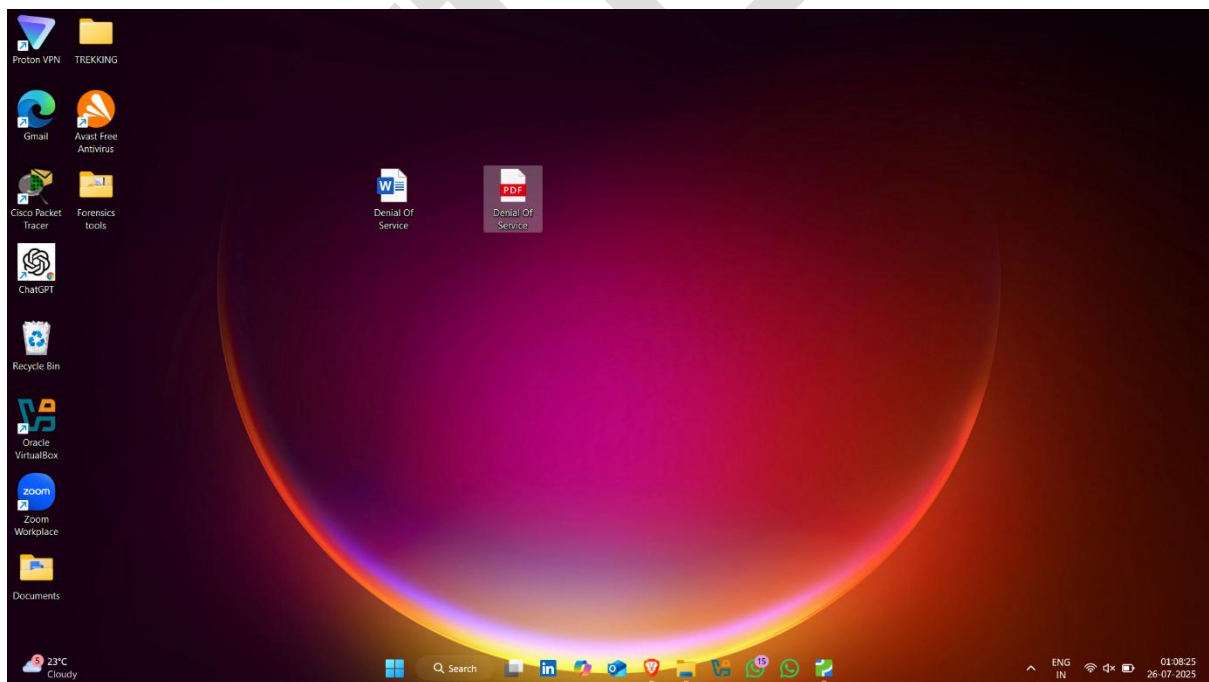
- Click on **ok**



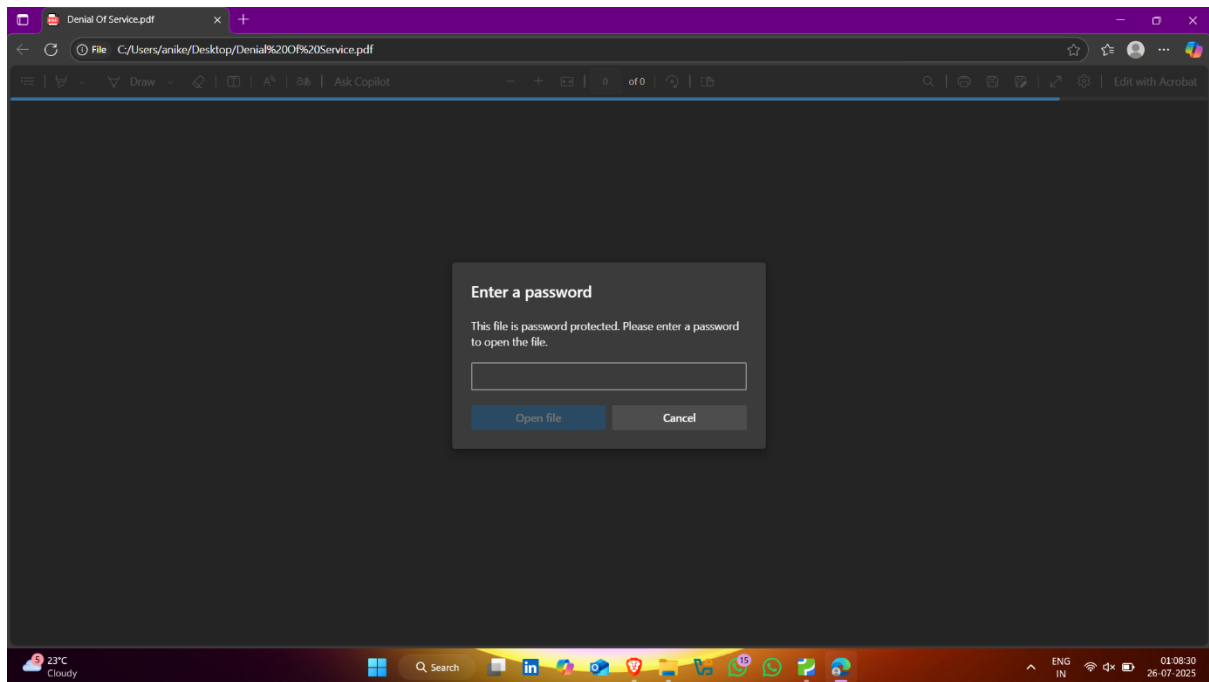
- Click on **save**



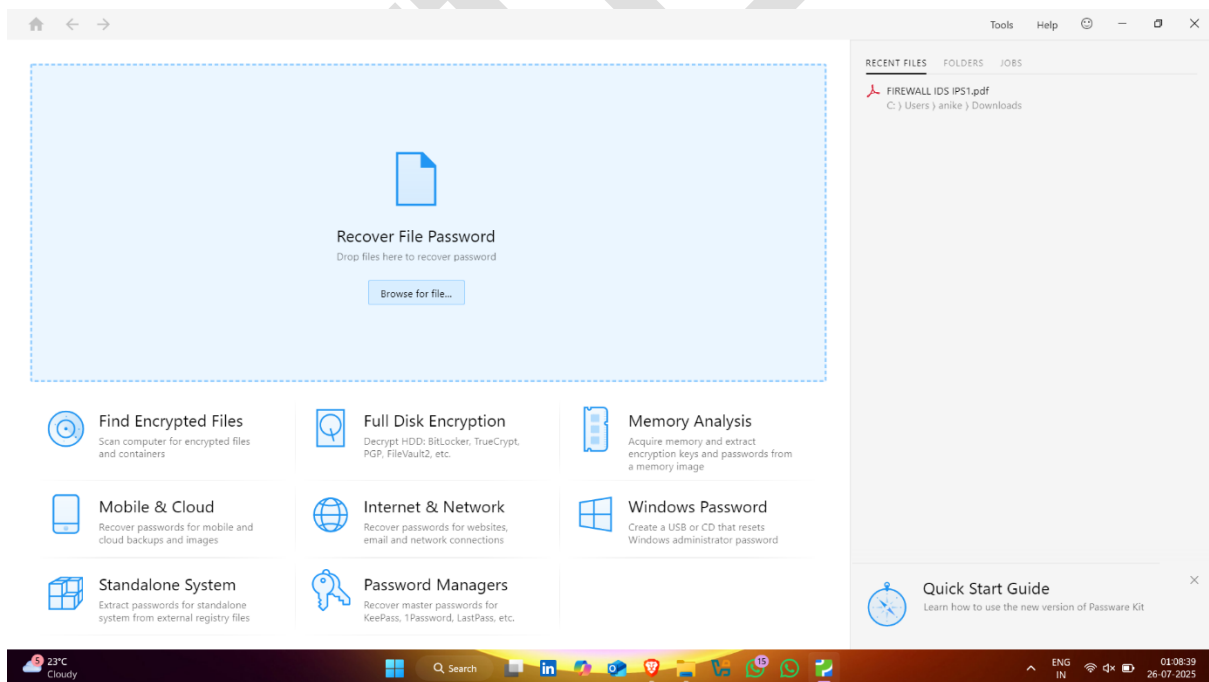
- Now open the file to check password is set or not



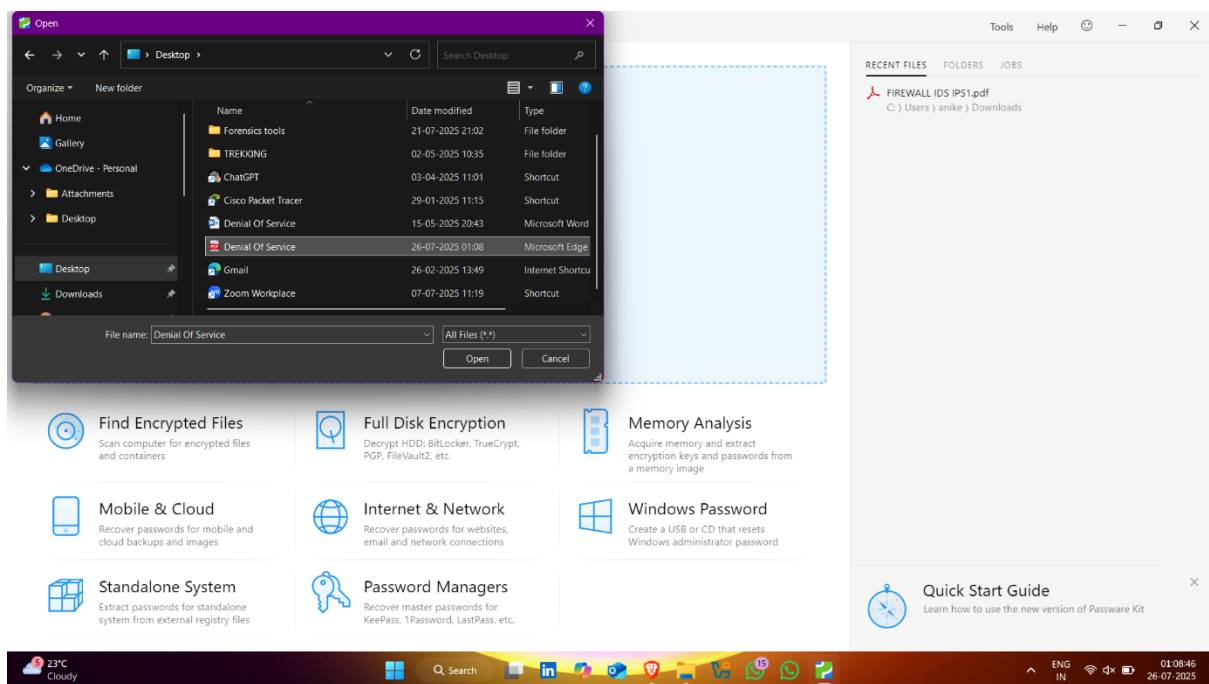
- Password Set



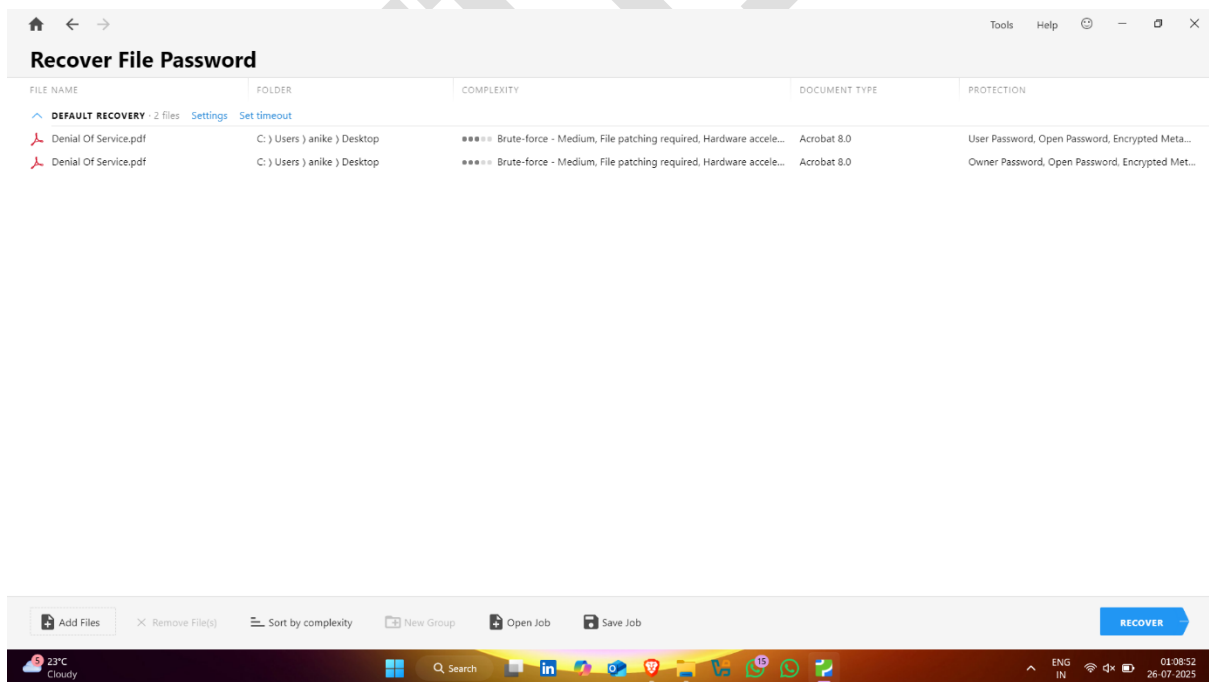
- Click on **Browse For File**



- Select File and click on open



- Click on **recover**



- Password Cracking Started

Recover File Password

Files | Passwords Found | Resources | Performance | Attacks | Log

FILE NAME	STATUS	PASSWORDS FOUND
DEFAULT RECOVERY		
Denial Of Service.pdf		—
Denial Of Service.pdf	Waiting	—

PASSWORDS FOUND
1

PASSWORDS ANALYZED
4

ESTIMATED TIME
—

TIME ELAPSED
2 seconds

PROCESSING FILE 1 OF 2
Denial Of Service.pdf

CURRENT ATTACK 10 OF 22
Acrobat Decryption attack

Skip Attack | Skip File | Skip Group

PAUSE | STOP

23°C Cloudy

ENG IN 01:09:01 26-07-2025

- Password Crack

Recover File Password

Files | Passwords Found | Resources | Performance | Attacks | Log

Denial Of Service.pdf

Folder: C:\Users\anike\Desktop
File Type: Acrobat 8.0 — User Password, Open Password, Encrypted Metadata, Permissions Password, AES Encryption, File patching required, Hardware acceleration possible
Complexity: Brute-force - Medium
MD5: 653F9F2A9FBC2752E282C2D817C01B97

Passwords:

File-Open: 123456
Permissions: Removed

Unprotected file:

C:\Users\anike\Desktop\Denial Of Service-unprotected.pdf
MD5: BA3E4D3B378669CB90A60EAB9178B08

Denial Of Service.pdf

Folder: C:\Users\anike\Desktop
File Type: Acrobat 8.0 — Owner Password, Open Password, Encrypted Metadata, Permissions Password, AES Encryption, File patching required, Hardware acceleration possible
Complexity: Brute-force - Medium
MD5: 653F9F2A9FBC2752E282C2D817C01B97

Passwords:

File-Open: 123456
Permissions: Removed

PASSWORDS FOUND
3

PASSWORDS ANALYZED
8

TIME ELAPSED
6 seconds

Print | Save Job

RESUME ATTACKS | SAVE REPORT | DONE

23°C Cloudy

ENG IN 01:09:34 26-07-2025

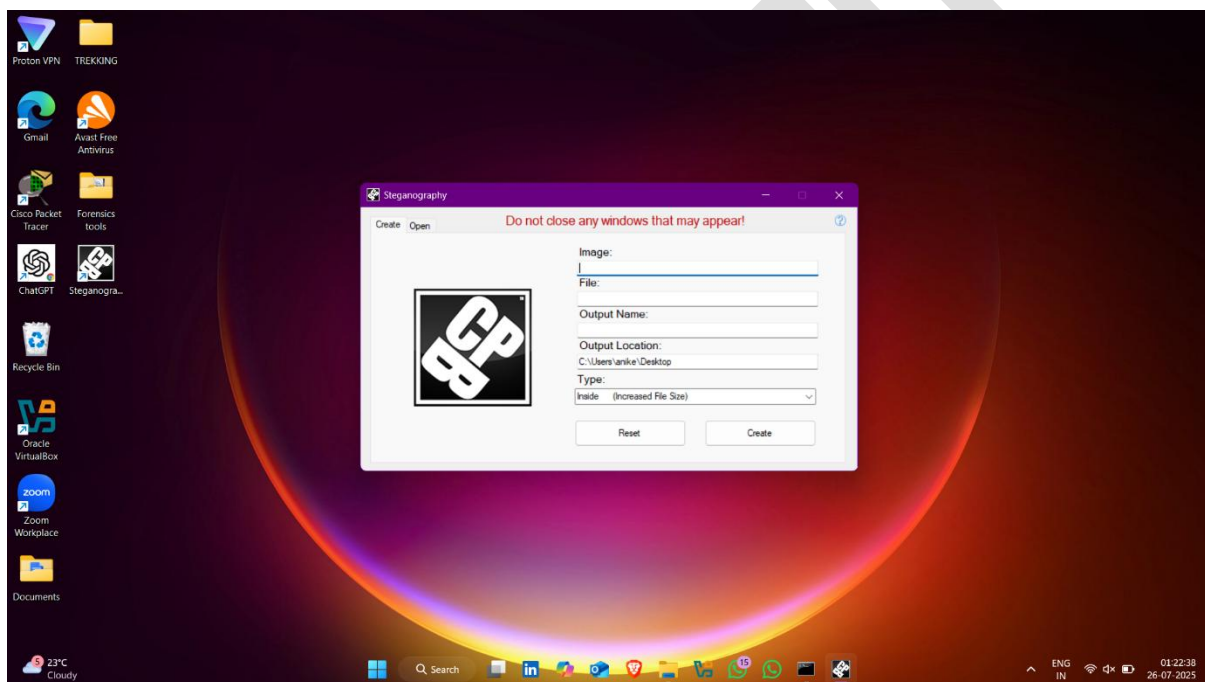
Lab-7 -: Detect Steganography

Steganography is the technique of **hiding secret information within an ordinary, non-secret file or message**, such that the presence of the hidden data is not obvious. Unlike encryption, which hides the content of a message, **steganography hides the existence of the message itself**.

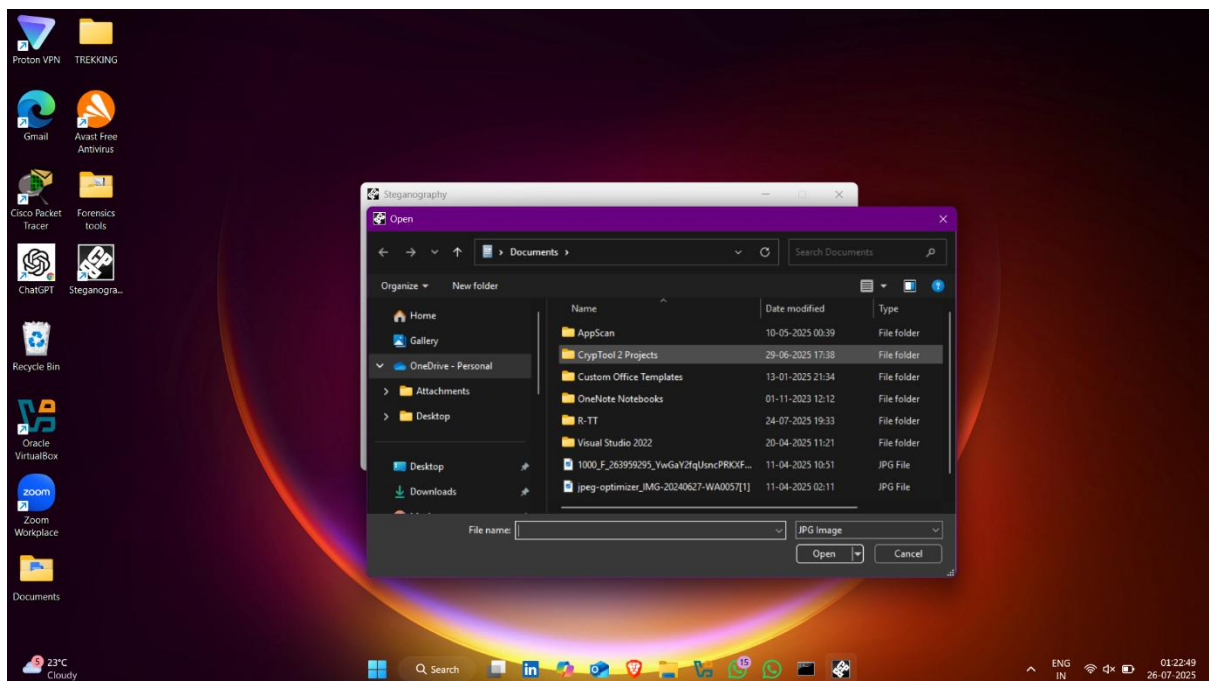
A) Create Steganography Image

How to use it -:

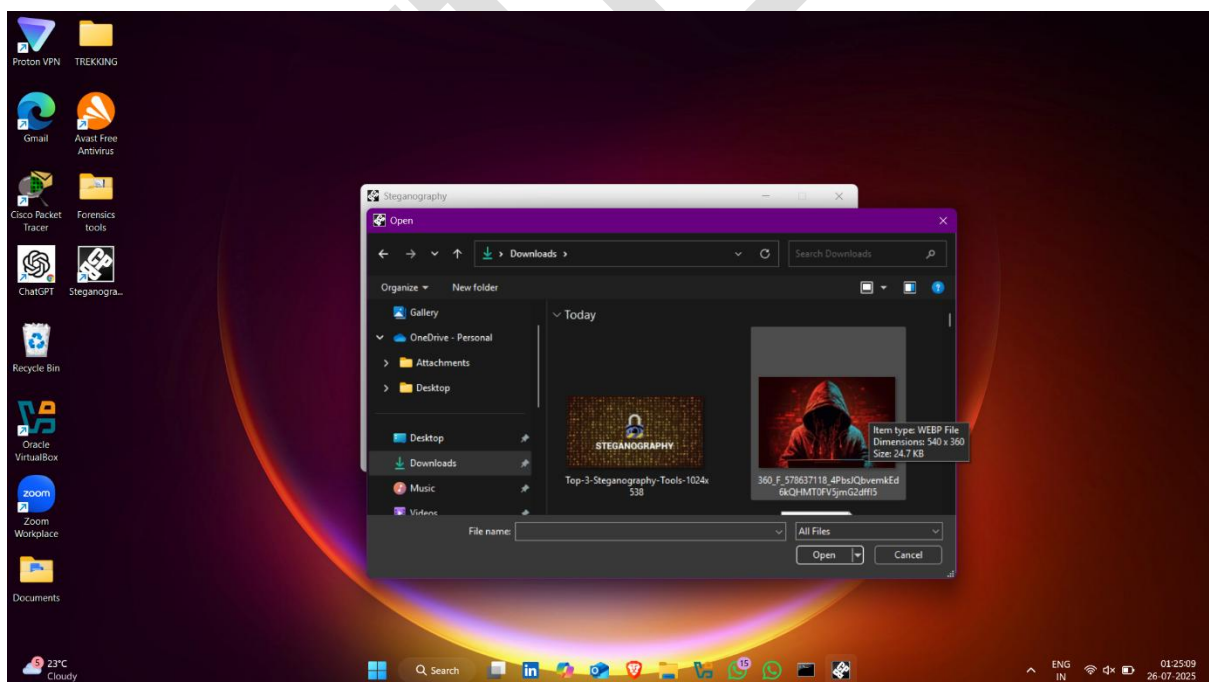
- Click on first image option



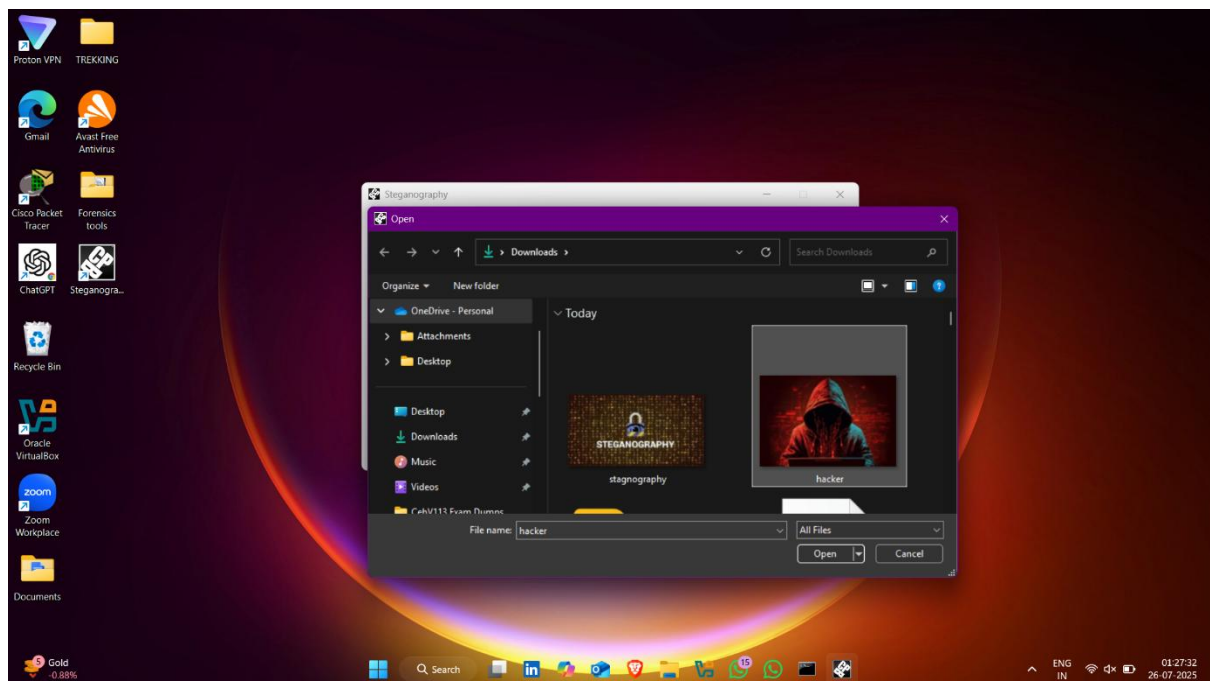
- Select image file



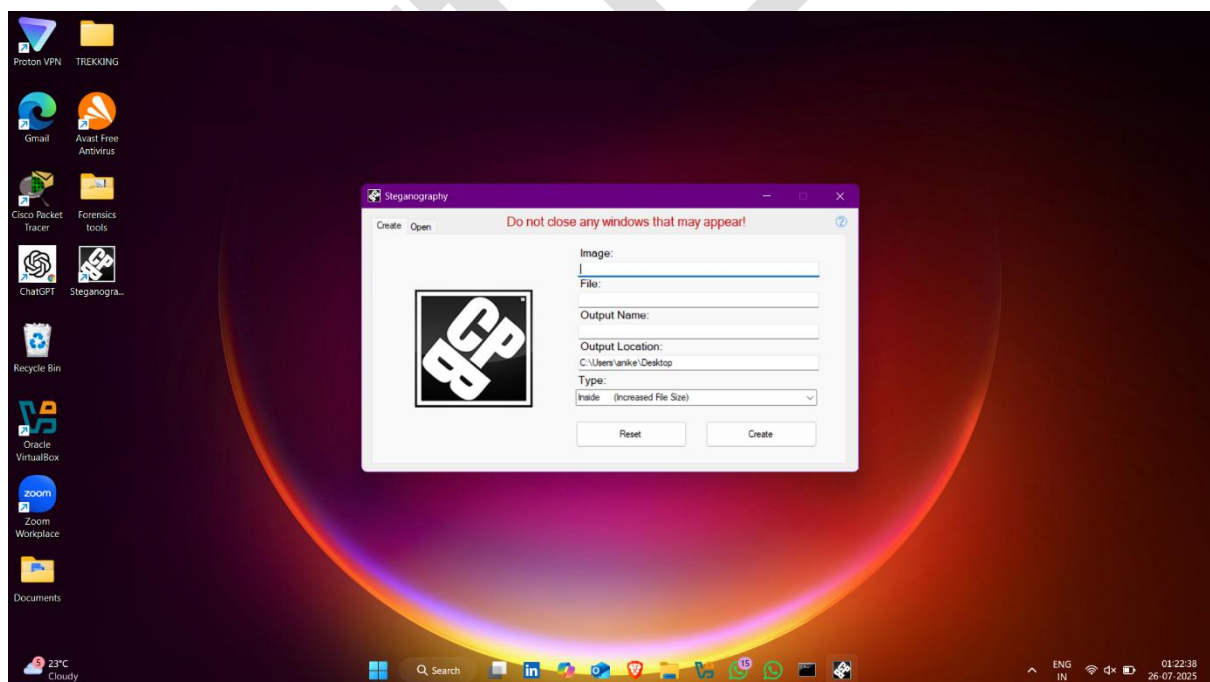
- Click on image that you want to add



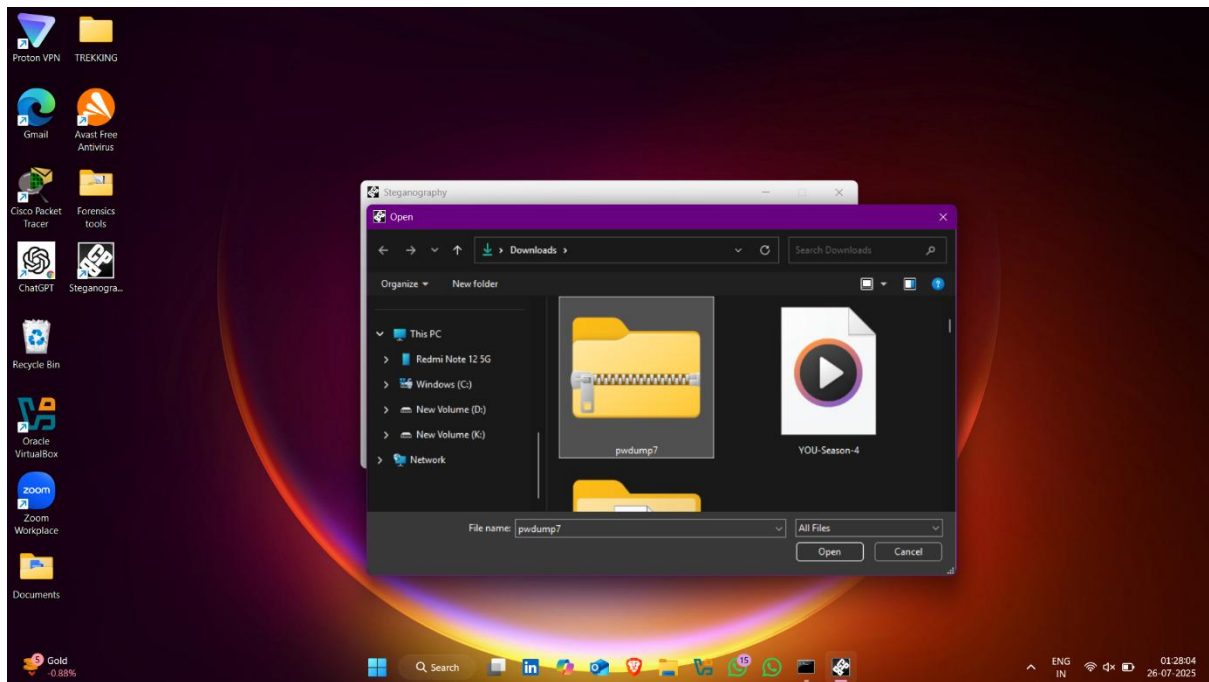
- Click on open



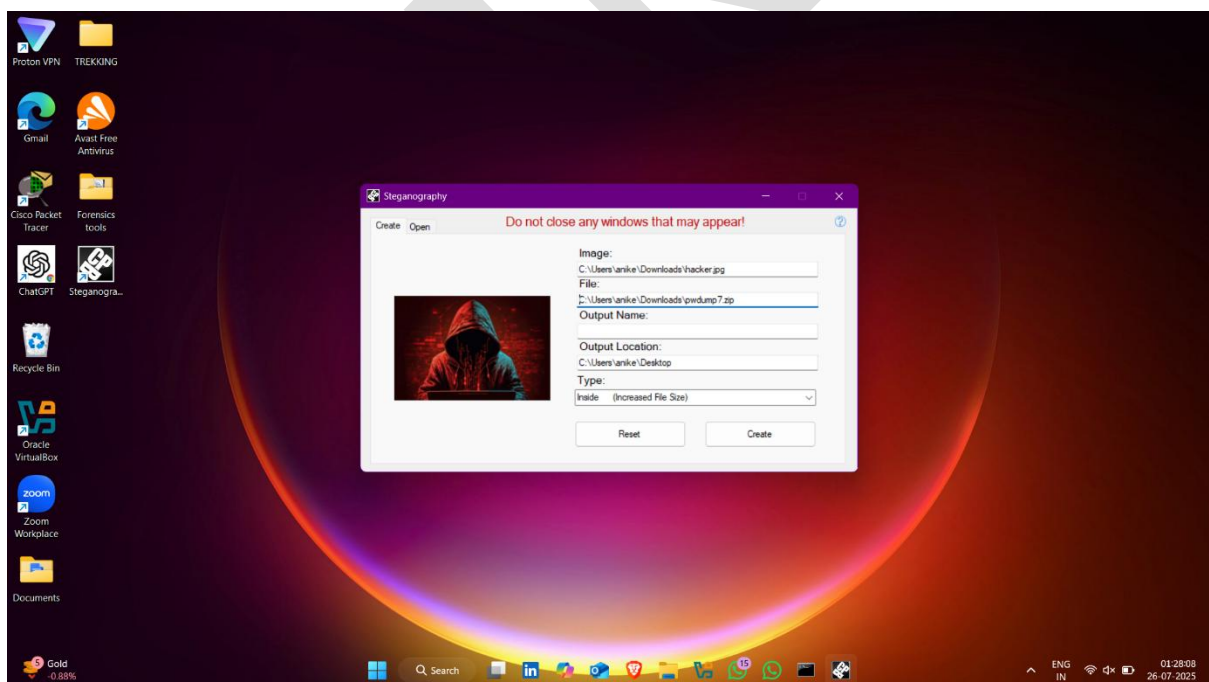
- Click on second file option



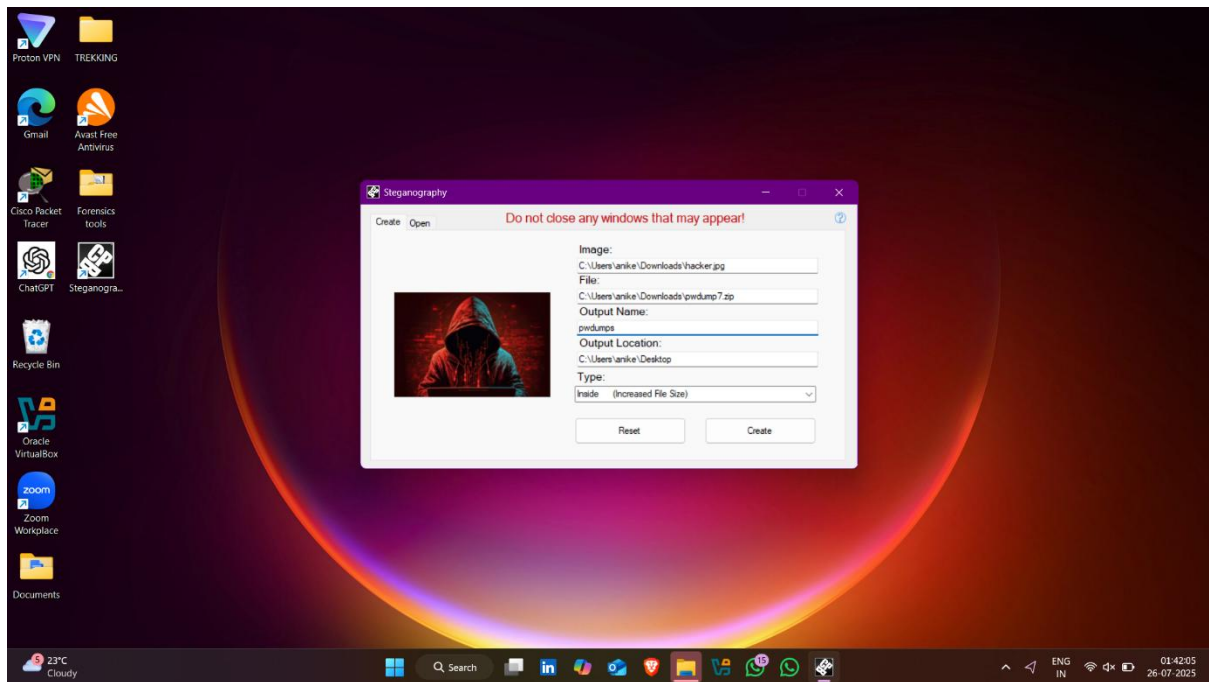
- Select zip file



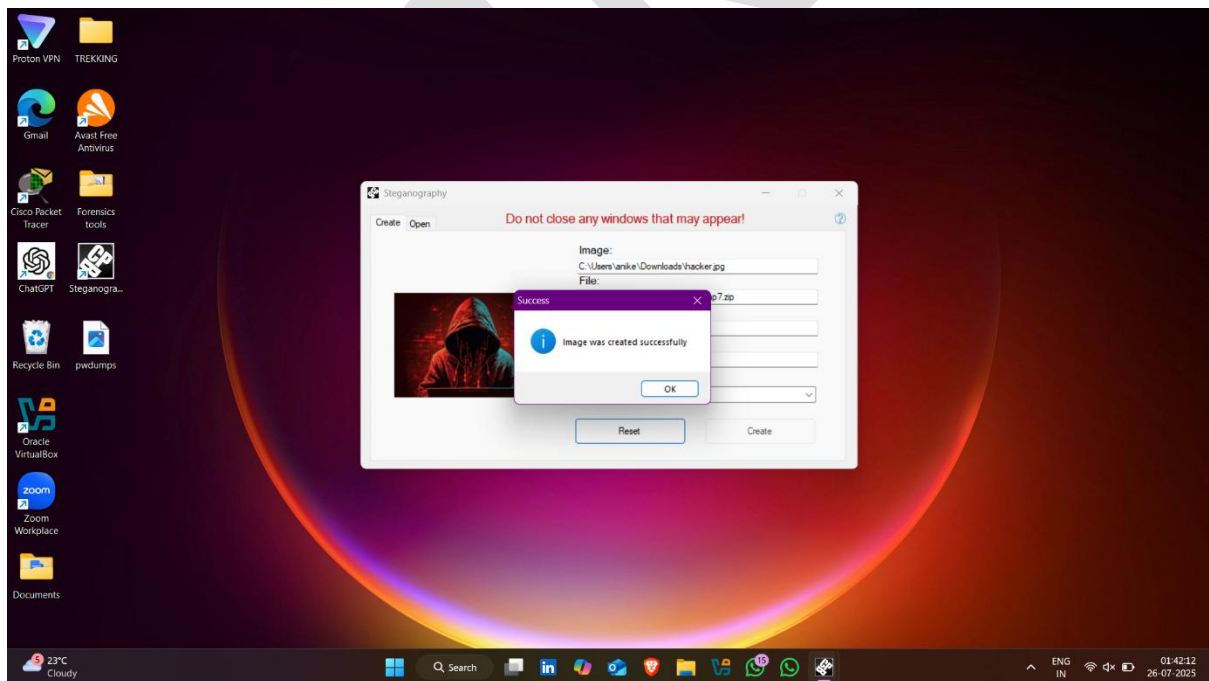
- Now set the output file name



- Click on create



- Image create successfully



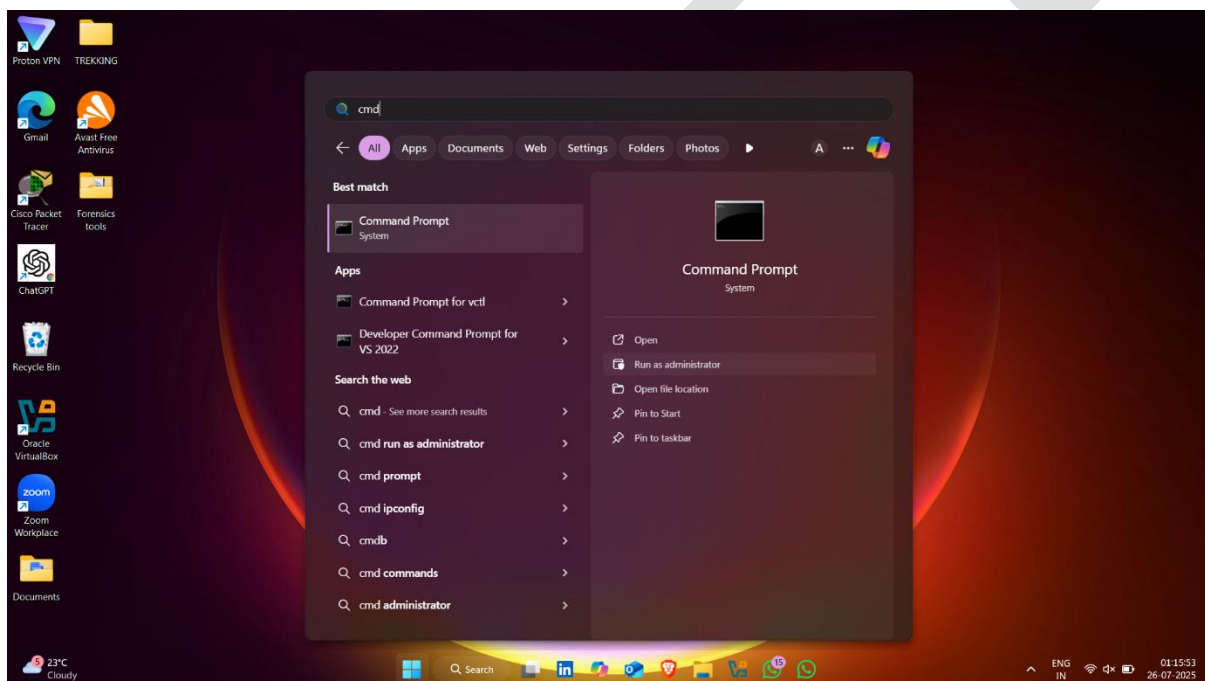
Lab-8 -: Alternate Data Streams

Alternate Data Streams (ADS) is a feature of the **NTFS (New Technology File System)** used in Windows, which allows **multiple data streams** to be associated with a single file. This means that a file can **store hidden data** alongside its main content **without changing its size or visibility in normal file listings**.

A) Create Alternate Data Stream File

How to use it -:

- Open **command prompt** and **run as administrator**

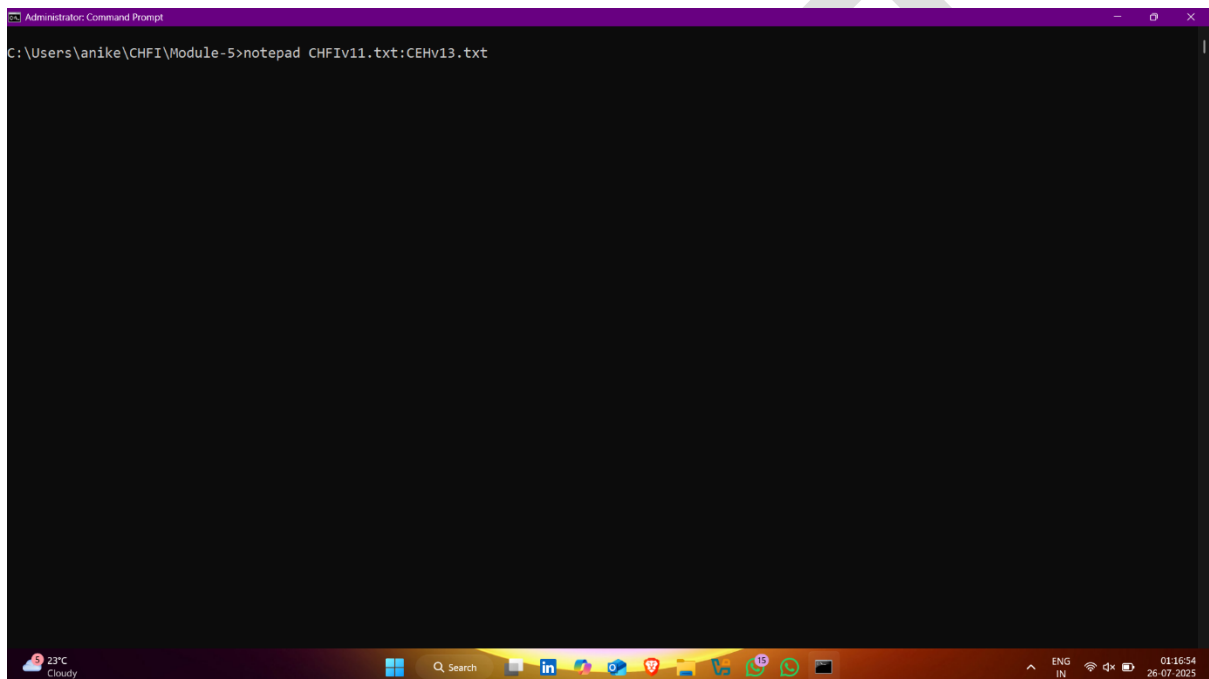


- Create a notepad file using following command

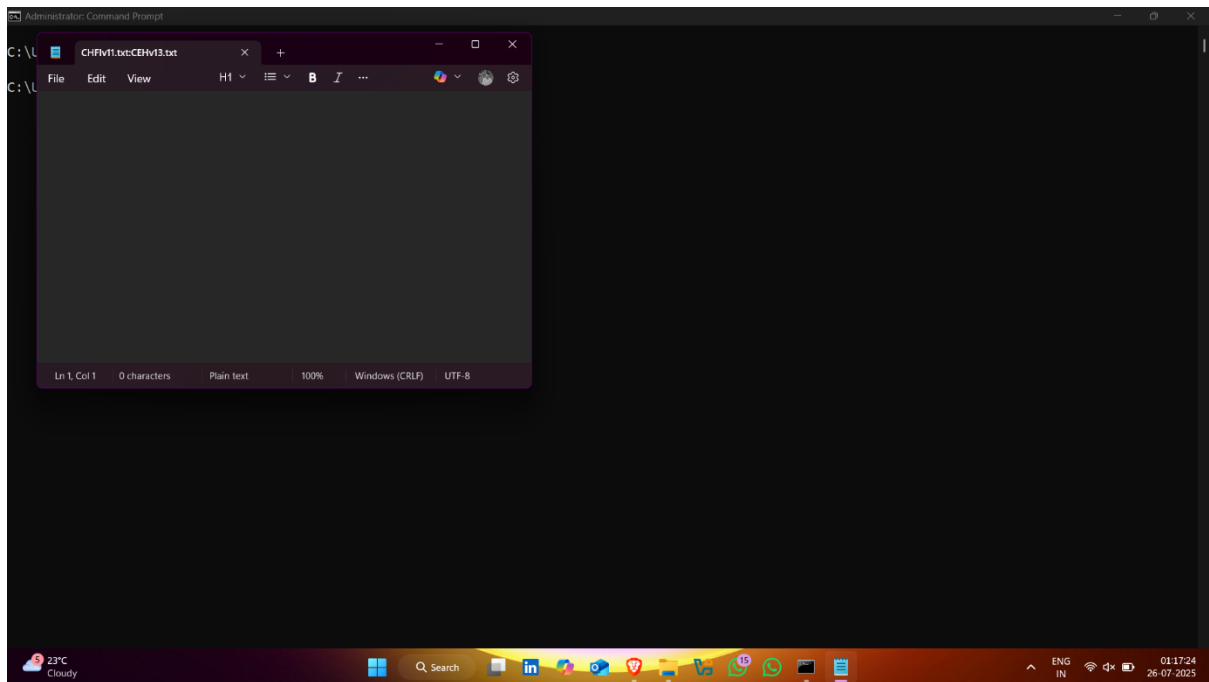
Command -: `notepad CHFIv11:CEHv13.txt`

Explanation -:

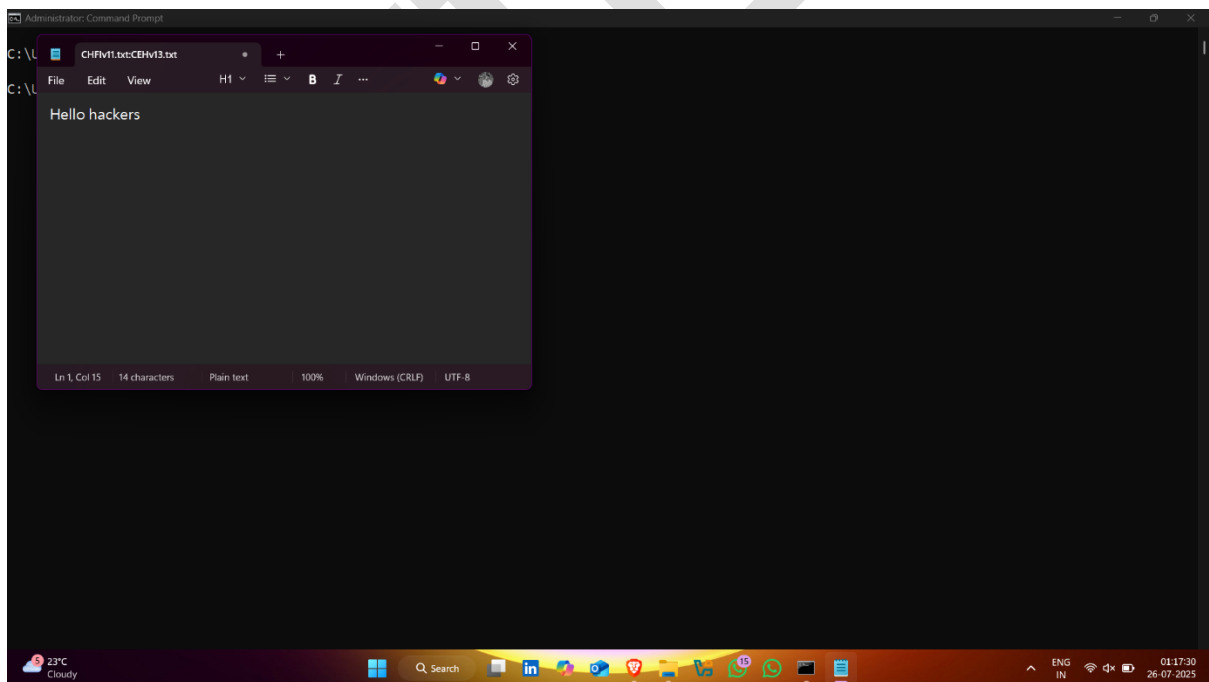
- **notepad:** This launches Notepad, a default text editor in Windows.
- **CHFIv11** is the main file (host file).
- **CEHv13.txt** is the **hidden alternate data stream** attached to CHFIv11.



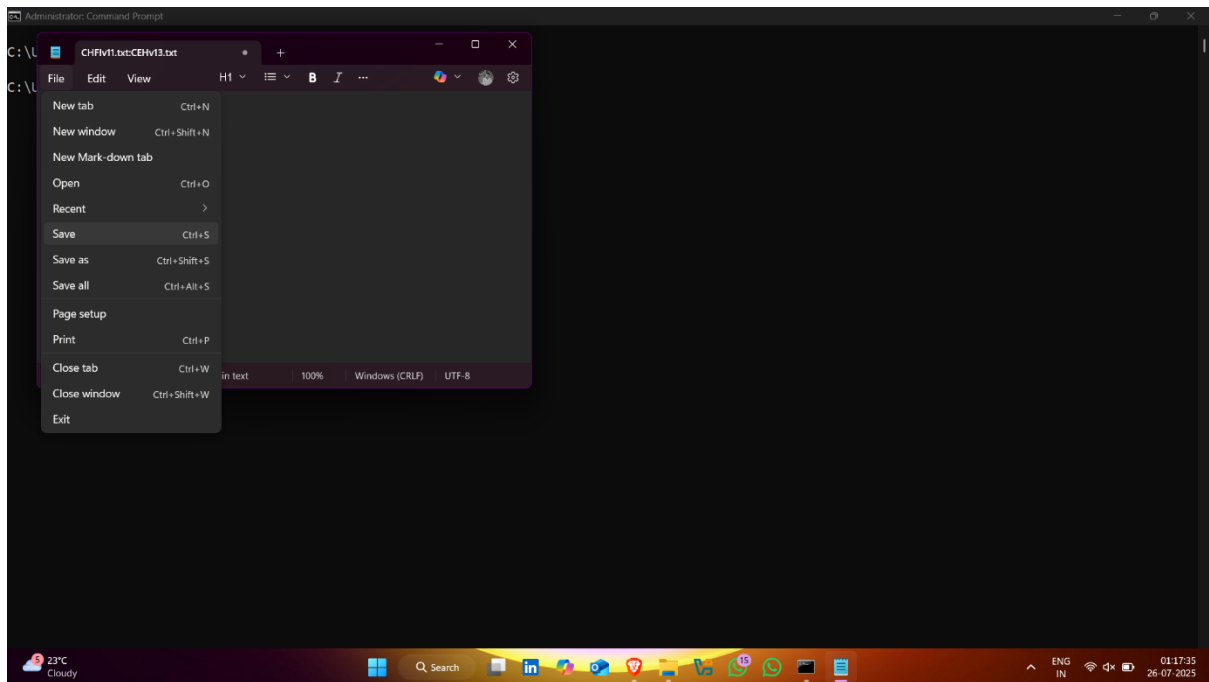
- File created



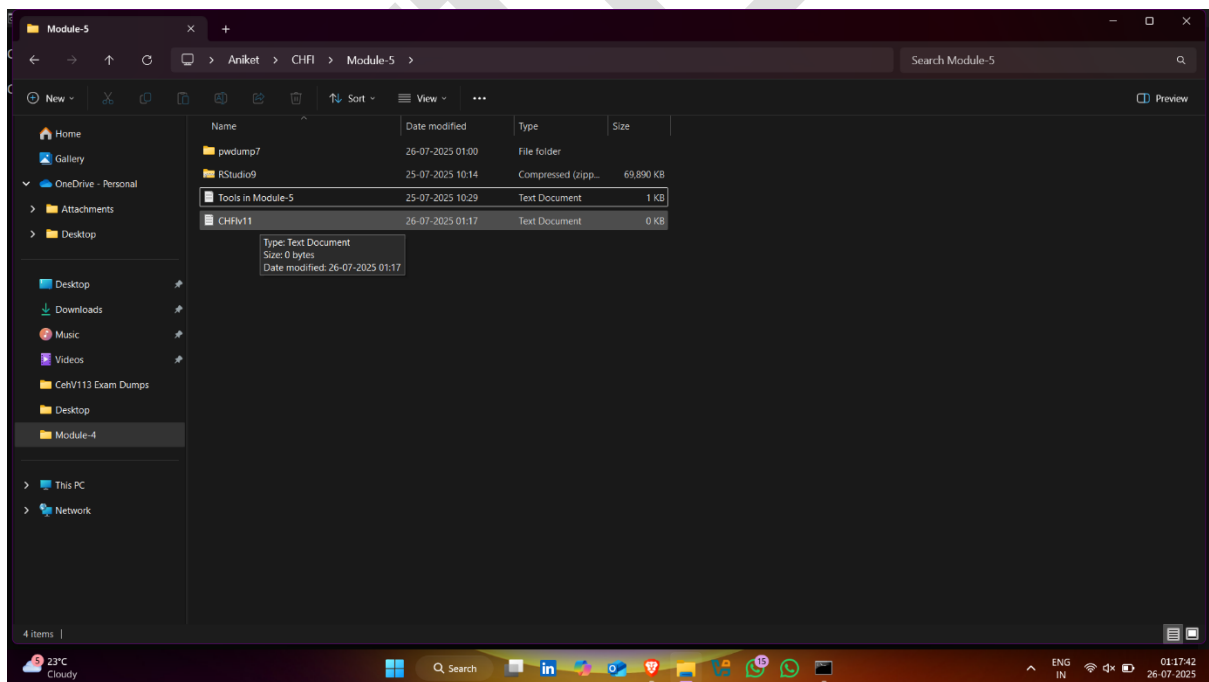
- Enter string



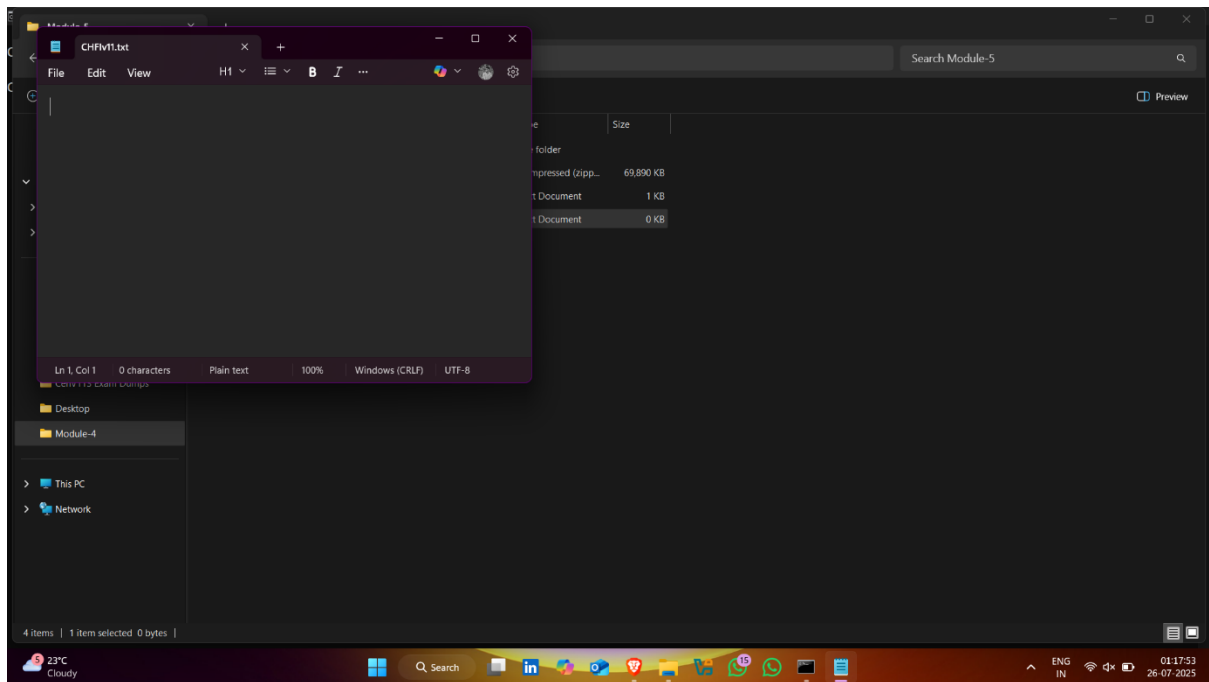
- Save this file



- Now open file



- File empty



B) Detect Alternate Data Stream File Using ADS Scanner

An **ADS Scanner tool** is a **forensic or security utility** used to **detect, list, and analyze Alternate Data Streams (ADS)** on files within the **NTFS file system** (used in Windows).

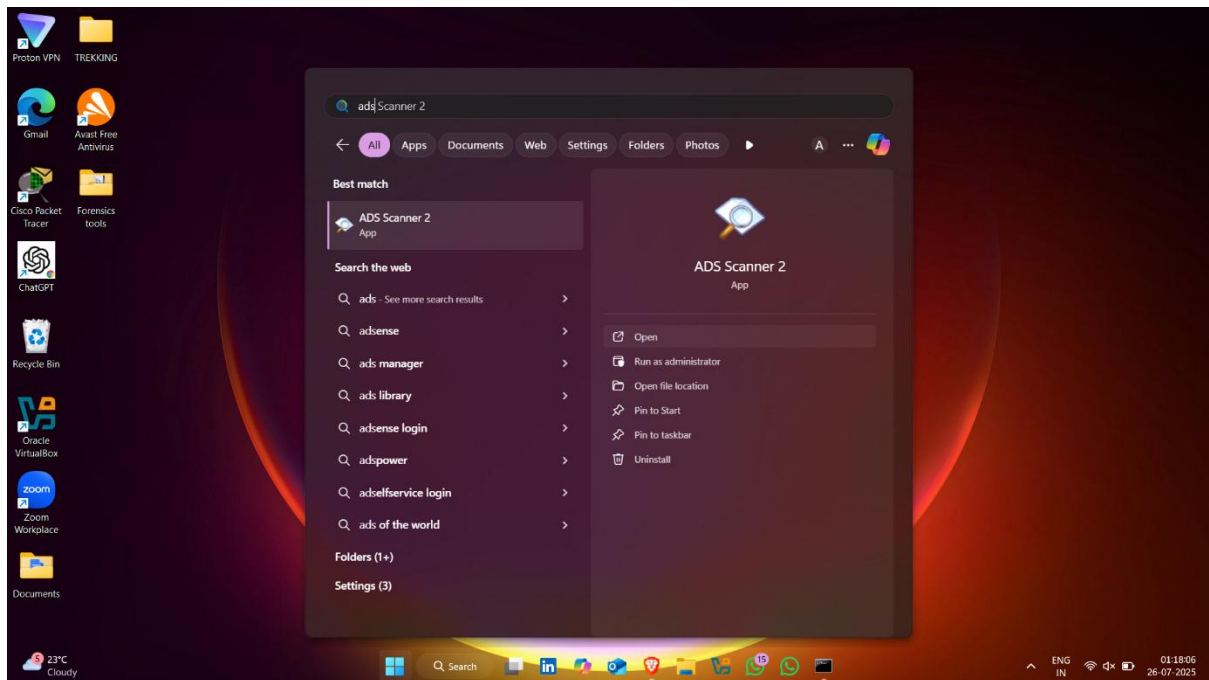
What is ADS (Alternate Data Streams)?

Alternate Data Streams are a feature of the NTFS file system that allows files to have **hidden data** associated with them, which is **not visible** in normal file browsing.

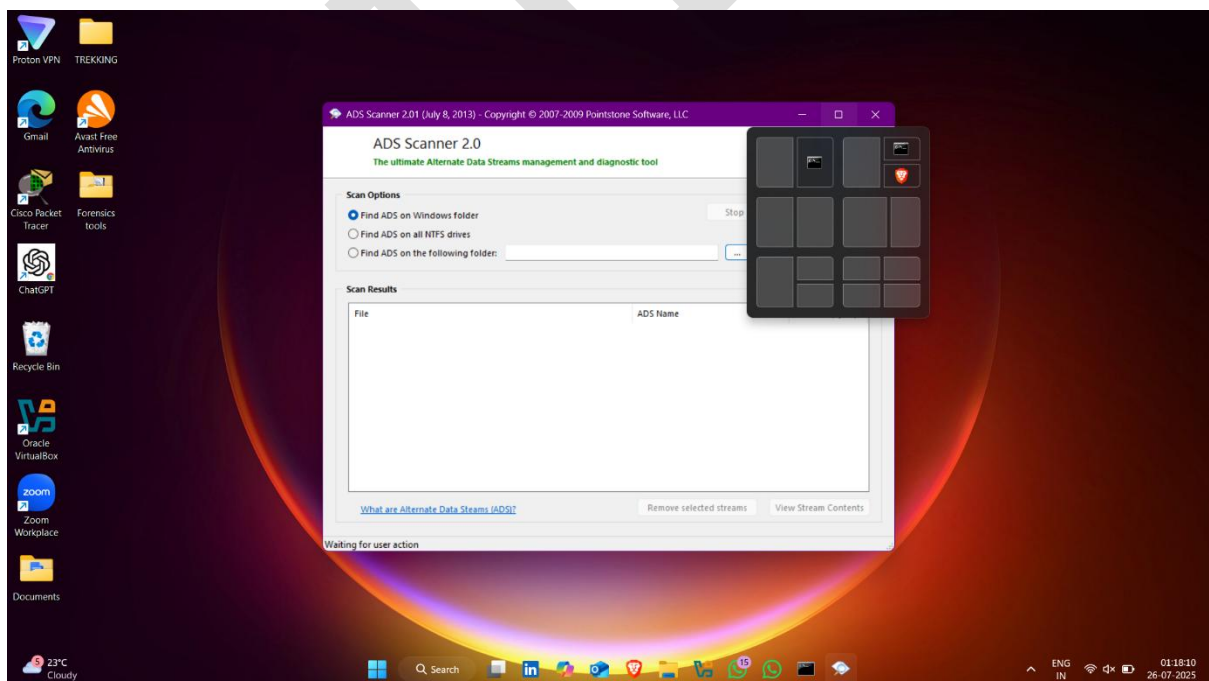
 ADS is often used to **hide malware, secret messages, or sensitive data** without modifying the main file's size or appearance.

How to use it :-

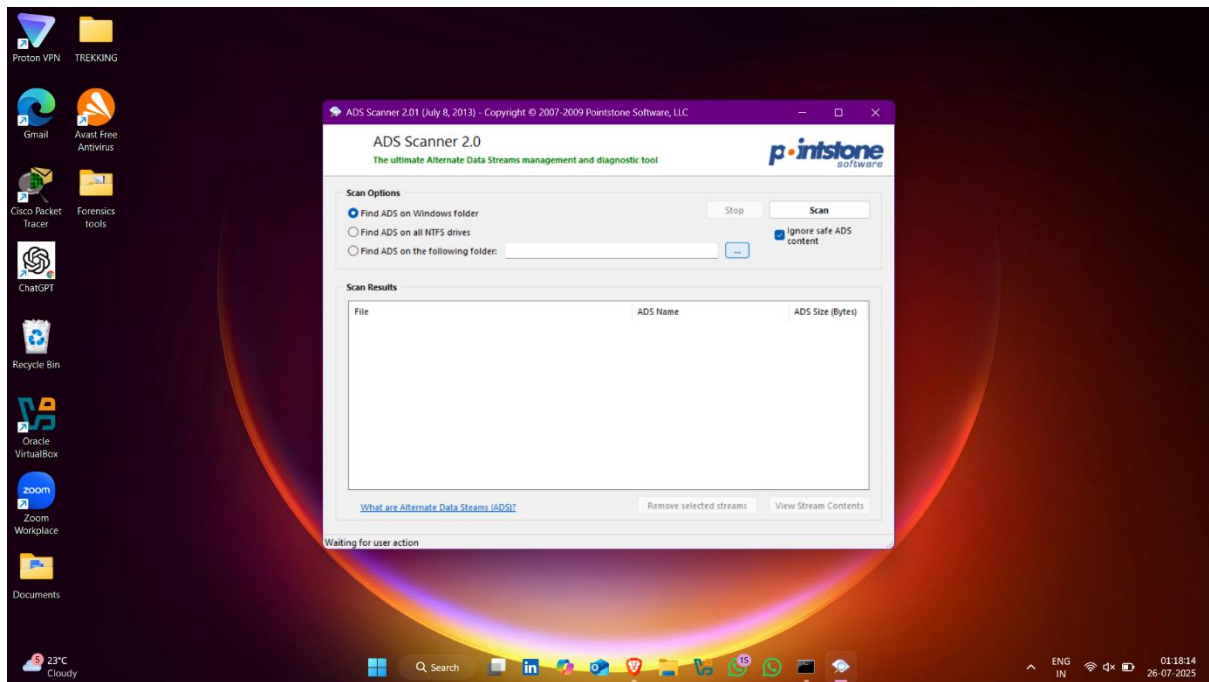
- Open Application



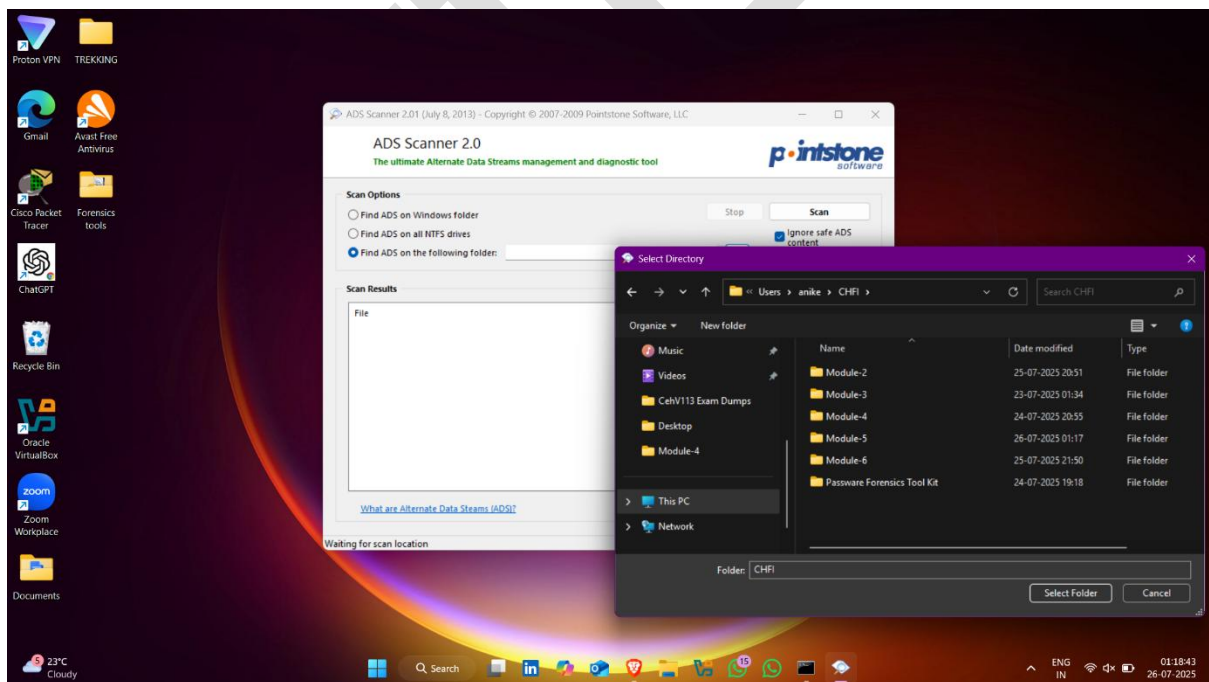
- Select third option –for specific folder



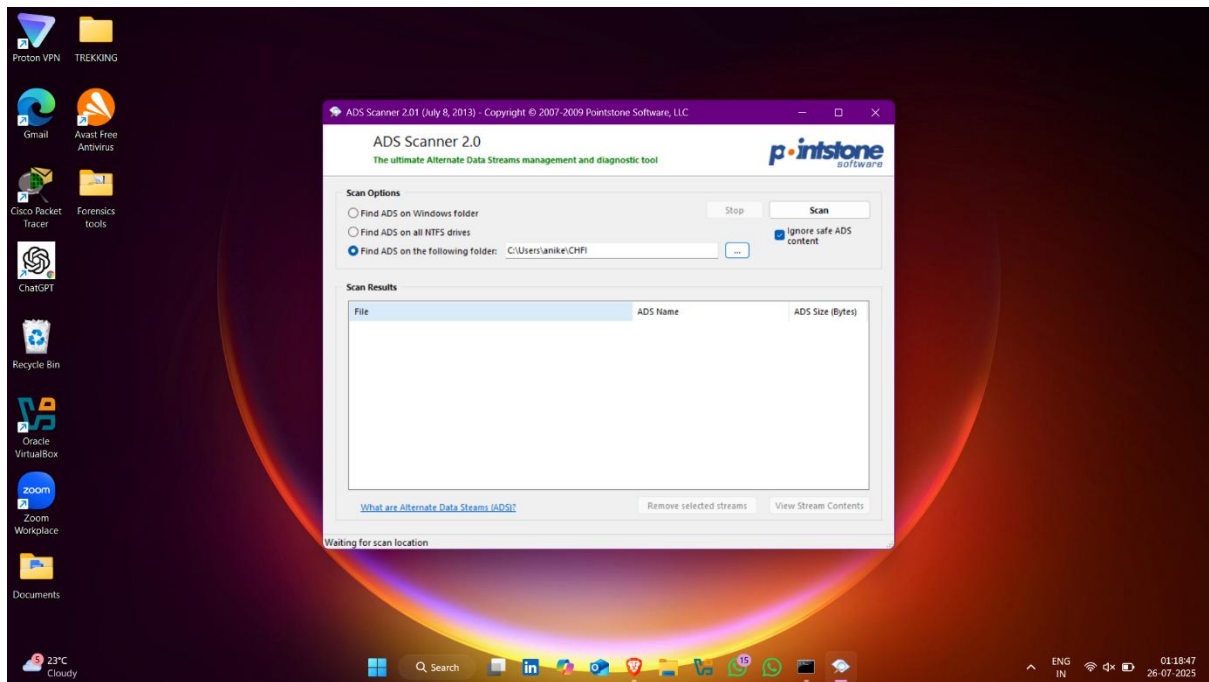
- Click on three dots



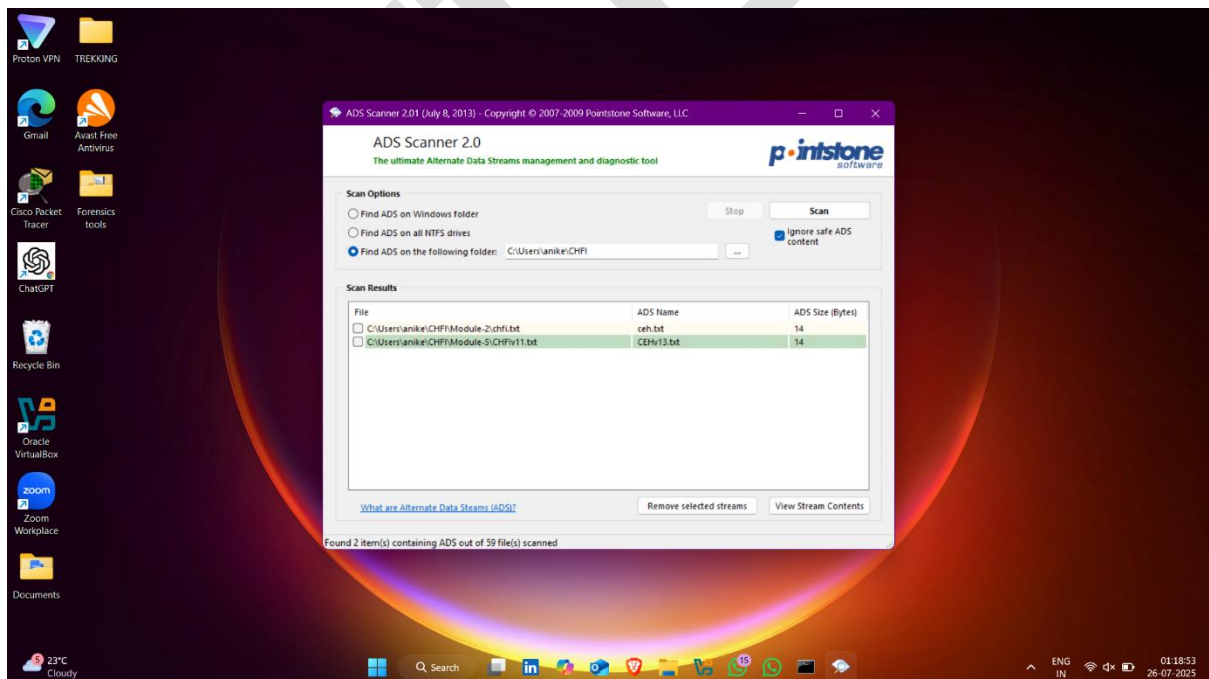
- Select Folder



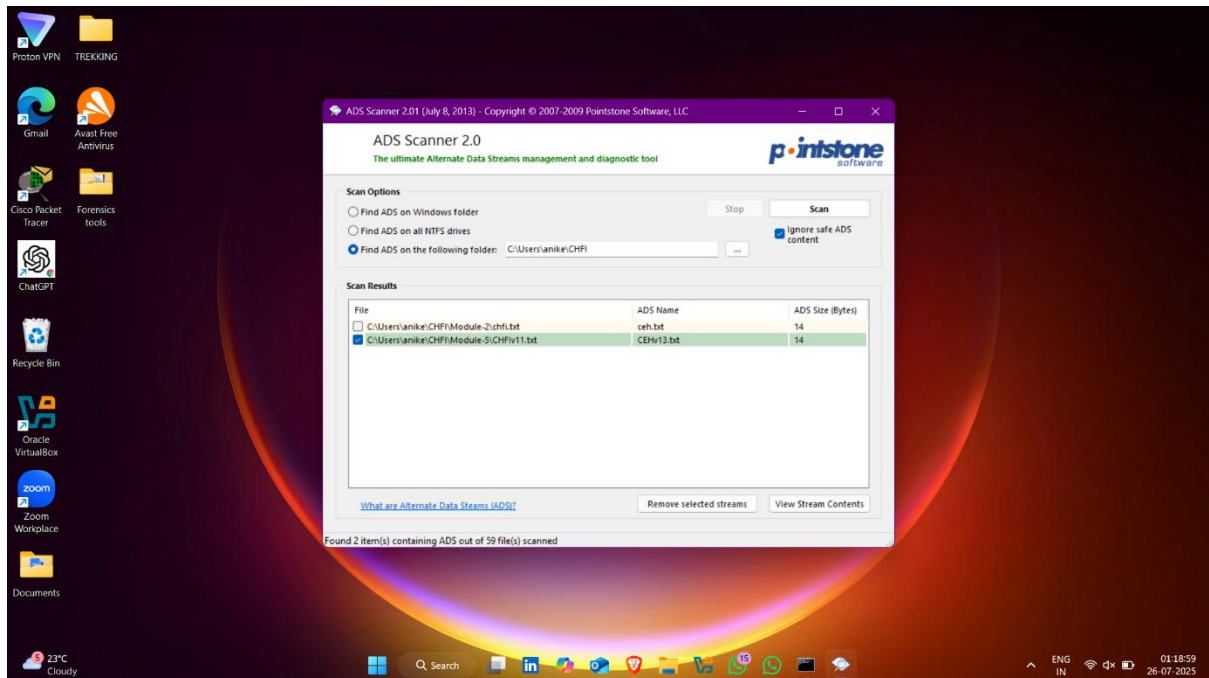
- Click on **scan**



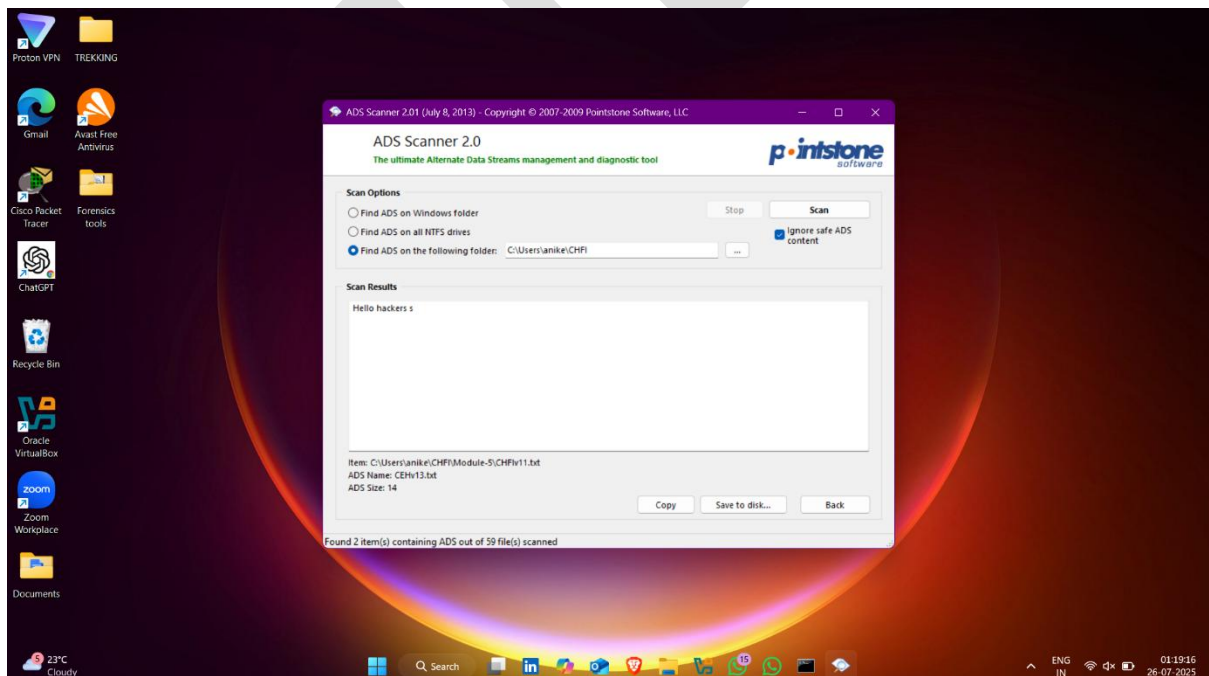
- detect hidden file ✓👉



- now view the content of the file
- click on file and then click on **View Stream Contents**



- Result ✓👉



Lab-9 -: Unpack Program Packets

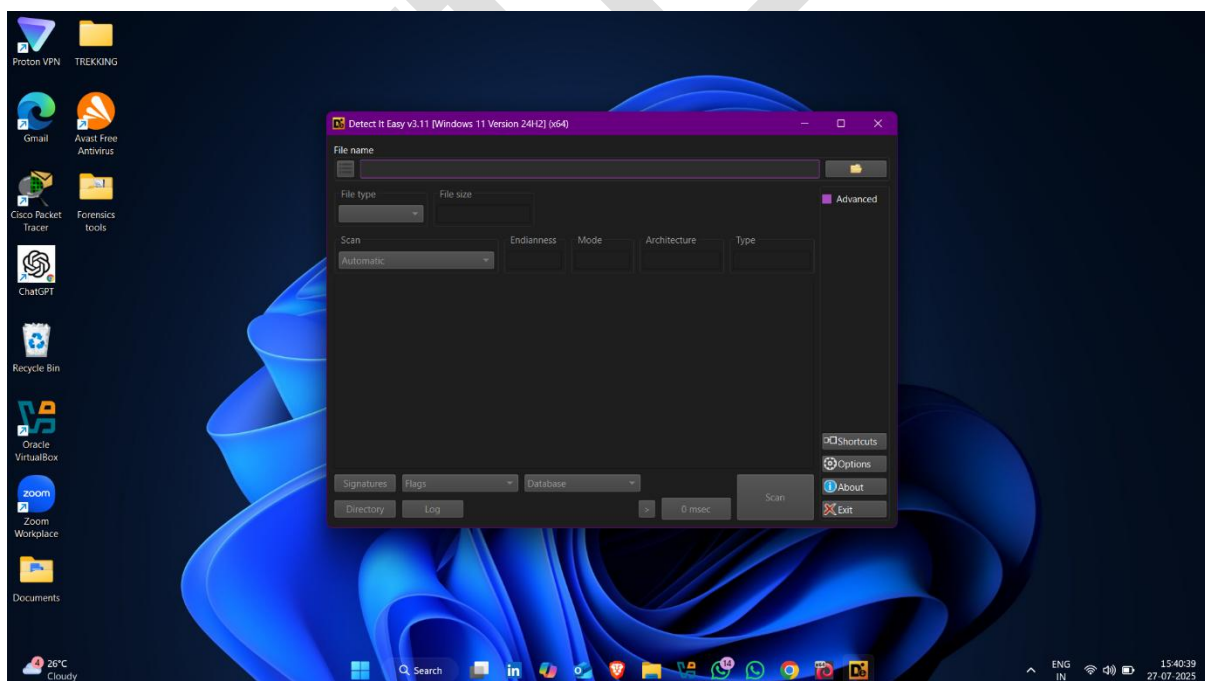
"Unpacking program packets" usually refers to **analyzing and extracting data from a packed or compressed executable** (often used in reverse engineering and malware analysis). It can also relate to **network packet analysis**, depending on the context. Below are both meanings clearly explained:

A) Unpack Program Packets Using Detect It Easy Tool

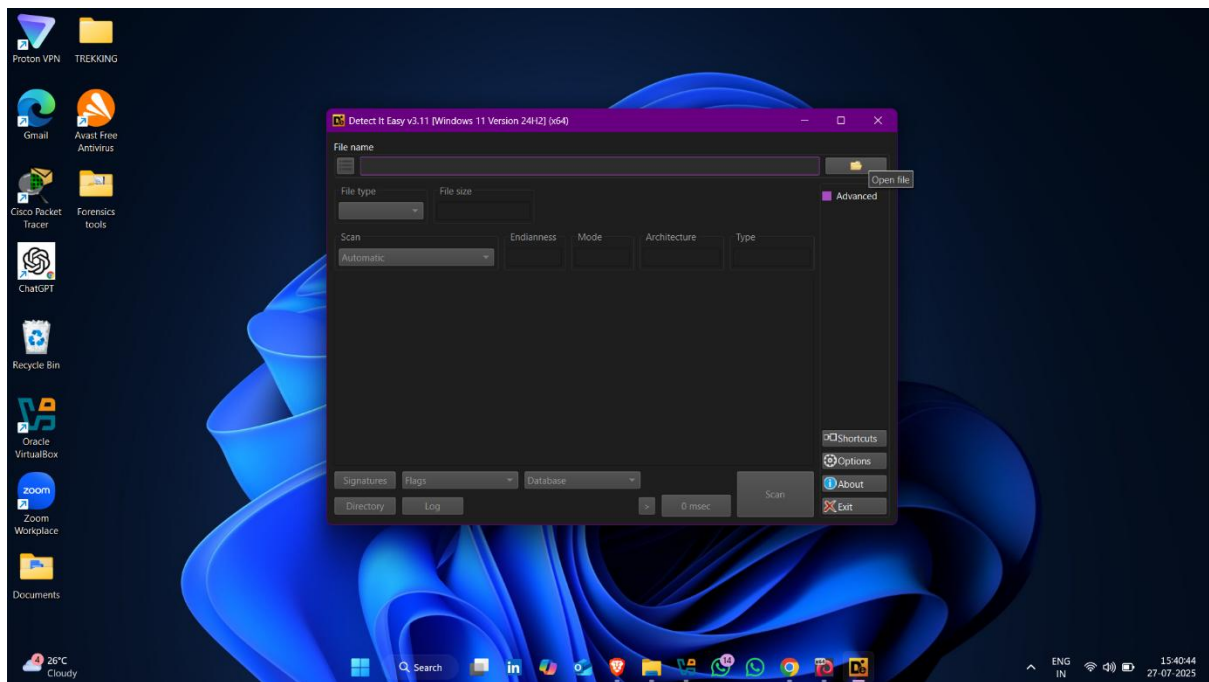
Detect It Easy (DIE) is a **portable GUI-based executable analysis tool** used by malware analysts, reverse engineers, and digital forensic investigators to identify:

How to use it -:

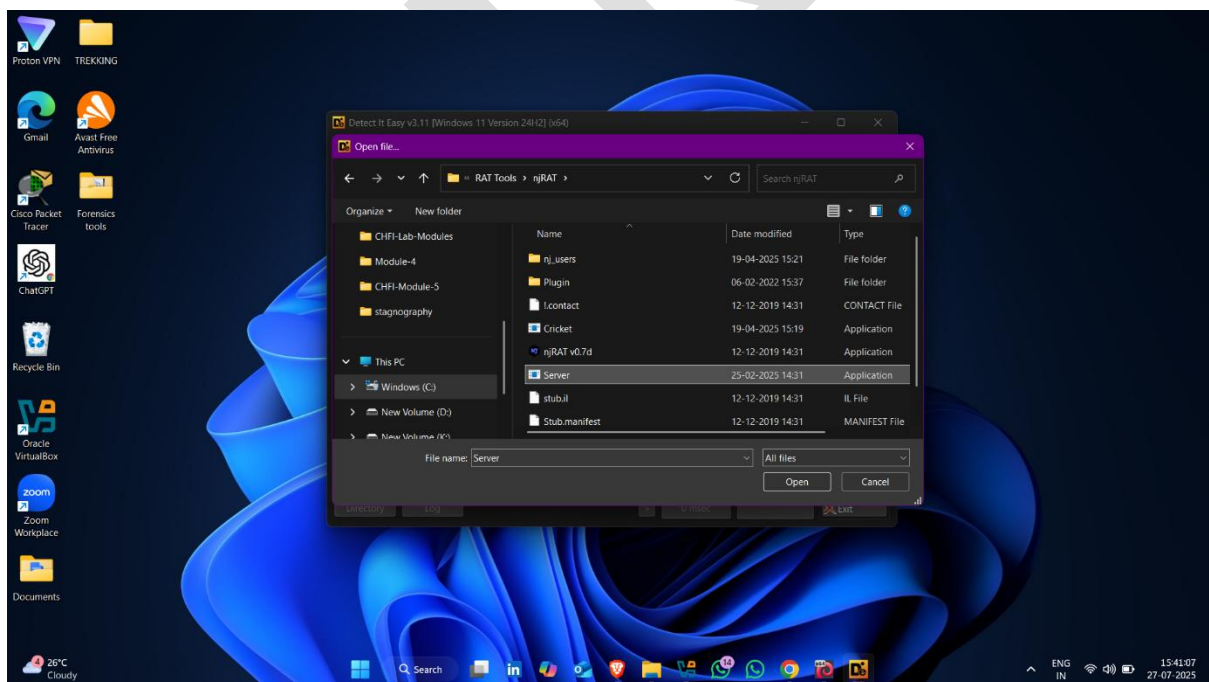
- Open Detect it easy Application



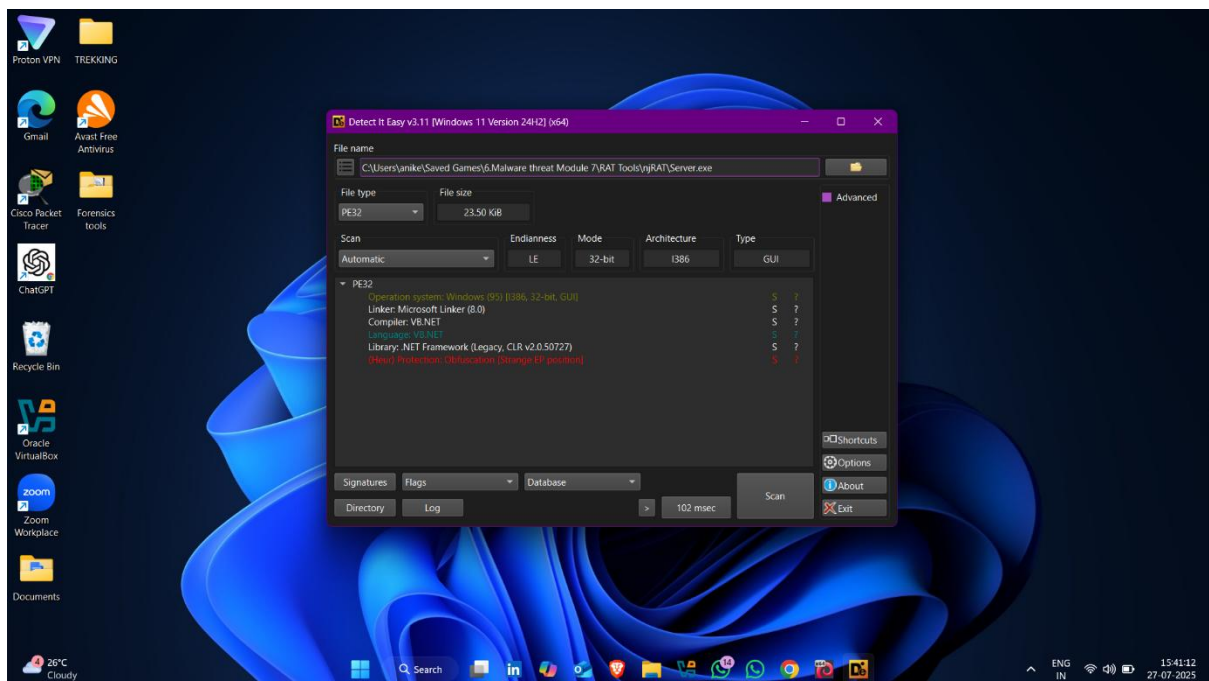
- Click on **open file**  option



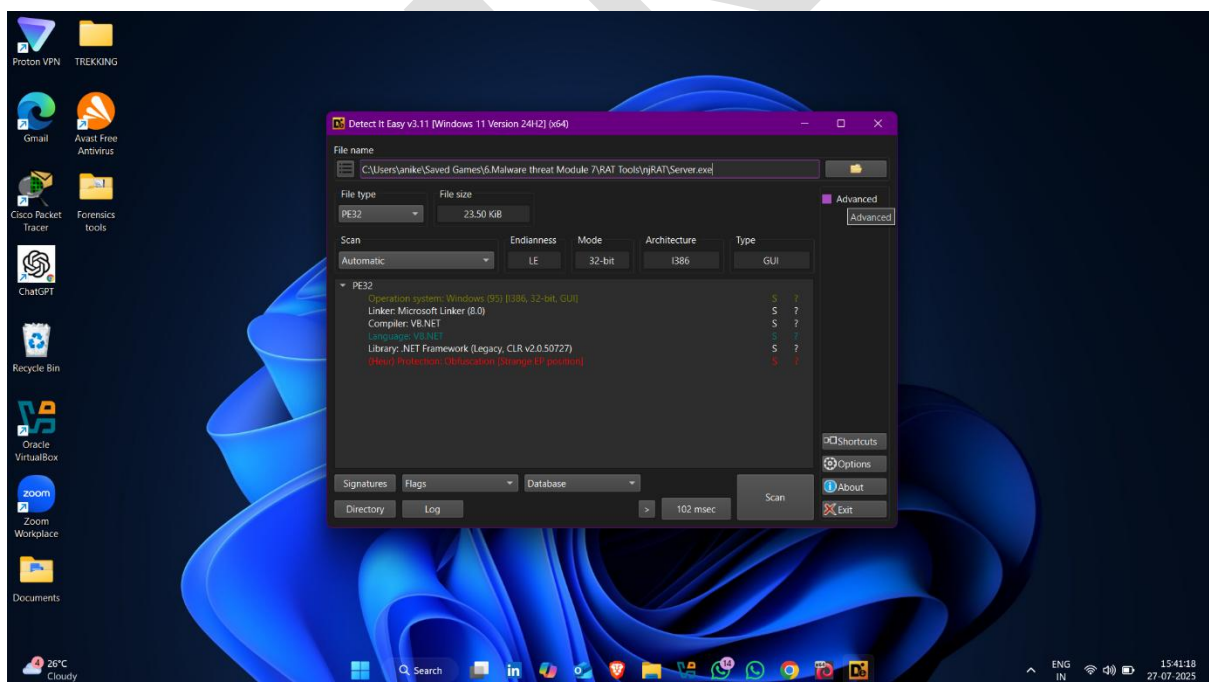
- Select file and click on **open**



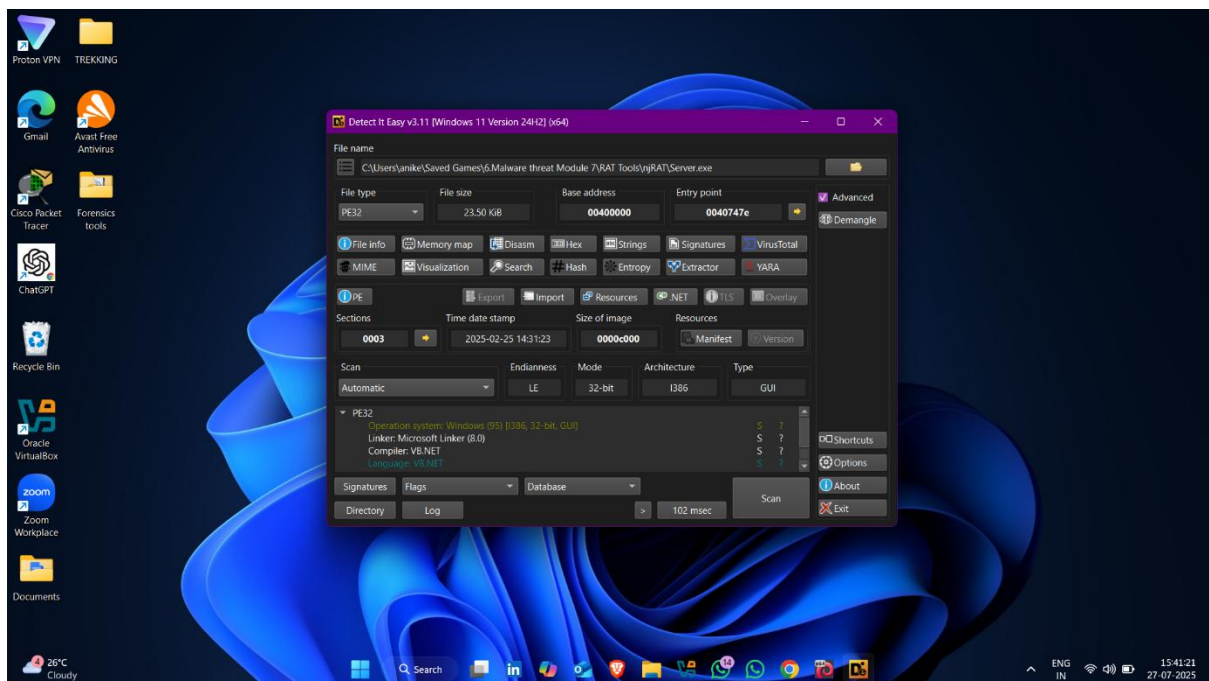
- File added



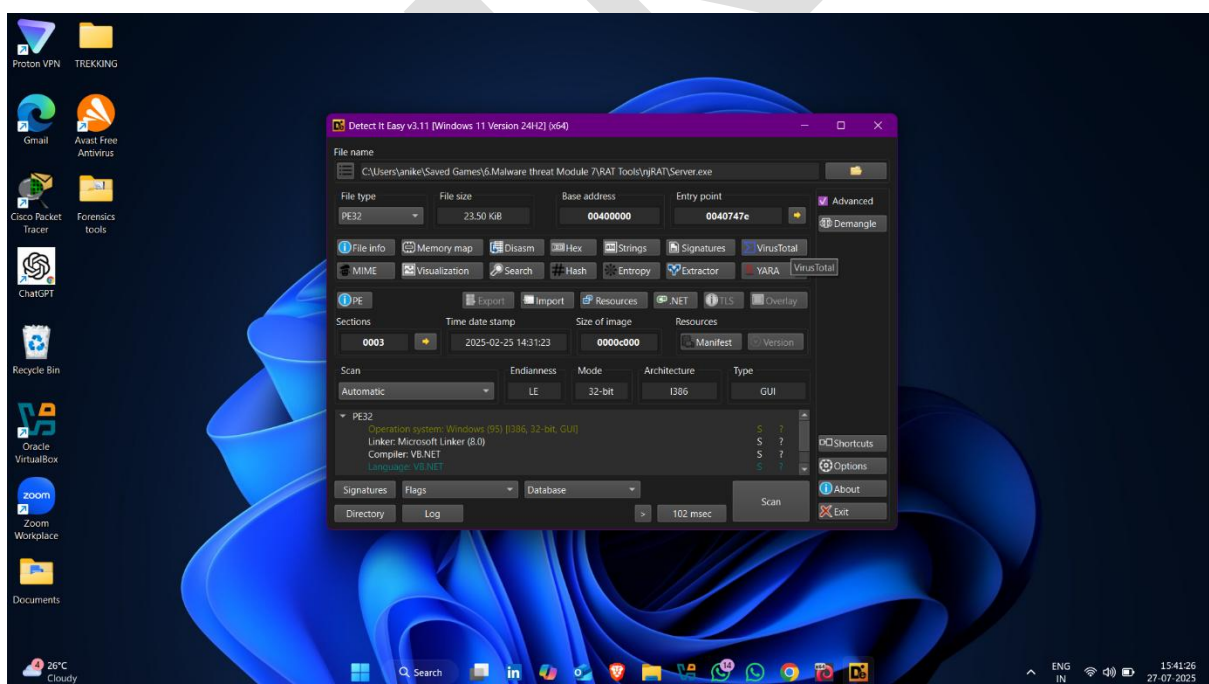
- Click on advanced option



- Multiple option are available for scan file



- Click on virus total



• Malicious file 🖱️✅

60 / 72

Community Score

60/72 security vendors flagged this file as malicious

f48e81873036e463e8a6353af4aac23381f5a7e977b951b3d8657ce1a2e4bae

Server.exe

Size: 23.50 KB

Last Analysis Date: 3 months ago

peexe assembly detect-debug-environment long-sleeps checks-user-input persistence

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 4

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label: trojan.bladabindi/msil

Threat categories: trojan dropper

Family labels: bladabindi msil njrat

Security vendors' analysis

Acronis (Static ML)	⚠️ Suspicious	AhnLab-V3	⚠️ Win-Trojan/Zbot.24064
AllCloud	⚠️ Backdoor:Win/Bladabindi.N(dyn)	ALYac	⚠️ Generic.MSIL.Bladabindi.A1DE1E9F
Antiy-AVL	⚠️ Trojan(Backdoor)/MSIL.Bladabindi.as	Arcabit	⚠️ Generic.MSIL.Bladabindi.A1DE1E9F
Arctic Wolf	⚠️ Unsafe	Avast	⚠️ MSIL-Agent-DRD [Trj]
AVG	⚠️ MSIL-Agent-DRD [Trj]	Avira (no cloud)	⚠️ TR/Dropper.Gen7

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